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(54) **QUANTITATIVE TRAIT LOCI ASSOCIATED WITH HERMAPHRODITISM IN CANNABIS**

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ABSTRACT

The invention relates to methods of characterizing a *Cannabis* spp. plant comprising quantitative trait loci (QTLs) associated with a hermaphroditism trait, and to methods of producing plants having a hermaphroditism trait of interest based on defined allelic states of polymorphisms defining the QTLs or the allelic state of a causal polymorphism provided herein. Also provided are *Cannabis* spp. plants with the hermaphroditism trait of interest comprising defined allelic states of polymorphisms and plants identified, characterized or produced by the methods described. Further provided are methods of marker assisted selection, genomic selection, marker assisted breeding, and genetic modification, for obtaining plants having a hermaphroditism trait of interest.

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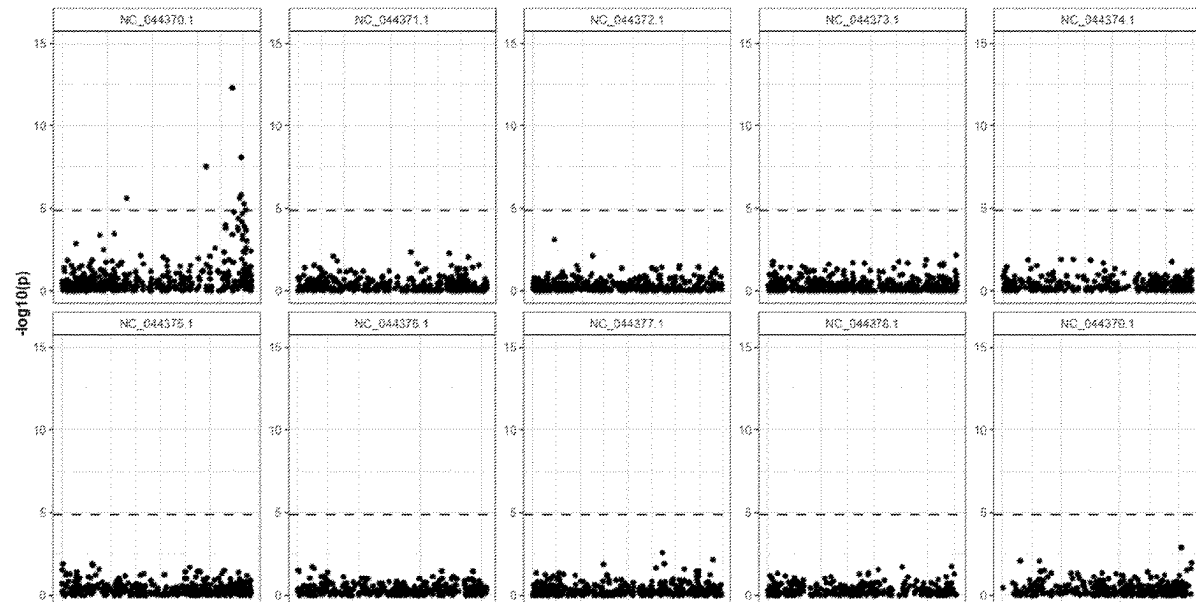
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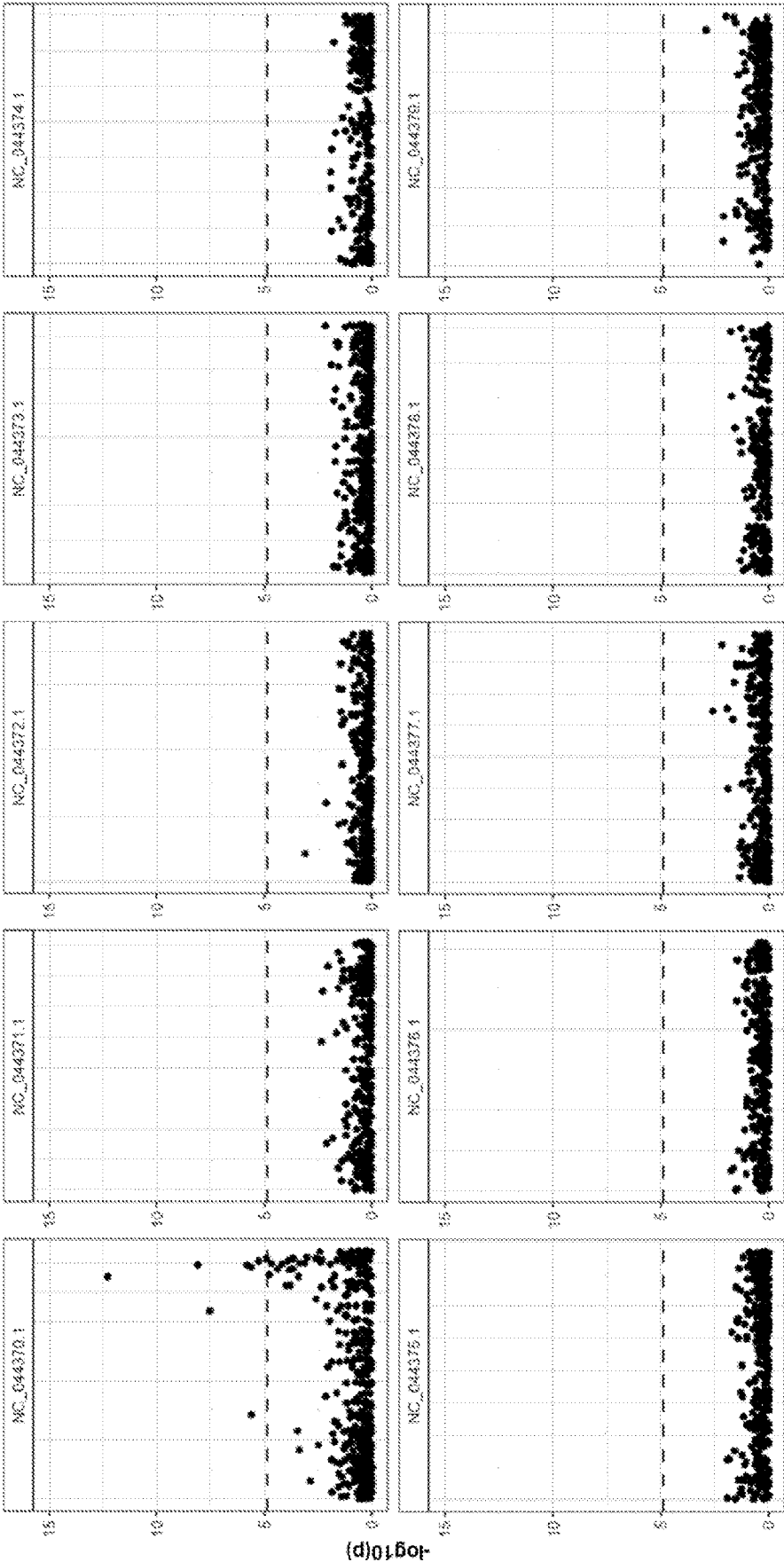


Figure 1

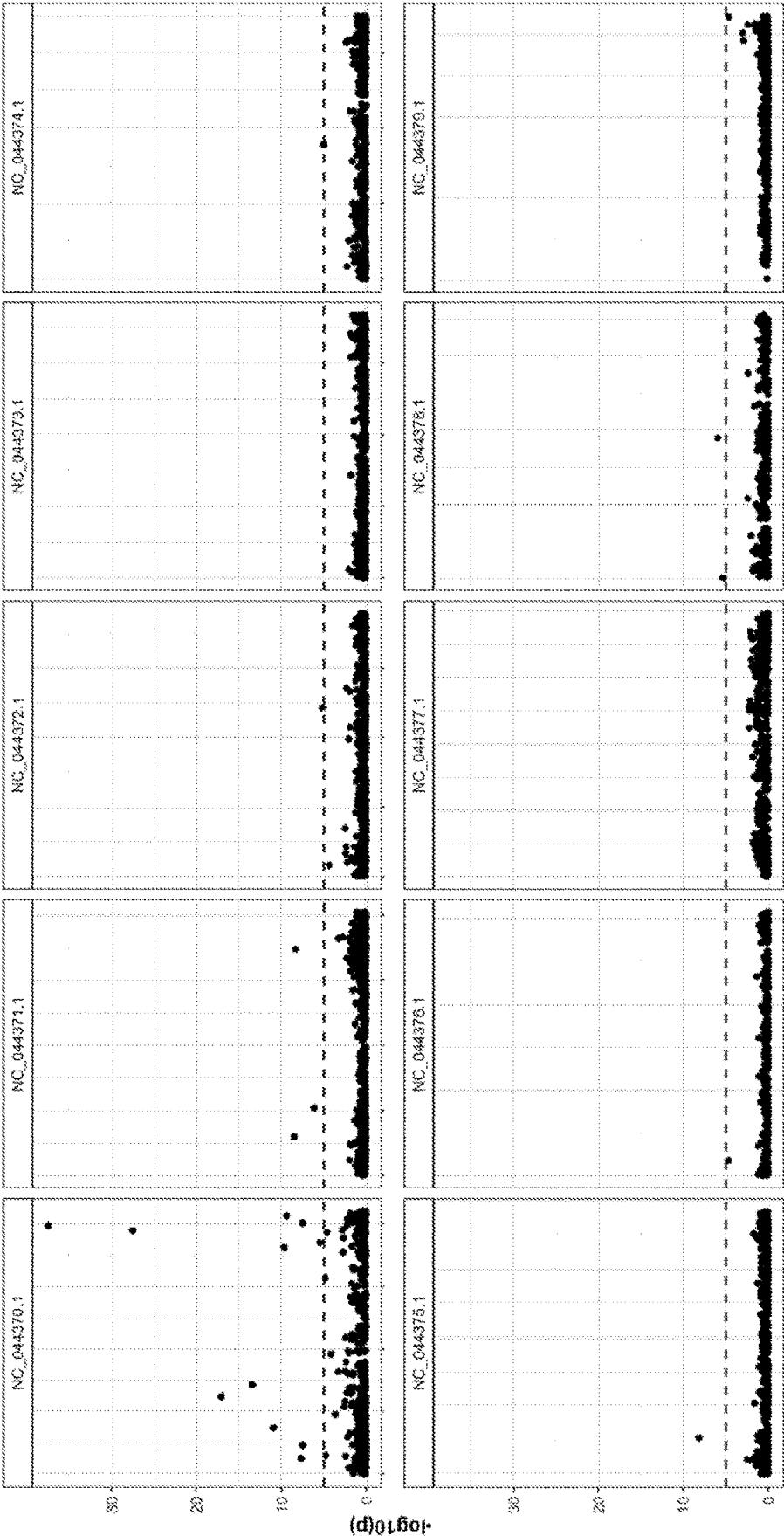


Figure 2

QUANTITATIVE TRAIT LOCI ASSOCIATED WITH HERMAPHRODITISM IN CANNABIS

BACKGROUND OF THE INVENTION

[0001] The invention relates to methods of characterizing a *Cannabis* spp. plant comprising quantitative trait loci (QTLs) associated with a hermaphroditism trait, and to methods of producing plants having a hermaphroditism trait of interest based on defined allelic states of polymorphisms defining the QTLs or the allelic state of a causal polymorphism provided herein. Also provided are *Cannabis* spp. plants with the hermaphroditism trait of interest comprising defined allelic states of polymorphisms and plants identified, characterized or produced by the methods described. The invention further provides for identification of a gene involved in the determination of the hermaphroditism trait. Further provided are methods of marker assisted selection, genomic selection, marker assisted breeding, and genetic modification, for obtaining plants having a hermaphroditism trait of interest.

[0002] Modern *Cannabis* is derived from the cross hybridization of three biotypes; *Cannabis sativa* L. ssp. *indica*, *Cannabis sativa* L. ssp. *sativa*, and *Cannabis sativa* L. ssp. *ruderalis*. *Cannabis* was divergently bred into two distinct, albeit tentative types, called Hemp and HRT (high-resin-type) *Cannabis*, respectively, which are used for different purposes. Hemp is primarily used for industrial purposes, for example in feed, food, seed, fiber, and oil production. Conversely, HRT cannabis is largely cultivated and bred for high concentrations of the pharmacological constituents, cannabinoids, derived from resin in the trichomes. Biomass, including the leaf and stem, of cannabis can also be an important source of cannabinoids.

[0003] *Cannabis* is the only species in the plant kingdom to produce phytocannabinoids. Phytocannabinoids are a class of terpenoid acting as antagonists and agonists of mammalian endocannabinoid receptors. The pharmacological action is derived from this ability of phytocannabinoids to disrupt and mimic endocannabinoids. Due to its psychoactive properties, one cannabinoid, delta-9-tetrahydrocannabinol (THC), the decarboxylation product of the plant-produced delta-9-tetrahydrocannabinolic acid (THCA), has received much attention in illegal or unregulated breeding programs, with modern HRT varieties having THC concentrations of 0.5% to 30%.

[0004] *Cannabis* is a dioecious plant, having male and female flowers produced on separate plants. In HRT cannabis production, only female plants are cultivated because unpollinated female flowers are valued for their high cannabinoid content. Male plants are a source of unwanted pollen and are excluded from production environments. Because there are no good methods for selecting female cannabis seedlings based on morphological indicators of gender, molecular markers would be useful for distinguishing male and female plants at the seedling stage.

[0005] In *Cannabis* sex determination is guided largely by sex chromosomes, where male plants have X and Y chromosomes, while female plants have two X chromosomes. Gender in cannabis is labile and can be influenced by a number of factors, including epigenetics, hormonal changes, and environmental stresses, such as changes in daylength. As such, *Cannabis* plants can produce male flowers in the absence of a Y chromosome. Hermaphroditic inflorescence, either staminate inflorescences produced on female plants or

the formation of anthers in pistillate flowers, can be triggered by environmental stresses such as prolonged periods of darkness during the flowering stage, inadequate nutrition, and low temperature. Sex determination is distinct from hermaphroditism as both processes are regulated by differing genetic mechanism, and where hermaphroditism can be influenced by environmental factors.

[0006] The emergence of hermaphroditic inflorescence can be a challenge to growers in female cannabis production systems because pollen from male flowers can result in production losses caused by the presence of seeds in flowers and because the hermaphroditic plants are of low or no value in these production systems. In cannabis production, hermaphroditic flowers are common, can be difficult to detect and their emergence can cause significant losses in HRT cannabis production. Additionally, the propensity for hermaphroditism may not be detected for several generations of selection as the genetic and environmental conditions may mask it, leading to an additional complication for breeders of cannabis.

[0007] Hermaphroditism in *Cannabis* can be induced by different chemical applications including silver nitrate (AgNO_3), silver thiosulfate solution (STS), and gibberellic acid (GA_3). There is indication that STS and GA_3 act as ethylene antagonists to promote the production of male flowers on female plants. This is particularly useful in producing all female progeny “feminized seeds” that are generated when male flowers are induced in a female plant in order to produce pollen to fertilize female flowers. Commercially, “feminized” seeds are more expensive to produce than normal dioecious seeds due to demand for all-female seeds and because they require technical expertise to produce.

[0008] The changes in phytohormone level that would accompany these stresses may underly the induction of hermaphroditic flower formation through signalling pathways connected to ethylene. However, there is nothing known about the precise signalling pathways or genetic control of hermaphroditism in cannabis. The identification of molecular markers to identify and select for and against cannabis plants that may be predisposed to producing hermaphroditic flowers would be a significant contribution to the cannabis industry in eradicating such flowers in female production systems. In addition, hermaphroditism in hemp may be advantageous in production systems for seed and fibre. Controlling the emergence of hermaphrodite flowers may also be important to developing novel breeding tools.

[0009] The present invention relates to markers and putative genes for the control of hermaphroditism in cannabis, with the aim of removing plants with an increased propensity for hermaphroditism from breeding populations.

SUMMARY OF THE INVENTION

[0010] The present invention relates to methods of characterizing a *Cannabis* spp. plant comprising one or more quantitative trait loci (QTLs) associated with a hermaphroditism trait, and to methods of producing plants having a hermaphroditism trait of interest based on defined allelic states of polymorphisms defining the QTLs or the allelic state of a causal polymorphism provided herein. Also provided are *Cannabis* spp. plants with the hermaphroditism trait of interest comprising QTLs defined by specific allelic states of polymorphisms as well as plants identified, characterized or produced by the methods described. Also pro-

vided are quantitative trait loci that control a hermaphroditic trait in *Cannabis* spp., wherein the quantitative trait loci are defined by single nucleotide polymorphisms defined herein, including a causal polymorphism for the trait, or genetic markers linked to the QTLs, as well as putative genes that control a hermaphroditism trait in a *Cannabis* spp. plant. Further provided are methods of marker assisted selection, genomic selection, marker assisted breeding, and genetic modification, for obtaining plants having a hermaphroditism trait of interest.

[0011] According to a first aspect of the present invention there is provided for a method for characterizing a *Cannabis* spp. plant with respect to a hermaphroditism trait, the method comprising the steps of: (i) genotyping at least one plant with respect to a hermaphroditism QTL by detecting (a) one or more polymorphisms associated with the hermaphroditism trait as defined in any one of Tables 1, 3 to 11 and 15; and/or (b) a polymorphism causal for the hermaphroditism trait which is an A/G SNP at position 99811627 on chromosome NC_044370.1 with reference to the cs10 reference genome; and (ii) characterizing the plant as having a hermaphroditism presence QTL or a hermaphroditism absence QTL based on the genotype at the polymorphism. In particular, the polymorphism may be selected from the group consisting of SNP markers “common_491”, “common_512”, “common_517”, “common_518”, “common_525”, “rare_57”, “rare_50”, “common_534”, “common_511”, “GBScompat_common_56”, “common_54”, “common_294”, and combinations thereof, as defined in any one of Tables 3 to 11 and 15. Further, the polymorphism may be the A/G SNP at position 99811627 on chromosome NC_044370.1 with reference to the cs10 reference genome (position 1 of SEQ ID NO:343), wherein the G allele at this position is causal for the hermaphroditism presence trait. In some embodiments, plants comprising a GG genotype at this position have an increased propensity for hermaphroditism, plants having an AA genotype at this position have a decreased propensity for hermaphroditism, and plants that are heterozygous (AG) at this position have an intermediate propensity for hermaphroditism.

[0012] In a first embodiment of the method for characterizing a *Cannabis* spp. plant with respect to a hermaphroditism trait, the genotyping may be performed by PCR-based detection, including using molecular markers, sequencing of PCR products containing the one or more polymorphisms, targeted resequencing, whole genome sequencing, or restriction-based methods, for detecting the one or more polymorphisms. While many suitable genotyping methods are known to those of skill in the art, in one embodiment, the genotyping may be performed using sequencing primers or similar molecular markers, wherein the molecular markers may be selected from the primer pairs as defined in Table 12 or 17 herein, which have been developed by the inventors of the present invention for detecting the polymorphisms provided in Tables 1, 3 to 11 and 15 herein.

[0013] According to a second embodiment of the method for characterizing a *Cannabis* spp. plant with respect to a hermaphroditism trait, the molecular markers may be designed for detecting polymorphisms at regular intervals within the hermaphroditism QTL such that recombination can be excluded.

[0014] In a third embodiment of the method for characterizing a *Cannabis* spp. plant with respect to a hermaphroditism trait, the molecular markers may be designed for

detecting polymorphisms at regular intervals within the hermaphroditism QTL such that recombination can be quantified to estimate linkage disequilibrium between a particular polymorphism and a hermaphroditic phenotype, or the absence thereof. For example, molecular markers may be for detecting polymorphisms such that recombination events can be detected to a resolution of 10'000 or 100'000 or 500'000 base pairs within the QTL.

[0015] According to a further embodiment of the method for characterizing a *Cannabis* spp. plant with respect to a hermaphroditism trait, the hermaphroditism QTL may be a quantitative trait locus having a sequence that corresponds to nucleotides 28544332 to 35677966 of NC_044370.1 with reference to the CS10 genome and defined by one or more polymorphisms associated with hermaphroditism as defined in any one of Tables 1, 3 to 11 and 15, or a genetic marker linked to the QTL; a quantitative trait locus having a sequence that corresponds to nucleotides 94129798 to 101726389 of NC_044370.1 with reference to the CS10 genome and defined by one or more polymorphisms associated with hermaphroditism as defined in any one of Tables 1, 3 to 11 and 15, or a genetic marker linked to the QTL; a quantitative trait locus defined by a sequence comprising a single nucleotide polymorphism at position 61687722 of NC_044372.1 with reference to the CS10 genome, or a genetic marker linked to the QTL; a quantitative trait locus defined by a sequence comprising a single nucleotide polymorphism at position 889040 of NC_044370.1 with reference to the CS10 genome, or a genetic marker linked to the QTL; and/or a quantitative trait locus having a sequence that corresponds to nucleotides 70255045 to 79818534 of NC_044370.1 with reference to the CS10 genome and defined by one or more polymorphisms associated with hermaphroditism as defined in any one of Tables 1, 3 to 11 and 15, or a genetic marker linked to the QTL.

[0016] According to a second aspect of the present invention there is provided for a method of producing a *Cannabis* spp. plant having a hermaphroditism trait of interest, the method comprising the steps of: (i) providing a donor parent plant having in its genome a hermaphroditism QTL characterized by (a) one or more polymorphisms associated with the hermaphroditism trait of interest as defined in any one of Tables 1, 3 to 11 and 15; and/or (b) a polymorphism causal for the hermaphroditism trait of interest which is an A/G SNP at position 99811627 on chromosome NC_044370.1 with reference to the cs10 reference genome; (ii) crossing the donor parent plant having the hermaphroditism QTL with at least one recipient parent plant to obtain a progeny population of cannabis plants; (iii) screening the progeny population of cannabis plants for the presence of the hermaphroditism QTL; and (iv) selecting one or more progeny plants having the hermaphroditism QTL, wherein the mature plant displays the hermaphroditism trait of interest.

[0017] In a first embodiment of the method of producing a *Cannabis* spp. plant having a hermaphroditism trait of interest, the method may further comprise: (v) crossing the one or more progeny plants with the donor recipient plant; or (vi) selfing the one or more progeny plants.

[0018] According to a second embodiment of the method of producing a *Cannabis* spp. plant having a hermaphroditism trait of interest, the step of screening may comprise genotyping at least one plant from the progeny population with respect to the hermaphroditism QTL by detecting one or more polymorphisms associated with the hermaphrodit-

ism trait of interest as defined in any one of Tables 1, 3 to 11 and 15 and/or for the polymorphism causal for the hermaphroditism trait of interest. In some embodiments, the progeny population of cannabis plants contains a minimum of 100, or 500, or 1000, or 10000 plants.

[0019] In a third embodiment of the method of producing a *Cannabis* spp. plant having a hermaphroditism trait of interest, the method may further comprise a step of genotyping the donor parent plant with respect to the hermaphroditism QTL by detecting one or more polymorphisms associated with the hermaphroditism trait of interest as defined in any one of Tables 1, 3 to 11 and 15 and/or the polymorphism causal for the hermaphroditism trait of interest, prior to step (i).

[0020] In another embodiment of the method of producing a *Cannabis* spp. plant having a hermaphroditism trait of interest, the genotyping may be performed by PCR-based detection using molecular markers, sequencing of PCR products containing the one or more polymorphisms, targeted resequencing, whole genome sequencing, or restriction-based methods, for detecting the one or more polymorphisms.

[0021] In some embodiments, the molecular markers may be for detecting polymorphisms at regular intervals within the hermaphroditism QTL such that recombination can be excluded or such that recombination can be quantified to estimate linkage disequilibrium between a particular polymorphism and the hermaphroditic trait of interest. For example, molecular markers may be for detecting polymorphisms such that recombination events can be detected to a resolution of 10'000 or 100'000 or 500'000 base pairs within the QTL. In some embodiments, genome sequencing, or marker-based PCR and resequencing of the QTL may be used for detecting a plurality of polymorphisms defined in any one of Tables 1, 3 to 11 and 15. In some embodiments, the molecular markers may be selected from the primer pairs provided in Table 12 or 17. In particular, the plant may be genotyped for a polymorphism selected from the group consisting of SNP markers "common_491", "common_512", "common_517", "common_518", "common_525", "rare_57", "rare_50", "common_534", "common_511", "GBScompat_common_56", "common_54", "common_294", and combinations thereof, as defined in any one of Tables 3 to 11 and 15. Further, the polymorphism may be the A/G SNP at position 99811627 on chromosome NC_044370.1 with reference to the cs10 reference genome (position 1 of SEQ ID NO:343), wherein the G allele at this position is causal for the hermaphroditism presence trait. In some embodiments, plants comprising a GG genotype at this position have an increased propensity for hermaphroditism, plants having an AA genotype at this position have a decreased propensity for hermaphroditism, and plants that are heterozygous (AG) at this position have an intermediate propensity for hermaphroditism.

[0022] In one embodiment, the hermaphroditism QTL is a hermaphroditism absence QTL, and the hermaphroditism trait of interest is a hermaphroditism absence trait.

[0023] According to a further embodiment of the method of producing a *Cannabis* spp. plant having a hermaphroditism trait of interest, the hermaphroditism QTL may be a quantitative trait locus having a sequence that corresponds to nucleotides 28544332 to 35677966 of NC_044370.1 with reference to the CS10 genome and defined by one or more polymorphisms associated with hermaphroditism as defined

in any one of Tables 1, 3 to 11 and 15, or a genetic marker linked to the QTL; a quantitative trait locus having a sequence that corresponds to nucleotides 94129798 to 101726389 of NC_044370.1 with reference to the CS10 genome and defined by one or more polymorphisms associated with hermaphroditism as defined in any one of Tables 1, 3 to 11 and 15, or a genetic marker linked to the QTL; a quantitative trait locus defined by a sequence comprising a single nucleotide polymorphism at position 61687722 of NC_044372.1 with reference to the CS10 genome, or a genetic marker linked to the QTL; a quantitative trait locus defined by a sequence comprising a single nucleotide polymorphism at position 889040 of NC_044370.1 with reference to the CS10 genome, or a genetic marker linked to the QTL; and/or a quantitative trait locus having a sequence that corresponds to nucleotides 70255045 to 79818534 of NC_044370.1 with reference to the CS10 genome and defined by one or more polymorphisms associated with hermaphroditism as defined in any one of Tables 1, 3 to 11 and 15, or a genetic marker linked to the QTL.

[0024] In a third aspect of the invention there is provided for a method of producing a *Cannabis* spp. plant comprising a hermaphroditism trait of interest, the method comprising introducing into a *Cannabis* spp. plant a hermaphroditism QTL: (a) characterized by one or more polymorphisms associated with the hermaphroditism trait of interest as defined in any one of Tables 1, 3 to 11 and 15, wherein said hermaphroditism QTL is associated with the hermaphroditism trait of interest in the plant; and/or (b) comprising a polymorphism causal for the hermaphroditism trait of interest which is an A/G SNP at position 99811627 on chromosome NC_044370.1 with reference to the cs10 reference genome (position 1 of SEQ ID NO:343).

[0025] In one embodiment of the method of producing a *Cannabis* spp. plant comprising a hermaphroditism trait of interest, introducing the hermaphroditism QTL comprises crossing a donor parent plant having the hermaphroditism QTL with a recipient parent plant.

[0026] In an alternative embodiment of the method of producing a *Cannabis* spp. plant comprising a hermaphroditism trait of interest, introducing the hermaphroditism QTL may comprise genetically modifying the *Cannabis* spp. plant. Several methods of genetic modification are known to those of skill in the art, including targeted mutagenesis, genome editing, and gene transfer. In particular the methods of genetically modifying a plant may be selected from the group consisting of CRISPR-Cas9 targeted gene editing, heterologous gene expression using various expression cassettes; TILLING, and non-targeted chemical mutagenesis using e.g., EMS. Alternatively, a cannabis spp. plant may be transformed with the hermaphroditism QTL or a part thereof, via any of the transformation methods known in the art. In one embodiment, the genetic modification may be by targeted mutagenesis and/or gene editing of guanine to adenine (G>A) at position 99811627 on chromosome NC_044370.1 with reference to the cs10 reference genome (position 1 of SEQ ID NO:343).

[0027] In a further embodiment of the method of producing a *Cannabis* spp. plant comprising a hermaphroditism trait of interest, introducing the hermaphroditism QTL may comprise introducing a hermaphroditism QTL into the *Cannabis* spp. plant, wherein the hermaphroditism QTL is a quantitative trait locus having a sequence that corresponds to nucleotides 28544332 to 35677966 of NC_044370.1 with

reference to the CS10 genome and defined by one or more polymorphisms associated with hermaphroditism as defined in any one of Tables 1, 3 to 11 and 15, or a genetic marker linked to the QTL; a quantitative trait locus having a sequence that corresponds to nucleotides 94129798 to 101726389 of NC_044370.1 with reference to the CS10 genome and defined by one or more polymorphisms associated with hermaphroditism as defined in any one of Tables 1, 3 to 11 and 15, or a genetic marker linked to the QTL; a quantitative trait locus defined by a sequence comprising a single nucleotide polymorphism at position 61687722 of NC_044372.1 with reference to the CS10 genome, or a genetic marker linked to the QTL; a quantitative trait locus defined by a sequence comprising a single nucleotide polymorphism at position 889040 of NC_044370.1 with reference to the CS10 genome, or a genetic marker linked to the QTL; and/or a quantitative trait locus having a sequence that corresponds to nucleotides 70255045 to 79818534 of NC_044370.1 with reference to the CS10 genome and defined by one or more polymorphisms associated with hermaphroditism as defined in any one of Tables 1, 3 to 11 and 15, or a genetic marker linked to the QTL.

[0028] In one embodiment of the method of producing a *Cannabis* spp. plant comprising a hermaphroditism trait of interest, introducing the hermaphroditism QTL may comprise introducing a hermaphroditism absence QTL into the plant, and the hermaphroditism trait of interest may be a hermaphroditism absence trait.

[0029] According to a further aspect of the present invention there is provided for a *Cannabis* spp. plant characterized according to any method of characterizing a *Cannabis* spp. plant described herein or produced according to any method of producing a *Cannabis* spp. plant described herein. In some embodiments, the *Cannabis* spp. plant thus characterized or produced is not exclusively obtained by means of an essentially biological process.

[0030] In another aspect of the invention there is provided for a *Cannabis* spp. plant comprising a hermaphroditism QTL: (a) characterized by one or more polymorphisms associated with the hermaphroditism trait of interest as defined in any one of Tables 1, 3 to 11 and 15, wherein said hermaphroditism QTL is associated with the hermaphroditism trait of interest in the plant; and/or (b) comprising a polymorphism causal for the hermaphroditism trait of interest which is an A/G SNP at position 99811627 on chromosome NC_044370.1 with reference to the cs10 reference genome. In some embodiments, the *Cannabis* spp. plant comprising a hermaphroditism QTL is not exclusively obtained by means of an essentially biological process.

[0031] In one embodiment, the *Cannabis* spp. plant comprising a hermaphroditism QTL comprises a hermaphroditism absence QTL and displays a hermaphroditism absence trait.

[0032] According to further aspect of the present invention there is provided for a quantitative trait locus that controls a hermaphroditic trait in *Cannabis* spp., wherein the quantitative trait locus is selected from one or more of: a quantitative trait locus having a sequence that corresponds to nucleotides 28544332 to 35677966 of NC_044370.1 with reference to the CS10 genome and defined by one or more polymorphisms associated with hermaphroditism as defined in any one of Tables 1, 3 to 11 and 15, or a genetic marker linked to the QTL; a quantitative trait locus having a sequence that corresponds to nucleotides 94129798 to

101726389 of NC_044370.1 with reference to the CS10 genome and defined by one or more polymorphisms associated with hermaphroditism as defined in any one of Tables 1, 3 to 11 and 15, or a genetic marker linked to the QTL; a quantitative trait locus defined by a sequence comprising a single nucleotide polymorphism at position 61687722 of NC_044372.1 with reference to the CS10 genome, or a genetic marker linked to the QTL; a quantitative trait locus defined by a sequence comprising a single nucleotide polymorphism at position 889040 of NC_044370.1 with reference to the CS10 genome, or a genetic marker linked to the QTL; and/or a quantitative trait locus having a sequence that corresponds to nucleotides 70255045 to 79818534 of NC_044370.1 with reference to the CS10 genome and defined by one or more polymorphisms associated with hermaphroditism as defined in any one of Tables 1, 3 to 11 and 15, or a genetic marker linked to the QTL. The invention further includes a genomic region defined by markers linked to the QTLs defined herein.

[0033] In yet a further aspect of the present invention there is provided for an isolated gene that controls a hermaphroditic trait in a *Cannabis* spp. plant, wherein the gene has the gene identity number LOC115696400 with reference to the CS10 genome and encodes a putative 1-aminocyclopropane-1-carboxylate synthase.

[0034] In one embodiment, the isolated gene comprises an A/G SNP at position 99811627 on chromosome NC_044370.1 with reference to the cs10 reference genome (position 1 of SEQ ID NO: 343). In one embodiment, the G allele at this position is causal for the hermaphroditism presence trait. In some embodiments, plants comprising a GG genotype at this position have an increased propensity for hermaphroditism, plants having an AA genotype at this position have a decreased propensity for hermaphroditism, and plants that are heterozygous (AG) at this position have an intermediate propensity for hermaphroditism.

[0035] In some embodiments, the gene may be selected from the group consisting of the genes as defined in Table 13 with reference to the CS10 genome.

[0036] In one embodiment, the isolated gene has the gene identity number LOC115715793 and encodes a DELLA protein RGL2, as defined in Table 13; or the isolated gene has the gene identity number LOC115702418 and encodes a nodulation-signaling pathway 2 protein, as defined in Table 13; or the isolated gene has the gene identity number LOC115719981 and encodes a FT-interacting protein 7, as defined in Table 13; or the isolated gene has the gene identity number LOC115698538 and encodes a *Clavata* 1 Receptor like kinase, as defined in Table 13; or the isolated gene has the gene identity number LOC115695629 and encodes a Myb/SANT-like DNA-binding domain, as defined in Table 13; or the isolated gene has the gene identity number LOC115695629 and encodes a Myb/SANT-like DNA-binding domain, as defined in Table 13; or the isolated gene has the gene identity number LOC115700622 and encodes an Ovate-Transcriptional repressor, as defined in Table 13.

BRIEF DESCRIPTION OF THE FIGURES

[0037] Non-limiting embodiments of the invention will now be described by way of example only and with reference to the following figures:

[0038] FIG. 1: GWA of hermaphroditic flowering in *Cannabis* in a mixed F2 population.

[0039] FIG. 2: GWA of hermaphroditic flowering in *Cannabis* in an expanded mixed F2 population.

SEQUENCES

[0040] The nucleic acid and amino acid sequences listed herein and in any accompanying sequence listing are shown using standard letter abbreviations for nucleotide bases, and the standard one- or three-letter abbreviations for amino acids. It will be understood by those of skill in the art that only one strand of each nucleic acid sequence is shown, but that the complementary strand is included by any reference to the displayed strand.

DETAILED DESCRIPTION OF THE INVENTION

[0041] The present invention will now be described more fully hereinafter with reference to the accompanying drawings, in which some, but not all embodiments of the invention are shown.

[0042] The invention as described should not be limited to the specific embodiments disclosed and modifications and other embodiments are intended to be included within the scope of the invention. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation.

[0043] As used throughout this specification and in the claims, which follow, the singular forms “a”, “an” and “the” include the plural form, unless the context clearly indicates otherwise.

[0044] The terminology and phraseology used herein is for the purpose of description and should not be regarded as limiting. The use of the terms “comprising”, “containing”, “having” and “including” and variations thereof used herein, are meant to encompass the items listed thereafter and equivalents thereof as well as additional items. It is, however, contemplated as a specific embodiment of the present disclosure that the term “comprising” encompasses the possibility of no further members being present, i.e., for the purpose of such an embodiment “comprising” is to be understood as having the meaning of “consisting of”.

[0045] Methods are provided herein for characterizing plants for a hermaphroditism trait and for obtaining plants that do not have a hermaphroditism trait prior to the plant displaying hermaphroditic inflorescence, using a molecular marker detection technique. The inventors of the present invention have further produced cannabis plants that are greatly reduced in hermaphroditic flowering by crossing plants that suppress hermaphroditic flowering to cannabis plants where hermaphroditic flowering occurs. Also demonstrated herein, the inventors were able to use genome wide association (GWA) to identify a QTL linked to hermaphroditism in *Cannabis*. The inventors were also able to identify single nucleotide polymorphisms (SNPs) associated with the hermaphroditism trait; these SNPs were verified as genetic markers for identifying plants carrying the hermaphroditism trait. The inventors used the methods described herein to identify candidate genes that are causative for the hermaphroditism trait, as well as a causative SNP in one of the candidate genes that regulates the hermaphroditism trait. This finding provides for the improvement of methods for producing plants that do not display hermaphroditism or plants that have a decreased likelihood of producing hermaphroditic inflorescences in *Cannabis* spp. plants. In addition,

this finding provides a method of prescreening a population for the hermaphroditism trait prior to the appearance of the trait.

[0046] At least two QTLs for hermaphroditism were identified and confirmed in the mixed populations tested and the 10 F2 populations tested.

[0047] Tables 1, 3 to 11 and 15 herein provide several single nucleotide polymorphisms (SNPs) which define the QTLs associated with the hermaphroditic trait and which can be used for characterizing a plant with respect to the hermaphroditism trait. In some embodiments one or more of the identified SNPs can be used to remove the hermaphroditic trait from a recipient plant, containing one or more of the QTLs associated with the trait. For example, the removal of the hermaphroditic phenotype may be performed by crossing a donor parent plant to a recipient parent plant to produce plants containing a haploid genome from both parents. Recombination of these genomes provides F1 progeny where each haploid complement of chromosomes, of the diploid genome, is comprised of genetic material from both parents.

[0048] In some embodiments, methods of identifying one or more QTLs that are characterized by a haplotype comprising of a series of polymorphisms in linkage disequilibrium are provided. The QTLs each display limited frequency of recombination within the QTLs. Preferably the polymorphisms are selected from any one of Tables 1, 3 to 11 and 15 herein, representing the hermaphroditism QTLs. Molecular markers may be designed for use in detecting the presence of the polymorphisms and thus the QTLs. Further, the identified QTL polymorphisms and the associated molecular markers may be used in a cannabis breeding program to predict the hermaphroditic trait of plants in a breeding population and can be used to produce cannabis plants that do not display the hermaphroditic trait, or which have a reduced propensity for the trait compared to the plants from which they are derived.

[0049] As used herein, reference to a “hermaphroditic inflorescence” or a variety with a “hermaphroditic inflorescence trait”, “hermaphroditic trait” or “hermaphroditism trait” refers to a plant or a variety in which pistillate flowers are accompanied by formation of anthers and/or where pistillate flowers and staminate flowers occur on the same cannabis plant at the time of harvest. The term “pistillate” refers to a flower that bears carpels, also referred to as the gynoecium, the female organs of a flower comprising the stigma, style, and ovary but with no stamens. The term “anthers” as referred to herein are the part of the stamen, the pollen-producing reproductive organ of a flower. The term “staminate” refers to a flower having only functional stamens and lacking functional carpels.

[0050] A “hermaphroditism trait of interest” refers to the state of the plant with respect to the hermaphroditism trait and includes the hermaphroditism absence trait and hermaphroditism presence trait.

[0051] A “hermaphroditism absence trait” is defined by the absence of hermaphroditism.

[0052] A “hermaphroditism presence trait” is defined by the presence of hermaphroditism.

[0053] The time of harvest is defined with respect to the maturity of the flower, where approximately greater than 50% of the pistils have turned brown in appearance. Alternatively, the time of harvest can also be determined by

initiation of flowering for hemp-type cannabis or by other agronomic criteria common in the art.

[0054] It is a particular aim of the present invention to identify and characterize a plant for the hermaphroditism trait of interest early in the plant lifecycle, particularly prior to the plant displaying the hermaphroditism trait, particularly to remove plants having the hermaphroditism trait from the breeding population early on. This can be achieved by genotyping the plant using molecular markers for detecting the QTL associated with the hermaphroditism trait of interest prior to the time of harvest.

[0055] As used herein a “quantitative trait locus” or “QTL” is a polymorphic genetic locus with at least two alleles that differentially affect the expression of a continuously varying phenotypic trait when present in a plant or organism which is characterised by a series of polymorphisms in linkage disequilibrium with each other.

[0056] As used herein, the term “hermaphroditism QTL” or “hermaphroditism quantitative trait locus” refers to a quantitative trait locus comprising part, or all, of the QTLs characterized by one or more polymorphisms having an allelic state associated with the hermaphroditism trait of interest as described in any one of Tables 1, 3 to 11 and 15, or characterized by combinations of said polymorphisms.

[0057] In some cases, it is particularly desirable to obtain a plant that does not display the hermaphroditism trait, or which has a decreased propensity for displaying the hermaphroditism trait. Thus, it is an objective of the invention to provide for cannabis plants having a hermaphroditism absence QTL as described herein. In some cases, this may be achieved by characterizing plants with respect to the hermaphroditism QTL to determine whether they contain a hermaphroditism absence QTL or a hermaphroditism presence QTL.

[0058] As used herein, “hermaphroditism absence QTL” or “hermaphroditism absence quantitative trait locus” refers to a quantitative trait locus characterized by one or more polymorphisms having an allelic state associated with the hermaphroditism absence trait, as described in Tables 1, 3 to 11 and 15.

[0059] As used herein, “hermaphroditism presence QTL” or “hermaphroditism presence quantitative trait locus” refers to a quantitative trait locus characterized by one or more polymorphisms having an allelic state associated with the hermaphroditism presence trait, as described in Tables 1, 3 to 11 and 15.

[0060] As used herein, “haplotypes” refer to patterns or clusters of alleles or single nucleotide polymorphisms that are in linkage disequilibrium and therefore inherited together from a single parent. The term “linkage disequilibrium” refers to a non-random segregation of genetic loci or markers. Markers or genetic loci that show linkage disequilibrium are considered linked.

[0061] As used herein, the term “hermaphroditic haplotype” refers to the subset of the polymorphisms contained within the hermaphroditism QTL which exist on a single haploid genome complement of the diploid genome, and which are in linkage disequilibrium with the hermaphroditic trait.

[0062] As used herein, the term “donor parent plant” refers to a plant having a hermaphroditic haplotype, or one or more hermaphroditic alleles, associated with the hermaphroditism trait of interest.

[0063] As used herein, the term “recipient parent plant” refers to a plant having a hermaphroditic haplotype, or one or more hermaphroditic alleles, not associated with the hermaphroditism trait of interest.

[0064] The term “hermaphroditic allele” refers to the haplotype allele within a particular QTL that confers, or contributes to, the hermaphroditic trait of interest, or alternatively, is an allele that allows the identification of plants with the hermaphroditic trait of interest that can be included in a breeding program, particularly to select for the hermaphroditic absence trait (“marker assisted breeding”, “marker assisted selection”, or “genomic selection”).

[0065] The term “crossed” or “cross” means the fusion of gametes via pollination to produce progeny (e.g., cells, seeds or plants). The term encompasses both sexual crosses (the pollination of one plant by another) and selfing (self-pollination, e.g., when the pollen and ovule are from the same, or genetically identical plant). The term “crossing” refers to the act of fusing gametes via pollination to produce progeny.

[0066] The term “GWAS” or “Genome wide association study” or “GWA” or “Genome wide association” as used herein refers to an observational study of a genome-wide set of genetic variants or polymorphisms in different individual plants to determine if any variant or polymorphism is associated with a trait, specifically the hermaphroditic trait of interest.

[0067] As used herein a “polymorphism” is a particular type of variance that includes both natural and/or induced multiple or single nucleotide changes, short insertions, or deletions in a target nucleic acid sequence at a particular locus as compared to a related nucleic acid sequence. These variations include, but are not limited to, single nucleotide polymorphisms (SNPs), indel/s, genomic rearrangements, and gene duplications.

[0068] As used herein, the term “LOD score” or “logarithm (base 10) of odds” refers to a statistical estimate used in linkage analysis, wherein the score compares the likelihood of obtaining the test data if the two loci are indeed linked, to the likelihood of observing the same data purely by chance. The LOD score is a statistical estimate of whether two genetic loci are physically near enough to each other (or “linked”) on a particular chromosome that they are likely to be inherited together. A LOD score of 3 or higher is generally understood to mean that two genes are located close to each other on the chromosome. In terms of significance, a LOD score of 3 means the odds are 1,000:1 that the two genes are linked and therefore inherited together.

[0069] As used herein, the term “quantile-quantile” or “Q-Q” refers to a graphical method for comparing two probability distributions by plotting their quantiles against each other. If the two distributions being compared are similar, the points in the Q-Q plot will approximately lie on the line $y=x$. If the distributions are linearly related, the points in the Q-Q plot will approximately lie on a line, but not necessarily on the line $y=x$. Q-Q plots can also be used as a graphical means of estimating parameters in a location-scale family of distributions.

[0070] As used herein, a “causal gene” is the specific gene having a genetic variant (the “causal variant”) which is responsible for the association signal at a locus and has a direct biological effect on the hermaphroditic trait. In the context of association studies, the genetic variants which are responsible for the association signal at a locus are referred

to as the “causal variants”. Causal variants may comprise one or more “causal polymorphisms” that have a biological effect on the phenotype.

[0071] The encompasses both ribonucleotides (RNA) and term “nucleic acid” deoxyribonucleotides (DNA), including cDNA, genomic DNA, isolated DNA and synthetic DNA. The nucleic acid may be double-stranded or single-stranded. Where the nucleic acid is single-stranded, the nucleic acid may be the sense strand or the antisense strand. A “nucleic acid molecule” or “polynucleotide” refers to any chain of two or more covalently bonded nucleotides, including naturally occurring or non-naturally occurring nucleotides, or nucleotide analogs or derivatives. By “RNA” is meant a sequence of two or more covalently bonded, naturally occurring or modified ribonucleotides. The term “DNA” refers to a sequence of two or more covalently bonded, naturally occurring or modified deoxyribonucleotides. By “cDNA” is meant a complementary or copy DNA produced from an RNA template by the action of RNA-dependent DNA polymerase (reverse transcriptase).

[0072] In some embodiments, the nucleic acid molecules of the invention may be operably linked to other sequences. By “operably linked” is meant that the nucleic acid molecules, such as those comprising the QTLs of the invention or genes identified herein, and regulatory sequences are connected in such a way as to permit expression of the proteins when the appropriate molecules are bound to the regulatory sequences. Such operably linked sequences may be contained in vectors or expression constructs which can be transformed or transfected into plant cells or plants for expression. A “regulatory sequence” refers to a nucleotide sequence located either upstream, downstream or within a coding sequence. Generally regulatory sequences influence the transcription, RNA processing or stability, or translation of an associated coding sequence. Regulatory sequences include but are not limited to: effector binding sites, enhancers, introns, polyadenylation recognition sequences, promoters, RNA processing sites, stem-loop structures, translation leader sequences and the like.

[0073] The term “promoter” refers to a DNA sequence that is capable of controlling the expression of a nucleic acid coding sequence or functional RNA. A promoter may be based entirely on a native gene or it may be comprised of different elements from different promoters found in nature. Different promoters are capable of directing the expression of a gene at different stages of development, or in response to different environmental or physiological conditions. An “inducible promoter” is promoter that is active in response to a specific stimulus. Several such inducible promoters are known in the art, for example, chemical inducible promoters, developmental stage inducible promoters, tissue type specific inducible promoters, hormone inducible promoters, environment responsive inducible promoters.

[0074] The term “isolated”, as used herein means having been removed from its natural environment. Specifically, the nucleic acid(s) or gene(s) identified herein may be isolated nucleic acids or gene(s), which have been removed from plant material where they naturally occur.

[0075] The term “purified”, relates to the isolation of a molecule or compound in a form that is substantially free of contamination or contaminants. Contaminants are normally associated with the molecule or compound in a natural environment, purified thus means having an increase in purity as a result of being separated from the other compo-

nents of an original composition. The term “purified nucleic acid” describes a nucleic acid sequence that has been separated from other compounds including, but not limited to polypeptides, lipids and carbohydrates which it is ordinarily associated with in its natural state.

[0076] The term “complementary” refers to two nucleic acid molecules, e.g., DNA or RNA, which are capable of forming Watson-Crick base pairs to produce a region of double-strandedness between the two nucleic acid molecules. It will be appreciated by those of skill in the art that each nucleotide in a nucleic acid molecule need not form a matched Watson-Crick base pair with a nucleotide in an opposing complementary strand to form a duplex. One nucleic acid molecule is thus “complementary” to a second nucleic acid molecule if it hybridizes, under conditions of high stringency, with the second nucleic acid molecule. A nucleic acid molecule according to the invention includes both complementary molecules.

[0077] As used herein a “substantially identical” or “substantially homologous” sequence is a nucleotide sequence that differs from a reference sequence only by one or more conservative substitutions, or by one or more non-conservative substitutions, deletions, or insertions located at positions of the sequence that do not destroy or substantially alter the activity of the polypeptide encoded by the nucleic acid molecule. Alignment for purposes of determining percent sequence identity can be achieved in various ways that are within the knowledge of those with skill in the art. These include using, for instance, computer software such as ALIGN, Megalign (DNASTAR), CLUSTALW or BLAST software. Those skilled in the art can readily determine appropriate parameters for measuring alignment, including any algorithms needed to achieve maximal alignment over the full length of the sequences being compared. In one embodiment of the invention there is provided for a polynucleotide sequence that has at least about 80% sequence identity, at least about 90% sequence identity, or even greater sequence identity, such as about 95%, about 96%, about 97%, about 98% or about 99% sequence identity to the sequences described herein.

[0078] Alternatively, or additionally, two nucleic acid sequences may be “substantially identical” or “substantially homologous” if they hybridize under high stringency conditions. The “stringency” of a hybridization reaction is readily determinable by one of ordinary skill in the art, and generally is an empirical calculation which depends upon probe length, washing temperature, and salt concentration. In general, longer probes required higher temperatures for proper annealing, while shorter probes require lower temperatures. Hybridization generally depends on the ability of denatured DNA to re-anneal when complementary strands are present in an environment below their melting temperature. A typical example of such “stringent” hybridization conditions would be hybridization carried out for 18 hours at 65° C. with gentle shaking, a first wash for 12 min at 65° C. in Wash Buffer A (0.5% SDS; 2×SSC), and a second wash for 10 min at 65° C. in Wash Buffer B (0.1% SDS; 0.5% SSC).

[0079] Nucleotide positions of polymorphisms described herein are provided with reference to the corresponding position on the *Cannabis sativa* (assembly cs10) representative genome, provided as RefSeq assembly accession:

GCF_900626175.2 on NCBI, loaded on 14 Feb. 2019, referred to herein as “cs10 reference genome” or “cs10 genome”.

Methods of Identifying a QTL or Haplotype Responsible for the Hermaphroditic Phenotype And Molecular Markers Therefor

[0080] In some embodiments, methods are provided for identifying a QTL or haplotype responsible for a hermaphroditism trait of interest and for selecting plants having the hermaphroditism trait of interest, thereby to identify the QTL or haplotype responsible for the trait. In some embodiments, the methods may comprise the steps of:

- [0081]** a. Identifying a plant that displays the hermaphroditic presence trait or the hermaphroditic absence trait within a breeding program.
- [0082]** b. Establishing a population by crossing the identified plant to itself (selfing) or a recipient parent plant.
- [0083]** c. Genotyping the resultant F1, or subsequent populations, for example by sequencing methods.
- [0084]** d. Performing association studies, including phenotyping and linkage analysis, to discover QTLs and/or polymorphisms contained within the QTL.
- [0085]** e. Optionally, identifying cannabis paralogs of previously characterized genes that may be involved in conferring the hermaphroditism trait of interest.
- [0086]** f. Developing molecular markers that detect one or more polymorphisms linked to QTLs, alleles within these QTLs, or existing or induced polymorphisms.
- [0087]** g. Validating the molecular markers by determining the linkage disequilibrium between the marker and the hermaphroditic trait.

Trait Development and Introgression

[0088] In some embodiments, methods are provided for marker assisted breeding (MAB) or marker assisted selection (MAS) of plants having the hermaphroditism absence QTL or displaying the hermaphroditic absence trait. The methods may comprise the steps of:

- [0089]** a. Identifying a plant that displays the hermaphroditism trait of interest or which contains a hermaphroditism QTL as defined herein.
- [0090]** b. Establishing a population by crossing the identified plant to itself (selfing) or another recipient parent plant.
- [0091]** c. Genotyping and phenotyping the resultant F1, or subsequent, populations, for example by sequencing methods.
- [0092]** d. Performing association studies, inputting phenotype and genotype information to identify genomic regions enriched with polymorphisms associated with the hermaphroditic trait, to discover QTLs and/or polymorphisms contained within the QTL.
- [0093]** e. Optionally, identifying cannabis paralogs of previously characterized genes that may be involved in the hermaphroditic phenotype.
- [0094]** f. Developing molecular markers that detect one or more polymorphisms linked to QTLs, alleles within these QTLs, or existing or induced polymorphisms.
- [0095]** g. Using the molecular markers when introgressing the QTLs or polymorphisms into new or existing

cannabis varieties to select plants where the hermaphroditic haplotype or the hermaphroditism trait is absent.

Qtl's and Marker Assisted Breeding

[0096] In some embodiments, during the breeding process, selection of plants displaying the hermaphroditism trait of interest may be based on molecular markers designed to detect polymorphisms linked to genomic regions that control the trait of interest. In some embodiments, QTLs containing such elements are identified using association studies. Knowledge of the mode-of-action is not required for the functional use of these genomic regions in a breeding program. Identification of regions controlling unidentified mechanisms may be useful in obtaining plants with the hermaphroditism absence trait, based on identification of polymorphisms that are either linked to, or found within QTLs that are associated with the hermaphroditic trait of interest using association studies.

Construction of Breeding Populations

[0097] Breeding populations are the offspring of sexual reproduction events between two or more parents. The parent plants (F0) are crossed to create an F1 population each containing a chromosomal complement of each parent. In a subsequent cross (F2), recombination has occurred and allows for mostly independent segregation of traits in the offspring and importantly the reconstitution of recessive phenotypes that existed in only one of the parental lines.

[0098] According to some embodiments, QTLs that lead to the hermaphroditism trait of interest identified within synthetic populations of plants capable of revealing dominant, recessive, or complex traits. In one embodiment of the invention, a genetically diverse population of cannabis varieties, that are used to produce the synthetic population are integrate them into a breeding program by unnatural processes. In some embodiments, these processes result in changes in the genomes of the plants. The changes may include, but are not limited to, mutations and rearrangements in the genomic sequences, duplication of the entire genome (polyploidy), or activation of movement of transposable elements which may inactivate, activate or attenuate the activity of genes or genomic elements. According to one embodiment of the invention, the methods employed to integrate the plants into a breeding program include some or all of the following:

- [0099]** a. Growing plants in rich media or soils under artificial lighting;
- [0100]** b. Cloning of plants, often through a multitude of sub-cloning cycles;
- [0101]** c. Introduction of plants into in vitro, sterile growth environments, and subsequent removal to standard growth conditions;
- [0102]** d. Exposure to mutagens such as EMS, colchicine, silver nitrate, ethidium bromide, dinitroanilines, high concentrations mono or poly-chromatic light sources;
- [0103]** e. Growing plants under highly stressful conditions which include restricted space, drought, pathogen, atypical temperatures, and nutrient stresses.

Hermaphroditism Trait of Interest Association Studies and QTL Identification

[0104] In some embodiments, the synthetic populations created are either the offspring of the sexual reproduction or clones of plants in the breeding program such that genetic material of individuals in the synthetic populations is derived from one, or two, or more plants from the breeding program.

[0105] In one embodiment, plants identified within the synthetic population as having a trait of interest, such as the hermaphroditic trait of interest, may be used to create a structured population for the identification of the genetic locus responsible for the trait. The structured population may be created by crossing one (selfing) or more plants and recovering the seeds from those plants.

[0106] Plants in the structured population may be fully genotyped using genome sequencing to identify genetic markers for use in the association study (AS) database. Association mapping is a powerful technique used to detect quantitative trait loci (QTLs) specifically based on the statistical correlation between the phenotype and the genotype. In this case the trait is the hermaphroditic phenotype. In a population generated by crossing, the amount of linkage disequilibrium (LD) is reduced between genetic marker and the QTL as a function of genetic distance in cannabis varieties with similar genome structures. Simple association mapping is performed by biparental crosses of two closely related lines where one line has a phenotype of interest and the other does not. In some embodiments, advanced population structures may be used, including nested association mapping (NAM) populations or multi-parent advanced generation inter-cross (MAGIC) populations, however it will be appreciated that other population structures can also be effectively used. Biparental, NAM, or MAGIC structured populations can be generated and offspring, at F1 or later generations, may be maintained by clonal propagation for a desired length of time. In some embodiments, QTLs may be identified using the high-density genetic marker database created by genotyping the founder lines and structured population lines. This marker database may be coupled with an extensive phenotypic trait characterization dataset, including, for example, the hermaphroditic phenotype of the plants. Using the association studies described herein, together with accurate phenotyping, this method is able to identify genomic regions, QTLs and even specific genes or polymorphisms responsible for the hermaphroditism trait of interest that are directly introduced into recipient lines. Polygenic phenotypes may also be identified using the methods described herein.

[0107] In one embodiment, the structured population is grown to the time of harvest. To characterize the phenotypes of the lines, they are clonally reproduced so the phenotypic data can be collected in feasible replicates.

Genomic Selection

[0108] In some embodiments, during the breeding process, selection of plants by genomic selection (GS) may be conducted. Genomic selection is a method in plant breeding where the genome wide genetic potential of an individual is determined to predict breeding values for those individuals. In some embodiments, the accuracy of genomic selection is affected by the data used in a GS model including size of the

training population, relationships between individuals, marker density, use of pedigree information, and inclusion of known QTLs.

[0109] In some embodiments, a QTL or a SNP known to be associated with a trait that contributes to selection criteria can improve the accuracy of genomic selection models. In some embodiments, a genomic selection model that incorporates the hermaphroditism phenotype can be improved by the inclusion of the hermaphroditism QTL in the GS model. In some embodiments, the SNPs described in any of Tables 1, 3 to 11 and 15 may be useful in a genomic selection model, for example where genotypes with unknown phenotypes are evaluated using an approach like a random forest algorithm for prediction of the hermaphroditism trait, and particularly in combination, to improve the predictive power of the model.

Molecular Markers to Detect Polymorphisms

[0110] As used herein, the term “marker” or “genetic marker” refers to any sequence comprising a particular polymorphism or haplotype described herein that is capable of detection. For example, a marker may be a binding site for a primer or set of primers that is designed for use in a PCR-based method to amplify and thus detect a polymorphism or haplotype. Alternatively, the marker may introduce a restriction enzyme recognition site, or result in the removal of a restriction enzyme recognition site. Plants can be screened for a particular trait based on the detection of one or more markers confirming the presence of the polymorphism. Marker detection systems that may be used in accordance with the present invention include, but are not limited to polymerase chain reaction (PCR) followed by sequencing, Kompetitive allele specific PCR (KASP), restriction fragment length polymorphisms (RFLPs) analysis, amplified fragment length polymorphisms (AFLPs), cleaved amplified polymorphic sequences (CAPS), or any other markers known in the art.

[0111] In some embodiments “molecular markers” refers to any marker detection system and may be PCR primers, such as those described in the examples below. For example, PCR primers may be designed that consist of a reverse primer and two forward primers that are homologous to the part of the genome that contains a polymorphism but differ in the 3' nucleotide such that the one primer will preferentially bind to sequences containing the polymorphism and the other will bind to sequences lacking it. The three primers are used in single PCR reactions where each reaction contains DNA from a cannabis plant as a template. Fluorophores linked to the forward primers provide, after thermocycling, a different relative fluorescent signal for homozygous and heterozygous alleles containing the polymorphism and for those lacking the polymorphism, respectively.

[0112] In some embodiments, allele-specific primers may each harbor a unique tail sequence that corresponds with a universal FRET (fluorescence resonant energy transfer) cassette. For example, the primer specific to the SNP may be labelled with a FAM and the other specific primer with a HEX dye. During the PCR thermal cycling performed with these primers, the allele-specific primer binds to the genomic DNA template and elongates, so attaching the tail sequence to the newly synthesized strand. The complement of the allele-specific tail sequence is then generated during subsequent rounds of PCR, enabling the FRET cassette to bind to the DNA. Alleles are discriminated through the

competitive binding of the two allele-specific forward primers. At the end of the PCR reaction a fluorescent plate is read using standard tools which may include RT-PCR devices with the capacity to detect florescent signals and is evaluated with commercial software.

[0113] If the genotype at a given polymorphism site is homozygous, one of the two possible fluorescent signals will be generated. If the genotype is heterozygous, a mixed fluorescent signal will be generated. By way of example, genomic DNA extracted from cannabis leaf tissue at seedling stage can be used as a template for PCR amplifications with reaction mixtures containing the three primers. Final fluorescent signals can be detected by a thermocycler and analyzed using standard software for this purpose, which discriminates between individuals that are heterozygotes or homozygotes for either allele.

[0114] In some embodiments, molecular markers to one, two or more of the SNPs in the haplotype can be used to identify the presence of the QTL and by association, the hermaphroditism trait of interest.

[0115] Further, the QTL may include a number of individual polymorphisms in linkage disequilibrium, which constitute a haplotype and which, with high frequency, can be inherited from a donor parent plant as a unit. Therefore, in some embodiments, molecular markers can be utilized which have been designed to identify numerous polymorphisms which are in linkage disequilibrium with other polymorphisms, any of which can be used to effectively predict the phenotype of the offspring for the hermaphroditism trait of interest.

[0116] According to some embodiments, any polymorphism in linkage disequilibrium with one or more of the hermaphroditism QTLs can be used to determine the hermaphroditism haplotype in a breeding population of plants, as long as the polymorphism is unique to the hermaphroditic trait of interest in the donor parent plant when compared to the recipient parent plant.

[0117] In some embodiments the desired trait is the hermaphroditism absence trait, and the donor parent plant may be a plant that has been genetically modified or selected to exclude a hermaphroditism absence QTL defined by a polymorphism associated with the hermaphroditic absence trait, for example any, some, or all of the polymorphisms defined in any of Tables 1, 3 to 11 and 15 associated with the trait.

[0118] Alternatively, the desired trait may be the hermaphroditism presence trait, and the donor parent plant may be a plant that has been genetically modified or selected to include a hermaphroditism presence QTL defined by a polymorphism associated with the hermaphroditic presence trait, for example any, some, or all of the polymorphisms defined in any of Tables 1, 3 to 11 and 15 associated with the trait.

[0119] In some embodiments, donor parent plants, as described above, are used as one of two parents to create breeding populations (F1) through sexual reproduction. Methods for reproduction that are known in the art may be used. The donor parent plant provides the trait of interest to the breeding population. The trait is made to segregate through the population (F2) through at least one additional crossing event of the offspring of the initial cross. This additional crossing event can be either a selfing of one of the offspring or a cross between two individuals, provided that

each plant used in the F1 cross contains at least one copy of a desired QTL allele or haplotype.

[0120] In some embodiments, the hermaphroditic allele or hermaphroditic haplotype in plants to be used in the F1 cross is determined using the described molecular markers. In some embodiments, the resulting F2 progeny is/are screened for any of the hermaphroditism polymorphisms associated with the hermaphroditism trait of interest described herein.

[0121] The plants at any generation can be produced by asexual means like cutting and cloning, or any method that yields a genetically identical offspring.

Production of *Cannabis* Spp. Plants Having the Hermaphroditism Absence Trait

[0122] It is a particular aim of the present invention to provide for the production of *Cannabis* spp. Plants that do not have the hermaphroditism trait. According, in some embodiments, a *Cannabis* spp. plant that has the hermaphroditism presence trait may be converted into a plant having a hermaphroditism absence trait according to the methods of the present invention by providing a breeding population where the donor parent plant contains a hermaphroditism absence QTL and the recipient parent plant either displays the hermaphroditism presence trait or contains a hermaphroditism presence QTL.

[0123] In some embodiments the hermaphroditism presence trait may be removed from a recipient parent plant by crossing it with a donor parent plant having the hermaphroditism absence QTL. In some embodiments, the donor parent plant does not have a hermaphroditism phenotype and contains a contiguous genomic sequence characterized by one or more of the polymorphisms of Tables 1, 3 to 11 and 15 associated with the hermaphroditism allele or hermaphroditism haplotype conferring the hermaphroditism absence trait.

[0124] In some embodiments, the donor parent plant is any *Cannabis* spp. variety that is cross fertile with the recipient parent plant.

[0125] In some embodiments, MAS or MAB may be used in a method of backcrossing plants carrying the hermaphroditism absence trait to a recipient parent plant. For example, an F1 plant from a breeding population can be crossed again to the recipient parent plant. In some embodiments, this method is repeated.

[0126] In some embodiments, the resulting plant population is then screened for the hermaphroditism absence trait using MAS with molecular markers to identify progeny plants that contain one or more polymorphism, such as any of those described in Tables 1, 3 to 11 and 15, indicating the presence of an allele of a QTL associated with the hermaphroditism absence phenotype. In another embodiment, the population of cannabis plants may be screened by any analytical methods known in the art to identify plants with desired characteristics, specifically hermaphroditism absence.

Production of *Cannabis* Spp. Plants Having the Hermaphroditism Presence Trait

[0127] In an alternative embodiment, a *Cannabis* spp. plant that does not have the hermaphroditism trait may be converted into a hermaphroditism plant according to the methods of the present invention by providing a breeding population where the donor parent plant contains a hermaphroditism presence QTL associated with the hermaphroditism

presence trait, and recipient parent plant either displays the hermaphroditism absence trait or contains the hermaphroditism absence QTL.

[0128] Stable hermaphrodite cannabis plants, in which hermaphroditism occurs irrespective of environmental conditions represent an expansion of the breeding potential and usefulness of cannabis plants in many production systems. Stable or inducible hermaphroditism in *Cannabis* can be used to facilitate inbreeding, without the need of lengthy and low-throughput chemical induction of male flowers and crossing procedures, in order to produce homozygous inbred lines that can be used as parents to generate hybrid lines that benefit from heterosis. Stable or inducible hermaphroditism may also aid in the generation of double haploid plants, where haploid cells undergo chromosome doubling, leading to plants with high homozygosity, an alternative and accelerated route to conventional inbreeding. The surplus of pollen from hermaphrodite plants versus sex-reversed female plants is an advantage for generating double haploid plants.

[0129] Stable or inducible hermaphroditism may also be useful for inbreeding in autoflowering cannabis plants, or other genetic backgrounds where propagation of clones is challenging or where chemical induction of male flowers is not practical or possible. In hemp, where seed yield is a trait of interest, introgression of a stable or inducible hermaphroditic trait into hemp-type has the potential to increase the yield of seed production by maximizing the number of plants that produce seeds, in contrast to the current conditions where both male and female plants are present and 50% of the plants in field are without seed. The generation of hermaphrodite hemp-type cannabis plants may also increase the yield of seed production. Such a system also ensures the synchronisation of flowering of both female and male flowers as compared to conventional dioecious varieties where male plants tend to flower earlier than their female counterparts.

[0130] Accordingly, in some embodiments the hermaphroditism presence trait may be introduced into a recipient parent plant by crossing it with a donor parent plant having the hermaphroditism presence QTL. In some embodiments the donor parent plant has a hermaphroditism phenotype and a contiguous genomic sequence characterized by one or more of the polymorphisms of any one of Tables 1, 3 to 11 and 15 associated with the hermaphroditism allele or hermaphroditism haplotype conferring the hermaphroditism presence trait.

[0131] In some embodiments, the donor parent plant is any *Cannabis* spp. variety that is cross fertile with the recipient parent plant.

[0132] In some embodiments, MAS or MAB may be used in a method of backcrossing plants carrying the hermaphroditism presence trait to a recipient parent plant. For example, an F1 plant from a breeding population can be crossed again to the recipient parent plant. In some embodiments, this method is repeated.

[0133] In some embodiments, the resulting plant population is then screened for the hermaphroditism presence trait using MAS with molecular markers to identify progeny plants that contain one or more polymorphisms, such as those described in any one of Tables 1, 3 to 11 and 15, indicating the presence of an allele of a QTL associated with the hermaphroditism presence phenotype. In another embodiment, the population of cannabis plants may be

screened by any analytical methods known in the art to identify plants with desired characteristics, specifically hermaphroditism presence.

Methods to Genetically Engineer Plants to Achieve the Hermaphroditism Trait of Interest Using Mutagenesis or Gene Editing Techniques

[0134] Identifying QTLs, and individual polymorphisms, that correlate with a trait when measured in an F1, F2, or similar, breeding population indicates the presence of one or more causative polymorphisms in close proximity the polymorphism detected by the molecular marker. In some embodiments, the polymorphisms associated with the presence or absence of the hermaphroditism trait are introduced into a plant by other means so that a trait can be introduced into plants that would not otherwise contain associated causative polymorphisms or removed from plants that would otherwise contain associated causative polymorphisms. For example, an A/G SNP at position 99811627 on chromosome NC_044370.1 with reference to the cs10 reference genome (position 1 of SEQ ID NO:343). Further, the polymorphisms detailed in Tables 1, 3 to 11 and 15 are molecular markers that can be used to indicate the presence of a possible causative polymorphism.

[0135] The entire QTLs or parts thereof which confer the hermaphroditism trait of interest described herein, or the genes or nucleic acid molecules described herein, may be introduced into the genome of a cannabis plant to obtain plants with a hermaphroditism trait of interest, through a process of genetic modification known in the art, for example, but not limited to, heterologous gene expression using an expression cassette including a sequence encoding the QTL(s) or part thereof, the gene(s), or the nucleic acids. The expression cassettes may contain all or part of the QTL(s) or gene(s), including possible causative polymorphisms.

[0136] The trait described herein may be introduced into, or removed from, the genome of a cannabis plant to obtain plants that include or exclude the causative polymorphisms and the potential to display a desired hermaphroditism trait of interest through processes of genetic modification known in the art, for example, but not limited to, CRISPR-Cas9 targeted gene editing, TILLING, non-targeted chemical mutagenesis using e.g., EMS.

[0137] The present invention further provides methods for producing a modified *Cannabis* spp. plant using genome editing or modification techniques. For example, genome editing can be achieved using sequence-specific nucleases (SSNs) the use of which results in chromosomal changes, such as nucleotide deletions, insertions or substitutions at specific genetic loci, particularly those associated with the hermaphroditism trait of interest, more particularly an A/G SNP at position 99811627 on chromosome NC_044370.1 with reference to the cs10 reference genome (position 1 of SEQ ID NO:343). Non limiting examples of SSNs include zinc finger nucleases (ZFNs), TAL effector nucleases (TALENs), meganucleases, and clustered regularly interspaced short palindromic repeats (CRISPR)/CRISPR-associated protein (Cas) system. In some embodiments, non-limiting examples of Cas proteins suitable for use in the methods of the present invention include Csn1, Cpf1, Cas9, Cas 12, Cas 13, Cas 14, CasX, and combinations thereof. In one embodiment, a modified *Cannabis* spp. plant having a hermaphroditism trait of interest is generated using CRISPR/Cas9

technology, which is based on the Cas9 DNA nuclease guided to a specific DNA target by a single guide RNA (sgRNA). For example, the genome modification may be introduced using guide RNA, e.g., single guide RNA (sgRNA) designed and targeted to introduce a polymorphism associated with the hermaphroditism trait of interest, such as a G>A SNP at position 99811627 on chromosome NC_044370.1 with reference to the cs10 reference genome (position 1 of SEQ ID NO:343).

[0138] DNA introduction into the plant cells can be performed using *Agrobacterium* infiltration, virus-based plasmid delivery of the genome editing molecules and mechanical insertion of DNA (PEG mediated DNA transformation, biolistics, etc.). In some embodiments, the Cas9 protein may be directly inserted together with a gRNA (ribonucleoprotein-RNP's) in order to bypass the need for in vivo transcription and translation of the Cas9+gRNA plasmid in planta to achieve gene editing. In one embodiment, a genome edited plant may be developed and used as a rootstock, so that the Cas protein and gRNA can be transported via the vasculature system to the top of the plant and create the genome editing event in the scion.

[0139] According to one embodiment of the present invention, the method of genetically modifying a plant may be achieved by combining the Cas nuclease (e.g., Cas9, Cpf 1) with a predefined guide RNA molecule (gRNA). The gRNA is complementary to a specific DNA sequence targeted for editing in the plant genome and which guides the Cas nuclease to a specific nucleotide sequence. The predefined gene-specific gRNAs may be cloned into the same plasmid as the Cas gene and this plasmid is inserted into plant cells as described above.

[0140] In some embodiments, once the gRNA molecule and Cas9 nuclease reach the specific predetermined DNA sequence, the Cas9 nuclease cleaves both DNA strands to create double stranded breaks leaving blunt ends. This cleavage site is then repaired by the cellular non homologous end joining DNA repair mechanism resulting in insertions or deletions which introduce a mutation at the cleavage site.

[0141] In one embodiment, a deletion form of the mutation may consist of at least 1 base pair deletion. As a result of this base pair deletion, the gene coding sequence for the putative gene(s) responsible for the hermaphroditism trait of interest, such as the genes described in Table 13, or more particularly a gene having the gene identity number LOC115696400 with reference to the CS10 genome and encoding a putative 1-aminocyclopropane-1-carboxylate synthase (SEQ ID NO: 343), is disrupted and the translation of the encoded protein is compromised by disruption of a start codon, introduction of a premature stop codon or disruption of a functional or structural property of the protein.

[0142] In another embodiment, the hermaphroditism trait of interest in *Cannabis* spp. plants may be introduced by generating gRNA with homology to a specific site of predetermined genes in the *Cannabis* genome or a QTL defined herein. In one embodiment the gene may be one or more of the genes described in Table 13 herein, or more particularly a gene having the gene identity number LOC115696400 with reference to the CS10 genome and encoding a putative 1-aminocyclopropane-1-carboxylate synthase (SEQ ID NO:343). This gRNA may be sub-cloned into a plasmid containing the Cas9 gene, and the plasmid inserted into the

Cannabis plant cells. In this way site specific mutations in the QTL are generated, including the SNPs associated with the hermaphroditism trait of interest described in Tables 1, 3 to 11 and 15, and in particular a causative polymorphism, more particularly the A/G SNP at position 99811627 on chromosome NC_044370.1 with reference to the cs10 reference genome (position 1 of SEQ ID NO:343), thus effectively introducing the hermaphroditism trait of interest into the genome edited plant.

[0143] In some embodiments, a modified *Cannabis* spp. plant exhibiting a hermaphroditism absence trait may be obtained using the targeted genome modification methods described above, wherein the plant comprises a targeted genome modification to introduce one or more polymorphisms associated with the hermaphroditism presence trait defined in Tables 1, 3 to 11 and 15, wherein the modification effects the hermaphroditism presence trait. In a preferred embodiment, the plant comprises a targeted genome modification to introduce a G>A SNP at position 99811627 on chromosome NC_044370.1 with reference to the cs10 reference genome (position 1 of SEQ ID NO:343), to obtain a modified *Cannabis* spp. plant exhibiting a hermaphroditism absence trait.

[0144] In some embodiments, for example where the hermaphroditism trait of interest is a hermaphroditism absence trait, the genetic modification may be introduced using gene silencing, a process by which the expression of a specific gene product is lessened or attenuated. Gene silencing can take place by a variety of pathways, including by RNA interference (RNAi), an RNA dependent gene silencing process. In one embodiment, RNAi may be achieved by the introduction of small RNA molecules, including small interfering RNA (siRNA), microRNA (miRNA) or short hairpin RNA (shRNA), which act in concert with host proteins (e.g., the RNA induced silencing complex, RISC) to degrade messenger RNA (mRNA) in a sequence-dependent fashion. In particular, RNAi may be used to silence one or more of the putative causative genes described in Table 13 herein, or more particularly a gene having the gene identity number LOC115696400 with reference to the CS10 genome and encoding a putative 1-aminocyclopropane-1-carboxylate synthase (SEQ ID NO:343). Such RNAi molecules may be designed based on the sequence of these genes. These molecules can vary in length (generally 18-30 base pairs) and may contain varying degrees of complementarity to their target mRNA in the antisense strand. Some, but not all, RNAi molecules have unpaired overhanging bases on the 5' or 3' end of the sense strand and/or the antisense strand. As used herein, the term "RNAi molecule" includes duplexes of two separate strands, as well as single strands that can form hairpin structures comprising a duplex region. The RNAi molecules may be encoded by DNA contained in an expression cassette and incorporated into a vector. The vector may be introduced into a plant cell using *Agrobacterium* infiltration, virus-based plasmid delivery of the vector containing the expression cassette and/or mechanical insertion of the vector (PEG mediated DNA transformation, biolistics, etc.).

[0145] Plants may be screened with the molecular markers as described herein to identify transgenic individuals with the hermaphroditism trait of interest or having a hermaphroditism QTL or polymorphism(s), following the genetic modification.

[0146] In some embodiments, *Cannabis* spp. plants having one or more of the polymorphisms of any one of Tables 1, 3 to 11 and 15 associated with the hermaphroditism QTLs or linked thereto are provided. More particularly, *Cannabis* spp. plants having a causative A/G SNP at position 99811627 on chromosome NC_044370.1 with reference to the cs10 reference genome (position 1 of SEQ ID NO:343) are provided. The polymorphisms, including the causative polymorphism, may be introduced, for example, by genetic engineering. In some embodiments the one or more polymorphisms associated with the hermaphroditism trait of interest or linked thereto are introduced into the plants by breeding, such as by MAS or MAB, for example as described herein.

[0147] The hermaphroditism QTLs described herein, or genes identified herein responsible for effecting the hermaphroditism trait, may be under the control of, or operably linked to, a promoter, for example an inducible promoter. Such QTLs or genes may be operably linked to the inducible promoter so as to induce or suppress the hermaphroditism trait or phenotype in the plant or plant cell.

[0148] Accordingly, in a further embodiment, *Cannabis* spp. plants comprising a hermaphroditism QTL described herein, including a hermaphroditism absence QTL or a hermaphroditism presence QTL, or one or more polymorphisms associated therewith, are provided. In some cases, such plants are provided for with the proviso that the plant is not exclusively obtained by means of an essentially biological process.

[0149] The following examples are offered by way of illustration and not by way of limitation.

Example 1

Genome-Wide Association Studies (GWAS) of Hermaphroditic Inflorescence in Mixed Population of Cannabis

[0150] During outdoor field trials in 2020 it was observed that several populations of cannabis plants were comprised of individuals with hermaphroditic inflorescences. To identify molecular markers for the appearance of hermaphroditism in Cannabis, the study initially focused on the apical inflorescence in a diverse population comprising 3220 individuals.

[0151] Through the 2020 field trial, *Cannabis sativa* genotypes were monitored for the emergence of hermaphroditic flowers, including pistillate flowers that showed the emergence of anthers, as well as the growth of staminate flowers along with pistillate flowers. Individual plants that were scored hermaphroditic were assigned as 1, those that were not coded as hermaphroditic were assigned as 0.

[0152] DNA was extracted from about 70 mg of leaf discs from all the plants evaluated using an adapted kit with “sbeadex” magnetic beads by LGC Genomics, which was automated on a KingFisher Flex with 96 Deep-Well Head by Thermo Fisher Scientific.

[0153] The extracted DNA served as a template for the subsequent library preparation for sequencing. The library pools were prepared according to the manufacturer’s instructions (AgriSeq™ HTS Library Kit-96 sample procedure from Thermo Fisher Scientific). Targeted sequencing of a custom SNP marker panel based on the *Cannabis Sativa* CS10 reference genome (NCBI GenBank assembly acces-

sion number GCA_900626175.2 as updated in April 2020 and accessed in February 2022) was carried out on the Ion Torrent system by Thermo Fisher Scientific. The primers for the SNPs identified are provided in Table 13. The library pool was loaded onto Ion 550 chips with Ion Chef and sequenced with Ion GeneStudio S5 Plus according to the manufacturer’s instructions (Ion 550™ Kit from Thermo Fisher Scientific).

[0154] From a population of 400 individuals, a genome-wide association study (GWAS) was performed to detect significant associations between genotypic information derived from targeted resequencing of the custom SNP marker panel described above and the appearance of hermaphroditism in the apical inflorescence. Individual plants that were scored hermaphroditic were assigned as 1, those that were not coded as hermaphroditic were assigned as 0.

[0155] The genotypic matrix was filtered for SNPs having more than 30% missing values within the population and a minor allele frequency lower than 5%. This resulted in 4815 SNP markers after filtering. The GWAS was performed using GAPIT version 3 (J. Wang & Zhang, 2021) with four statistical models: General Linear Model (GLM), Mixed Linear Model (MLM), FarmCPU and Blink (model=c (“GLM”, “MLM”, “FarmCPU”, “Blink”). A quantile-quantile plot (QQ plot) was used to evaluate the statistical models. The FarmCPU model performed the best by our evaluation and was used for the analysis. SNPs surpassing a LOD (−log₁₀ (p-value)) value of 5 were considered to have a significant association with trait variation.

[0156] SNPs showing a significant association with hermaphroditic inflorescence, with an LOD value greater than 5, were found on chromosome NC_044370.1 and NC_044372.1 with reference to the *Cannabis Sativa* CS10 genome and are listed in Table 1 below. The homozygous allele of the SNPs in Table 1 that are associated with hermaphroditic inflorescence are listed along with their position and reference sequence. Only one SNP was found associated with hermaphroditic inflorescence on NC_044372.1 defining a QTL there, while four SNP markers were found associated with hermaphroditic inflorescence on NC_044370.1. These comprise three distinct QTLs associated with hermaphroditism defined by the SNPs in Table 1. On Chromosome NC_044370.1 we found three QTLs, the first is defined by the SNP “common_10” at position 889040, the second by SNPs “rare_30” and “GBScompat_common_54” from position 28544332-32763867, and a third at position 99761993 defined by the SNP “common_524”. For all the SNPs identified in Table 1 there is an indication that the allele state that indicates the least likelihood of hermaphroditic inflorescences may be useful in screening and/or breeding plants that have a highly decreased likelihood of producing hermaphroditic inflorescence in environmentally controlled and outdoor environments. Because, hermaphroditism is in part a stress induced trait, conducting this trial in an outdoor exposed environment likely facilitated the identification of the relevant markers.

TABLE 1

SNPs associated with hermaphroditic inflorescence in Cannabis a field trial on Chromosomes NC_044372.1 and NC_044370.1. The presence of the hermaphroditic inflorescence is predicted by the occurrence of the indicative allele (marked with *). The positions of the SNPs are provided with reference to the CS10 reference genome as described herein. "Homo_1" denotes the average phenotypic value associated with homozygous allele 1 based on a score from 0-1, as described above, where 1 indicates a plant with hermaphroditic inflorescence and 0 indicates one without, "Homo_2" denotes the average phenotypic value associated with homozygous allele 2 based on a score from 0-1, as described above, where 1 indicates a plant with hermaphroditic inflorescence and 0 indicates one without and "Hetero" denotes the average phenotypic value associated with the heterozygous allele state based on a score from 0-1, as described above, where 1 indicates a plant with hermaphroditic inflorescence and 0 indicates one without. BP refers to the nucleotide position of the SNP.

SNP Marker	Chromosome	BP	LOD	Allele1	Allele2	Homo_1	Hetero	Homo_2
common_1527	NC_044372.1	61687722	9.57142	A	G*	0.143	0.219	0.405
common_524	NC_044370.1	99761993	7.58089	A*	G*	0.25	0.416	0.124
rare_30	NC_044370.1	28544332	5.42871	A*	C*	0.308	0.364	0.05
common_10	NC_044370.1	889040	5.17368	A*	C*	0.09	0.426	0.423
GBScompat_common_54	NC_044370.1	32763867	4.99593	A*	C	0.309	0.259	0

Example 2

Genome-Wide Association Studies (GWAS) of Hermaphroditic Inflorescence in Mixed F2 Population in Cannabis

[0157] To better understand the segregation of the hermaphroditic trait and to confirm the QTLs identified in Example 1, several targeted crosses were made between genotypes displaying low rates of hermaphroditic inflorescence, those with intermediate degrees of hermaphroditic inflorescence, and those with higher rates of inflorescence. The progeny of these crosses were selfed to generate 10 F2 populations (Table 2).

[0159] DNA was extracted from about 70 mg of leaf discs from all the plants evaluated in these 10 F2 populations. Using an adapted kit with "sbeadex" magnetic beads by LGC Genomics, which was automated on a KingFisher Flex with 96 Deep-Well Head by Thermo Fisher Scientific.

[0160] The extracted DNA served as a template for the subsequent library preparation for sequencing. The library pools were prepared according to the manufacturer's instructions (AgriSeq™ HTS Library Kit-96 sample procedure from Thermo Fisher Scientific). Targeted sequencing of a custom SNP marker panel based on the *Cannabis Sativa* CS10 reference genome was carried out on the Ion Torrent system by Thermo Fisher Scientific. The library pool was

TABLE 2

Genotype identification and populations used. "GPA" refers to the identity of the Grandparent Pollen Acceptor and "GPD" refers to the identity of the Grandparent Pollen Donor.

GID of F2	Total # of Plants	F2 Percentage Hermaphroditic	GPA	GPA Percent Hermaphroditic	GPD	GPD Percent Hermaphroditic	# of SNP Markers After Filtering	Significant GWA
21 002 003 0000	146	27.40	20 000 027 0000	24.54%	20 000 017 0000	22.49%	3227	Y
21 002 004 0000	137	20.44	20 000 081 0000	18.46%	20 000 017 0000	22.49%	3885	Y
21 002 012 0000	153	11.76	20 000 099 0000	11.76%	20 000 020 0000	19.33%	NA	N
21 002 014 0000	96	20.83	20 000 098 0000	12.96%	20 000 020 0000	19.33%	3728	Y
21 002 028 0000	123	16.26	20 000 081 0000	18.46%	20 000 070 0000	19.46%	4429	Y
21 002 035 0000	163	16.56	20 000 104 0000	16.56%	20 000 072 0000	14.76%	NA	N
21 002 036 0000	129	12.40	20 000 426 0000	16.41%	20 000 072 0000	14.76%	4212	Y
21 002 038 0000	127	20.47	20 000 426 0000	16.41%	20 000 072 0000	14.76%	3250	Y
21 002 041 0000	99	10.10	20 000 061 000	10.28%	20 000 072 0000	14.76%	NA	N
21 002 046 0000	115	1.74	20 000 006 0000	1.74%	20 000 083 0000	10.44%	NA	N

[0158] The inventors observed the emergence of hermaphroditic inflorescence in an outdoor field trial of each of the 10 F2 populations described in Table 2. In order to identify genetic regions associated with hermaphroditism in cannabis flowers they scored these 10 F2 populations for the presence or absence of hermaphroditic inflorescence by visually inspecting female flowers for the appearance of stamens or the growth of staminate flowers alongside pistillate flowers. This was used to calculate the percent of the population in which hermaphroditic inflorescence emerged. The inventors note that the evaluation of the trait expression and segregation patterns is complicated by the influence of environmental factors on hermaphroditic flowering. By conducting these experiments in a randomized field trial, they sought to minimize positional effects in the field.

loaded onto Ion 550 chips with Ion Chef and sequenced with Ion GeneStudio S5 Plus according to the manufacturer's instructions (Ion 550™ Kit from Thermo Fisher Scientific).

[0161] Targeted DNA sequenced from all 10 F2 populations segregating for the hermaphroditic trait, a population of 1267 individual, was used in a genome-wide association analysis (GWAS) to detect significant associations between genotypic information derived from targeted resequencing of the custom SNP marker panel described above and the appearance of hermaphroditic flowers, individual plants that were scored hermaphroditic were assigned as 1, while those that were not hermaphroditic were assigned as 0.

[0162] The genotypic matrix was filtered for SNPs having more than 30% missing values within the population and a minor allele frequency lower than 5%. This resulted in 3858

SNP markers after filtering. The GWAS was performed using GAPIT version 3 (J. Wang & Zhang, 2021) with four

tion is confirmation of the other QTLs identified in the 2020 field experiment from Example 1.

TABLE 3

SNPs associated with hermaphroditic inflorescence in Cannabis from a mixed F2 population on Chromosome NC_044370.1. The presence of the hermaphroditic inflorescence is predicted by the occurrence of the indicative allele (marked with *). The positions of the SNPs are provided with reference to the CS10 reference genome as described herein. "Homo_1" denotes the average phenotypic value associated with homozygous allele 1 based on a score from 0-1, as described above, where 1 indicates a plant with hermaphroditic inflorescence and 0 indicates one without, "Homo_2" denotes the average phenotypic value associated with homozygous allele 2 based on a score from 0-1, as described above, where 1 indicates a plant with hermaphroditic inflorescence and 0 indicates one without, and "Hetero" denotes the average phenotypic value associated with the heterozygous allele state based on a score from 0-1, as described above, where 1 indicates a plant with hermaphroditic inflorescence and 0 indicates one without. BP refers to the nucleotide position of the SNP.							
SNP Marker	BP	LOD	Allele1	Allele2	Homo_1	Hetero	Homo_2
common_491	94129798	9.130452	A	G *	0.01	0.08	0.50
common_517	99121582	7.410864	A	G *	0.02	0.07	0.32
common_518	99186615	7.227338	A	G *	0.01	0.04	0.26
rare_50	79534090	6.695617	A	G *	0.01	0.09	0.48
common_534	100736341	5.744335	A *	G	0.25	0.06	0.01
common_525	99830512	5.541966	C *	G	0.31	0.08	0.01
common_511	97379141	5.357322	A	G *	0.01	0.08	0.26
GBScompat_common_56	35677966	5.241665	A	G *	0.05	0.12	0.23
common_546	101726389	5.172549	A *	C	0.21	0.12	0.03

statistical models: General Linear Model (GLM), Mixed Linear Model (MLM), FarmCPU and Blink (model=c ("GLM", "MLM", "FarmCPU", "Blink")). A quantile-quantile plot (QQ plot) was used to evaluate the statistical models. The MLM model performed the best by the inventors' evaluation as it best accounted for population structure and was used for the analysis. SNPs surpassing a Bonferroni-corrected LOD (−log 10 (0.05/number of markers)) were considered to have a significant association with trait variation.

[0163] SNPs showing a significant association with hermaphroditism, with an LOD value greater than 5, were found only on chromosome NC_044370.1 with reference to the *Cannabis Sativa* CS10 genome and are listed in Table 3. The homozygous allele of the SNPs in Table 1 that can distinguish the likelihood that a hermaphroditic inflorescence will emerge during growth are listed along with their position and reference sequence. The alternative allele will in this case indicate plants that on average have 0 or close to 0 likelihood of producing hermaphroditic inflorescence.

[0164] From the results of the GWA, the inventors identified three QTLs based on the SNPs identified as associated with hermaphroditic inflorescence in the mixed F2 population listed in Table 3. The QTLs are defined by the SNPs on chromosome NC_044370.1. The first QTL is defined by the SNP "GBScompat_common_56" at position 35677966, the second can be defined by the SNP "rare_50" at position 79534090, and the third is at position 94129798-101726389 demarcated by SNPs "common_491" and "common_546".

[0165] The QTL on NC_044372.1 identified in Example 1 was not detected here, nor was the QTL defined by SNP "common_10" on chromosome NC_044370.1. However, the possibility cannot be ruled out that these QTLs and the SNPs that define them are associated with the hermaphroditic inflorescence phenotype. It may be the case that the source of these QTLs is not present in the targeted F2 populations designed in Example 2. The mixed F2 popula-

Example 3

Genome-Wide Association Studies (GWAS) of Hermaphroditic Inflorescence in Individual F2 Population in Cannabis

[0166] The inventors then looked at each of the F2 populations from Example 2 individually to assess if associations were masked by looking at the whole population together.

[0167] Targeted DNA sequenced from all 10 F2 populations segregating for the hermaphroditic trait, were evaluated separately in a genome-wide association analysis (GWAS) to detect significant associations between genotypic information derived from targeted resequencing of the custom SNP marker panel described in Example 2 above and the appearance of hermaphroditic flowers, evaluated as the percent hermaphroditic (plants with hermaphroditic flowers/ total plants).

[0168] The experimental design followed the same methodology as conducted in the mixed F2 populations. The number or identity of individuals in each individual population is listed in Table 2 above.

[0169] The genotypic matrix was filtered for SNPs having more than 30% missing values within the population and a minor allele frequency lower than 5% for each population individually the results of the SNP filtering is listed in Table 2 for each individual F2 population. The GWAS was performed using GAPIT version 3 (J. Wang & Zhang, 2021) with four statistical models: General Linear Model (GLM), Mixed Linear Model (MLM), FarmCPU and Blink (model=c ("GLM", "MLM", "FarmCPU", "Blink")). A quantile-quantile plot (QQ plot) was used to evaluate the statistical models. The MLM model performed the best by evaluation in all individual F2 populations as it best accounted for population structure and was used for the analysis. SNPs surpassing a Bonferroni-corrected LOD (−log 10 (0.05/number of markers)) were considered to have a significant association with trait variation.

[0170] In 6 out of 10 of the F2 populations, the inventors detected significant associations between genotypic information and the presence of hermaphroditic inflorescences Table 2. Surprisingly, even though all F2 populations showed the occurrence of hermaphroditic inflorescence, four of the populations, those with the lowest percent hermaphroditic levels did not show significant association from the GWA experiment. This shows that the population design was important in the identification of SNPs associated with hermaphroditic inflorescence that may otherwise not have been identified. Indeed, conducting this trial in any environment may also not have brought out the expression of the hermaphroditic trait strongly. Conducting this experiment in the field in non-ideal growth conditions may have also strongly contributed to the experiment identifying molecular markers for this trait.

[0171] Looking individually at each of F2 population (Tables 5, 6, 7, 8 and 9) did not identify additional QTLs, rather it identified additional associated SNP markers for, as well as supporting, each of the three QTLs defined in Example 2, that are useful in predicting the likelihood that a plant will display hermaphroditic inflorescence.

[0172] The inventors detected 1 SNP, termed “common_491”, found in 5 of the 6 individual F2 populations where a significant association was found. The inventors detected 5 additional SNPs, termed “common_512”, “common_517”, “common_518”, “common_525”, and “rare_57”, found in 4 of the 6 individual F2 populations where a significant association was found. Together these 6 SNPs represent ideal markers that are able to identify both the homozygous allele state that increases the likelihood that a particular plant will display the hermaphroditic phenotype. These 6 markers are equally useful in selecting plants that contain the homozygous allele state with no or a very low chance of the occurrence of hermaphroditic inflorescence. The significance of a SNP marker in the association studies does not indicate the presence or absence of that SNP in the population. In all populations the SNP markers or closely linked markers are present.

[0173] For F2 population GID 21 002 003, the SNP markers showing a significant association with hermaphroditism, with an LOD value greater than 5, found on chromosome NC_044370.1 with reference to the *Cannabis Sativa* CS10 genome are listed in Table 4.

[0174] For F2 population GID 21 002 004, the SNP markers showing a significant association with hermaphro-

ditism, with an LOD value greater than 5, found on chromosome NC_044370.1 with reference to the *Cannabis Sativa* CS10 genome are listed in Table 5. Here while only four SNPs were found associated, they define two QTLs.

[0175] For F2 population GID 21 002 014, the SNP markers showing a significant association with hermaphroditism, with an LOD value greater than 5, found on chromosome NC_044370.1 with reference to the *Cannabis Sativa* CS10 genome are listed in Table 6.

[0176] For F2 population GID 21 002 028, the SNP markers showing a significant association with hermaphroditism, with an LOD value greater than 5, found on chromosome NC_044370.1 with reference to the *Cannabis Sativa* CS10 genome are listed in Table 7.

[0177] For F2 population GID 21 002 035, the SNP markers showing a significant association with hermaphroditism, with an LOD value greater than 5, found on chromosome NC_044370.1 with reference to the *Cannabis Sativa* CS10 genome are listed in Table 8.

[0178] For F2 population GID 21 002 038, the SNP markers showing a significant association with hermaphroditism, with an LOD value greater than 5, found on chromosome NC_044370.1 with reference to the *Cannabis Sativa* CS10 genome are listed in Table 9.

[0179] The homozygous allele of the SNPs in Tables 5, 6, 7, 8, and 9 that can distinguish the likelihood that a hermaphroditic inflorescence will emerge during growth are listed along with their position.

[0180] In each of Tables 4 to 9 presented in this Example 3 below, the presence of the hermaphroditic inflorescence is predicted by the occurrence of the indicative allele (marked with *). The positions of the SNPs are provided with reference to the CS10 reference genome as described herein. “Homo_1” denotes the average phenotypic value associated with homozygous allele 1 based on a score from 0-1, as described above, where 1 indicates a plant with hermaphroditic inflorescence and 0 indicates one without, “Homo_2” denotes the average phenotypic value associated with homozygous allele 2 based on a score from 0-1, as described above, where 1 indicates a plant with hermaphroditic inflorescence and 0 indicates one without and “Hetero” denotes the average phenotypic value associated with the heterozygous allele state based on a score from 0-1, as described above, where 1 indicates a plant with hermaphroditic inflorescence and 0 indicates one without. BP refers to the nucleotide position of the SNP.

TABLE 4

SNPs associated with hermaphroditic inflorescence in Cannabis from F2 population GID: 21 002 003 on Chromosome NC_044370.1.							
SNP Marker	BP	LOD	Allele1	Allele2	Homo_1	Hetero	Homo_2
common_525	99830512	6.315994077	C *	G	0.67	0.12	0.10
common_517	99121582	6.220406615	A	G *	0.12	0.12	0.66
common_511	97379141	5.93898384	A	G *	0.05	0.15	0.62
common_512	97440573	5.891622544	A	T *	0.05	0.17	0.63
common_523	99748182	5.792407444	A *	G	0.60	0.12	0.11
GBScompat_	89565596	5.405502008	A *	G	0.56	0.18	0.05
common_91							
common_531	100304872	5.208939775	A	G *	0.10	0.13	0.56
common_500	95278820	5.170970412	A	G *	0.04	0.18	0.59
common_494	94521889	5.133641208	A *	G	0.57	0.19	0.04
common_484	91453227	4.82881167	A	G *	0.06	0.20	0.55

TABLE 5

SNPs associated with hermaphroditic inflorescence in Cannabis from F2 population GID: 21 002 004 on Chromosome NC_044370.1.							
SNP Marker	BP	LOD	Allele1	Allele2	Homo_1	Hetero	Homo_2
common_491	94129798	6.349096748	A	G *	0	0.03	0.66
rare_50	79534090	5.767453912	A	G *	0	0.07	0.67
common_497	95062684	5.676689017	A	G *	0	0.08	0.66
common_532	100388338	5.021729572	C *	G	0.68	0.11	0

TABLE 6

SNPs associated with hermaphroditic inflorescence in Cannabis from F2 population GID: 21 002 014 on Chromosome NC_044370.1.							
SNP Marker	BP	LOD	Allele1	Allele2	Homo_1	Hetero	Homo_2
rare_57	98263682	6.63	C *	G	0.79	0.08	0.00
common_497	95062684	6.50	A	G *	0.00	0.04	0.71
common_491	94129798	6.18	A	G *	0.11	0.04	0.71
common_532	100388338	5.82	C *	G	0.74	0.11	0.00
common_518	99186615	5.76	A	G *	0.00	0.05	0.78
common_531	100304872	4.88	A	G *	0.00	0.11	0.63

TABLE 7

SNPs associated with hermaphroditic inflorescence in Cannabis from F2 population GID: 21 002 028 on Chromosome NC_044370.1.							
SNP Marker	BP	LOD	Allele1	Allele2	Homo_1	Hetero	Homo_2
common_512	97440573	10.17989	A	T *	0.00	0.01	0.82
common_517	99121582	9.9980648	A	G *	0.00	0.01	0.78
common_511	97379141	9.9831618	A	G *	0.00	0.01	0.78
common_525	99830512	9.4120983	C *	G	0.77	0.01	0.00
common_518	99186615	9.1416083	A	G *	0.00	0.01	0.74
common_528	100088557	8.8877226	A *	G *	0.00	0.50	0.01
common_522	99561159	8.5214968	A	G *	0.00	0.01	0.73
common_521	99499651	8.5022921	C *	G	0.73	0.01	0.00
common_526	99965933	8.3072658	A	C *	0.00	0.01	0.73
rare_63	101144766	8.198918	A	T *	0.00	0.03	0.68
rare_57	98263682	8.1612221	C *	G	0.67	0.04	0.00
common_488	92041720	7.9867277	A	G *	0.00	0.04	0.62
common_491	94129798	7.9038878	A	G *	0.00	0.04	0.67
common_496	94684180	7.8273302	A	G *	0.00	0.03	0.64
common_479	90357946	7.825793	A	G *	0.00	0.04	0.62
common_473	89460138	7.7791312	A *	C	0.62	0.04	0.00
common_533	100626851	7.7177988	A *	C	0.63	0.02	0.00
GBScompat_	89565596	7.7084491	A *	G	0.62	0.04	0.00
common_91							
rare_50	79534090	7.680025	A	G *	0.00	0.04	0.62
common_472	89337710	7.5776055	A	G *	0.00	0.04	0.62
common_470	89227662	7.495949	A *	G	0.59	0.04	0.00
common_474	89539255	7.4869673	A *	G	0.62	0.04	0.00
common_484	91453227	7.4869673	A	G *	0.00	0.04	0.62
common_492	94257304	7.4338084	A *	G	0.65	0.04	0.00
common_534	100736341	7.3968511	A *	G	0.59	0.02	0.00
common_520	99379219	7.2076383	A *	T	0.76	0.08	0.00
common_545	101637128	7.0443121	A	G *	0.00	0.05	0.59
common_487	91953671	7.0214515	A *	G	0.58	0.04	0.00
common_527	99976278	6.9930683	A	G *	0.00	0.01	0.63
common_476	89639165	6.9753622	A	C *	0.00	0.04	0.58
GBScompat_	79818534	6.974447	A *	G	0.60	0.06	0.00
common_84							
common_483	90999267	6.9736607	A	G *	0.00	0.04	0.58
GBScompat_	101069452	6.9590037	A	G *	0.00	0.03	0.63
common_98							
common_489	92434073	6.9175851	A *	C	0.58	0.05	0.00
common_478	89807220	6.8841185	A	C *	0.00	0.04	0.61
GBScompat_	92378793	6.8553686	A *	G	0.58	0.05	0.00
common_94							

TABLE 7-continued

SNPs associated with hermaphroditic inflorescence in Cannabis from F2 population GID: 21 002 028 on Chromosome NC_044370.1.							
SNP Marker	BP	LOD	Allele1	Allele2	Homo_1	Hetero	Homo_2
common_544	101523781	6.6533006	A *	G	0.58	0.05	0.00
common_452	81689735	6.6344269	A	G *	0.00	0.06	0.56
GBScompat_101752789	101752789	6.5396364	A	G *	0.00	0.07	0.58
common_99							
common_552	102179472	6.5090881	A	T *	0.00	0.06	0.58
common_445	79634333	6.4699565	A *	G	0.56	0.04	0.00
common_539	101113276	6.4224429	A *	G	0.61	0.03	0.00
GBScompat_98559392	98559392	6.422416	A	C *	0.00	0.04	0.62
common_96							
common_486	91877425	6.3471412	A *	G	0.56	0.04	0.00
common_546	101726389	6.3370943	A *	C	0.56	0.07	0.00
common_432	70956672	6.1907734	A	C	0.04	0.40	0.00
common_449	79968518	6.1808976	A *	G	0.56	0.06	0.00
common_425	70407835	5.9928012	A *	C	0.54	0.06	0.00
common_453	81762017	5.881149	A *	G	0.58	0.04	0.05
common_541	101321815	5.8621967	A *	G	0.59	0.05	0.00
rare_66	102037098	5.7869904	A *	T	0.60	0.11	0.00
GBScompat_100988215	100988215	5.7831367	A	G *	0.00	0.03	0.57
common_97							
common_455	81901321	5.760267	C *	G	0.56	0.04	0.05
common_428	70624527	5.6943233	A *	G	0.56	0.04	0.05
common_426	70445826	5.6341286	C *	G	0.56	0.03	0.07
common_553	102227038	5.6149247	A *	G	0.54	0.07	0.00
GBScompat_75985317	75985317	5.5856371	A	T *	0.00	0.06	0.50
rare_10							
GBScompat_70516405	70516405	5.5418631	A	C *	0.06	0.04	0.56
common_79							
common_431	70913971	5.5281841	C *	G	0.54	0.04	0.05
common_439	72435815	5.4974102	C *	G	0.52	0.04	0.00
common_437	71800647	5.419316	A	C *	0.00	0.04	0.52
common_313	34512948	5.3609989	A	G *	0.00	0.06	0.48
common_375	57469730	5.2550177	A *	G	0.54	0.04	0.05
common_454	81831758	5.2448814	C	G *	0.05	0.04	0.56
common_436	71705041	5.1968557	A *	G	0.54	0.04	0.05
common_288	30449533	5.1792437	A *	G	0.52	0.03	0.00
common_448	79909671	5.0247419	C *	G	0.52	0.06	0.00
common_424	70255045	4.9725568	A	G *	0.05	0.04	0.52
common_434	71490671	4.9725568	A	C *	0.05	0.04	0.52
common_514	98401235	4.9531923	A *	G	0.55	0.03	0.00
common_459	82560029	4.9431435	C	G *	0.05	0.04	0.52

TABLE 8

SNPs associated with hermaphroditic inflorescence in Cannabis from F2 population GID: 21 002 035 on Chromosome NC_044370.1.							
SNP Marker	BP	LOD	Allele1	Allele2	Homo_1	Hetero	Homo_2
common_522	99561159	7.078061636	A	G *	0.00	0.07	0.56
common_517	99121582	6.585512305	A	G *	0.00	0.09	0.56
common_525	99830512	6.199958929	C *	G	0.57	0.10	0.00
common_491	94129798	6.106741417	A	G *	0.00	0.10	0.57
common_544	101523781	5.877345351	A *	G	0.52	0.09	0.00
common_510	97283326	5.8642263	C *	G	0.51	0.08	0.00
common_512	97440573	5.77051714	A	T *	0.00	0.10	0.53
common_518	99186615	5.506996042	A	G *	0.00	0.07	0.53
rare_57	98263682	5.415374722	C *	G	0.56	0.12	0.00
common_550	102006836	5.41426113	A *	C	0.50	0.10	0.00
common_545	101637128	5.338539557	A	G *	0.02	0.09	0.53
common_483	90999267	5.277138081	A	G *	0.00	0.09	0.52
common_539	101113276	5.213997161	A *	G	0.52	0.11	0.00
common_534	100736341	5.17925627	A *	G	0.50	0.11	0.00
common_549	101942035	5.088653633	A	T *	0.02	0.11	0.50
common_515	98481067	4.993112835	A	G *	0.00	0.09	0.56
common_533	100626851	4.948369261	A *	C	0.50	0.12	0.00
common_546	101726389	4.943779589	A *	C	0.52	0.09	0.02

TABLE 9

SNPs associated with hermaphroditic inflorescence in Cannabis from F2 population GID: 21 002 038 on Chromosome NC_044370.1.							
SNP Marker	Position	LOD	Allele1	Allele2	Homo_1	Hetero	Homo_2
common_533	100626851	8.509493763	A *	C	0.72	0.07	0.00
common_517	99121582	8.026552008	A	G *	0.00	0.05	0.65
common_532	100388338	7.455291164	C *	G	0.73	0.10	0.00
common_511	97379141	7.398320886	A	G *	0.00	0.09	0.63
common_534	100736341	6.926696529	A *	G	0.72	0.05	0.00
common_545	101637128	6.815330759	A	G *	0.00	0.07	0.65
common_527	99976278	6.738436035	A	G *	0.00	0.04	0.68
common_512	97440573	6.589382242	A	T *	0.00	0.08	0.59
common_525	99830512	6.386528883	C *	G	0.67	0.10	0.00
common_521	99499651	6.376695383	C *	G	0.62	0.02	0.00
GBScompat_	101752789	6.235536438	A	G *	0.00	0.07	0.63
common_99							
common_539	101113276	6.172028988	A *	G	0.67	0.07	0.00
common_491	94129798	6.156854894	A	G *	0.00	0.10	0.63
common_546	101726389	5.978964179	A *	C	0.59	0.08	0.00
common_518	99186615	5.828160881	A	G *	0.00	0.04	0.67
common_522	99561159	5.612304618	A	G *	0.00	0.02	0.62
rare_57	98263682	5.566979184	C *	G	0.65	0.15	0.00
common_553	102227038	5.518969152	A *	G	0.60	0.09	0.00
common_486	91877425	5.416506194	A *	G	0.64	0.11	0.00
common_474	89539255	5.299968402	A *	G	0.61	0.12	0.05
common_544	101523781	5.293858536	A *	G	0.61	0.09	0.00
GBScompat_	75985317	5.158044141	A	T *	0.05	0.11	0.59
rare_10							
rare_50	79534090	5.12903219	A	G *	0.04	0.13	0.59
common_514	98401235	5.119430229	A *	G	0.67	0.11	0.00
common_472	89337710	4.989467159	A	G *	0.05	0.10	0.59
common_203	20652741	4.976037505	A	T *	0.06	0.11	0.59
common_552	102179472	4.915029864	A	T *	0.00	0.13	0.59
common_300	32721609	4.902771546	A	C *	0.05	0.13	0.59

Example 4

Validation of SNP Markers for Hermaphroditic Inflorescence in Cannabis

[0181] The inventors sought to validate the QTLs identified and the broad use of the SNP markers found to distinguish allele states in Cannabis that determine the likelihood that a plant will produce hermaphroditic inflorescences. Representative markers of the four QTLs identified were selected based on the strength of their association with hermaphroditic inflorescence and their commonality in multiple individual F2 populations, Table 10.

[0182] The mixed F2 population was used, plants were filtered for those having a genotype call at the markers tested. After filtering, the average phenotypic value associated with each allele state based on a score from 0-1, where 1 indicates a plant with hermaphroditic inflorescence and 0 indicates one without, was calculated (Table 10).

[0183] The findings in Table 10, support the broad applicability of these markers, and those in linkage disequilibrium to them, and support the QTLs identified. Specifically, the inventors have provided the allelic state for each SNP that can be used to distinguish plants with a high likelihood of being hermaphroditic. Even though in some of the individual F2 populations a marker in Table 10 was not found, based on this experiment all of the markers show the ability to distinguish the allelic states for the hermaphroditic phenotype in the mixed population. SNP marker “common_491” performed the best at predicting the presence of hermaphroditic plants based on the allelic states in Table 10.

On average only 1% of plants in the mixed population of F2 plants examined were found to be hermaphroditic when the homozygous allele state of SNP marker “common_491” is AA. This contrasts with the homozygous allele state for this SNP of GG where on average half of the plants having this marker state are hermaphroditic. There may be other genetic and environmental factors influencing hermaphroditism that are not under the control of the QTLs found here which can explain some of the variation.

[0184] The inventors also looked at pairwise combinations of the SNPs that comprise independent QTLs to determine if combinatorial allelic states had an effect on the expected phenotypic outcome. They found no SNP combinations that altered the expected phenotypic outcome as compared with a single SNP alone. There may be combinatorial effects related to these loci however they may be masked by the variable stress induced nature of the hermaphroditic phenotype.

[0185] The reference sequences for each of the markers identified in Tables 1 and 3-10 are provided in Table 11 below. The reference or context sequence is based on the CS10 reference genome. The alternative allele “Alt” in Table 11 indicates plants that have 0 or close to 0 likelihood of producing hermaphroditic inflorescence.

[0186] Targeted sequencing primers for each of the markers identified in Tables 1 and 3-10 are provided in Table 12. The context sequence and locations are provided with reference to the CS10 reference genome for cannabis.

TABLE 10

Validation of SNPs associated with hermaphroditic inflorescence in Cannabis from all F2 populations on Chromosome NC_044370.1. The presence of the hermaphroditic inflorescence is predicted by the occurrence of the indicative allele (marked with *). The positions of the SNPs are provided with reference to the CS10 reference genome as described herein. "Homo_1" denotes the average phenotypic value associated with homozygous allele 1 based on a score from 0-1, as described above, where 1 indicates a plant with hermaphroditic inflorescence and 0 indicates one without, "Homo_2" denotes the average phenotypic value associated with homozygous allele 2 based on a score from 0-1, as described above, where 1 indicates a plant with hermaphroditic inflorescence and 0 indicates one without and "Hetero" denotes the average phenotypic value associated with the heterozygous allele state based on a score from 0-1, as described above, where 1 indicates a plant with hermaphroditic inflorescence and 0 indicates one without. BP refers to the nucleotide position of the SNP.

SNP Marker	BP	LOD	Allele1	Allele2	Homo_1	Hetero	Homo_2
common_491	94129798	9.130452	A	G *	0.01	0.08	0.5
common_517	99121582	7.410864	A	G *	0.02	0.07	0.32
common_518	99186615	7.227338	A	G *	0.01	0.04	0.26
rare_50	79534090	6.695617	A	G *	0.01	0.09	0.48
common_534	100736341	5.744335	A *	G	0.25	0.06	0.01
common_525	99830512	5.541966	C *	G	0.31	0.08	0.01
common_511	97379141	5.357322	A	G *	0.01	0.08	0.26
GBScompat_	35677966	5.241665	A	G *	0.05	0.12	0.23
common_56							
common_546	101726389	5.172549	A *	C	0.21	0.12	0.03

TABLE 11

Detailed information of each of the SNPs associated with hermaphroditic inflorescence in Cannabis as provided in Tables 1 and 3 to 10. The reference allele "Ref" is the allele that is associated with hermaphroditism, while the alternative allele "Alt" indicates the allele that when homozygous for that allele results in plants that have 0 or close to 0 likelihood of producing hermaphroditic inflorescence. The positions of the SNPs as well as context sequences are provided with reference to the CS10 reference genome as described herein. BP refers to the nucleotide position of the SNP. All of the sequences and alleles are provided with reference to the plus strand.

SNP Marker	Chromosome	BP	Ref	Alt	Context sequence
common_491	NC_044370.1	94129798	T	C	TTGATTCGAGAAAGGTCATAAAGAAAAATACACAGAAAGAACTGAAGATCCCAAA TTCGAAAGTCCTAAATGCTCAGTTACAGAAACTGCAATATGCAATTGGCTAAAGTCAT GCTCCATGTTTCAATAATGAATAAACACAGATAGTAATATATATGCTATCGTAAAGAGATA ACCACCTGATTAGACTGATAGT/C/GTGTCCAGTCACGCTGTGTAAACAGTAGGTACC ACCTAAACATAAACAGCAACCTTGTATCTGAGAGCTTAATAGATTTCCCTTATATGCTCT AGTATTAGGTATCAATTTGTGCGTGTGTTTCAAGAAATGAATCCAAAGGACTTACCTTG ATAAATACTGATACATCCCATAGGTGCGCTCGACCATTTGCAC (SEQ ID NO: 1)
common_512	NC_044370.1	97440573	T	A	TTGGGAAAGAGAAACTGTTAATTTTCATCAACAAATCTCTGAATGTAAGACTTGAT TCGAGACTTAAAGGGAAGATGATGAAGTCTCTCGAGGCCCATCTCAAGCGGAA AGAGATCTTTTGAAGTGCTTACTAAAGGGAAGTGTTCGGGAAACACGCTCCAAGCA ACTTTGGCTCAAAAGAGGAGATAGT/A/AAATAGTAGTATTTCCATGTTTTCGAGTGC TCGTAAACGCCAAATACGATTCATAGCTTCAAGATTACAATGGGAGTTGGCTTGATT GGGAGCTCCGCTCATGATCATATGTCGCTTCTAATCGAAGTTATTTTCACTTCTGCAC TTCATACCGATATTTTGATGTTGCTGCTGCTTACTTCTTGAATGA (SEQ ID NO: 2)
common_517	NC_044370.1	99121582	T	C	TGTGTGTAGATCACAAGACCAACCGTACGAGCTGTTACAGATTTGCAATGAAGATCA ATGCTTTATCTACAAATTAGATAGCATTTGGCCAAATTCACCCAAAAGTGTGCTGAAGG ATTTGCTTCAGCAGACCGCGTGACGGTGTGCGAGTGGGTATAGAGATGTGGCGA GGCGTTGGAGGAGGATCACGGTTGGT/C/TCATGCTTAAAGTGTGTTAACTGCGAGA TTTGGCGGCCAGGCCCAAGCCCAACAAAGGCCCAAAAGGCAGATATTCAGGTAG TACTAGCGTTCATGATAATCGAAAGAGATCATGACGAAAGGTGATGATGATAAA GAACTTTAGCCGATTTGGTATATAGCGAAGCTGCGAAAGTGGTTTGGGAAAGAGA (SEQ ID NO: 3)
common_525	NC_044370.1	99830512	C	G	TCTGCGTGATATTTTGAATTTTAGGTTTGTTCCTGTAATTTGATCGTTGTACATAA TTTGTAGCTTAAAGTTTATGATTTTGTCTGAACAGCTTGGGTGATACAGCTGCTGCA GCGAATATGATGAGTTTGGTGATTTGACGGAGAGGTTTGGATTGAAGCTCAAGGTA AATCGCTCCAATGGCGGCTTC/C/GAAGAGATCTGTCAATAGTAACGATGGCCAGAT TGGAAATTCGGGTTGATTCGGGTAAACCCAACTTTTCATCTTTTTCAGACGATCG GAACGGTTTTATCAATCCGGCAAGATAGGTAGCTCAGAACAGCGGAGCTTGAATA TTATGATGATATATTTGGTGACCTAGTAGCAATCGGTAGTCATGGT (SEQ ID NO: 4)
common_511	NC_044370.1	97379141	A	G	ATTTGTTACTTCTTCGTGCTTGTATGTCGCAAGATAGCGAAAGATACATAGATCATTCGTT CATTTGAGATGCTAGATTGCTTGAATCCGAGATGCTTAGAGAAAGAGACTGCTTAA AGACTATTTCTTAGGACCTATGGGAATTTATGTTTCATAGGCAATTTATTAAGGAAGT CTATAACAATGATTTTACCGA/G/TTTGTGTTGAGACTGAGAACATTAATGGGATTGC TGAGGTATGGAGATCTATGGGAGTATCATTAGTGGATTGTCATTACCTTTAAAGGAGG

TABLE 11-continued

Detailed information of each of the SNPs associated with hermaphroditic inflorescence in Cannabis as provided in Tables 1 and 3 to 10. The reference allele "Ref" is the allele that is associated with hermaphroditism, while the alternative allele "Alt" indicates the allele that when homozygous for that allele results in plants that have 0 or close to 0 likelihood of producing hermaphroditic inflorescence. The positions of the SNPs as well as context sequences are provided with reference to the CS10 reference genome as described herein. BP refers to the nucleotide position of the SNP. All of the sequences and alleles are provided with reference to the plus strand.									
SNP Marker	Chromosome	BP	Ref	Alt	Context sequence				
common_518	NC_044370.1	99186615	C	T	AGCATAAGATATTCTTGCAGAGGTTTTGATTCTCTGTCATAGGCCAAAGTCTTTGGGG				
					GTTTACTTCCAACAACCTCTCTACTGTGTTACACAATTTATAGAGAG (SEQ ID NO: 5)				
common_522	NC_044370.1	99561159	C	T	CCCATATGATACATATGAATGTAACACACTTACCCGCTCTTCAGGAAAGGAACCTTGAGA				
					ATCCGAATATTTTACAGAAAATACATTGCAAGTTTTCGGTTTGAGTCTTGAACCAA				
common_533	NC_044370.1	100626851	C	A	GTTCACACTCTTTGGAGCTTATTTTATCACTTACTGACCCCTTACTTTTATTTATGAC				
					AGCAGTAACGTTACTCTGCTG[C/T]GATGATTTGCTGCAGTTGCCATAATGTTGACATC				
common_534	NC_044370.1	100736341	A	G	AGTCGTGTTGATTCTCGGTTTTTTCCTTTTCTCCGCCAATGGTAATGGCAATCCAAAT				
					TCCACCTTTATGCTACCAATAAAGGATTAGATGGCATCGGATGGAACAGCAGAGA				
common_539	NC_044370.1	101113276	A	G	CTCTAGCTTGATTAGACTGGTCAATCCTGTAATGCTCTCTCAAGA (SEQ ID NO: 6)				
					TCGGATGTTTATTCACCCTCGTCTTTTACTTCGAATCTCAGTGTCTGTAGAGTTTA				
common_539	NC_044370.1	101113276	A	G	TTAGAGCTGCTTCGCCCTCAGACATAGTGTCTGTAATATCAAGATGATGATGTT				
					TTATTGGCTTAAGCAACACATATGTTTATCAATTAATCACTACTCAGCAGCATATAAGCTCATCTTA				
common_539	NC_044370.1	101113276	A	G	GGCAGAGTAACCAAGTTC[C/T]ACTGAGGCAATTAAGTACTTACCCTGCTTTGG				
					TATAAGTAAGGAAGCCACAATGTTGGAATAAATCCCAAAAGTTAATGCTCTGA				
common_539	NC_044370.1	101113276	A	G	AAACCCAAAAGATAAATACAAAATTAAGAGTGTTCACGGATTGGTTTAGTCTAGATT				
					TTTAATTTTTTCAAAATAAACCACTATTACGGTTTTTTTAAAAAT (SEQ ID NO: 7)				
common_539	NC_044370.1	101113276	A	G	TTTCCAGTGTATACGAACCTTCGACTCTTATCAATGTTGTGGTCCATTGTTATCTAAA				
					CGTTTATCATGTTGTGCTTTTGGAGTACTATCTTTGGATATTACATATTTAAGTCCTCT				
common_539	NC_044370.1	101113276	A	G	CTTGATAGCCATCTACTGGATCTGCTATGGAACAAGTACTGGGTGAATACACTTCTTCT				
					TCACCCCTACTGGGTAAAG[C/A]GATTATGTTGACGAGCAAAATTCAGACTTAGGTACC				
common_539	NC_044370.1	101113276	A	G	TAAATGTTAAGTACAGAAATTTTTTTTAAAGTCGCTGCTAGCTTTCTGTAATGTTATC				
					ATGTTAATAAATCAATTAAGTGTGTTTACTTTTATCATTTATGAGCTGAAAAAGTTGGA				
common_539	NC_044370.1	101113276	A	G	GCAAGCAGAGAACCAATTGTCACATTCGCCCTTTGGGCCATT (SEQ ID NO: 8)				
					TTGACTTCATAACTATTAGTAAGTATTATTCTTCTACTTCCACATCTTGTGGTTGC				
common_539	NC_044370.1	101113276	A	G	TTATTATGATGTTGTGTTTCTTCTCTCTAAAGAGAGCAAGATTCTGAAGTTTATA				
					GGCTTATGTTGGAATCCTTTGCTCTGGGTATGGAAGTCTGCTATAATGGCTATTGC				
common_539	NC_044370.1	101113276	A	G	CTTGGCAATAGTGGTGGAA[A/G]CCTCCGGATTGGCAGGACTTTGTTGTTATTATG				
					TCCTGCTGATCATCACTCACTATCAGTTTTTATTGAAGAAAAACAATGCTGGTAATGACAG				
common_539	NC_044370.1	101113276	A	G	CAGTGTCTCATGGCTGGTCTTGCTCCAAAGACTAAGGTAAGAAACACAGCCTCACCAT				
					CACCATCAGACTTGTGTTGTTTATGCTTTTATGGTATTTTTTTTAA (SEQ ID NO: 9)				
common_539	NC_044370.1	101113276	A	G	CTACTCTCTCTCTCCATTTCCATCTCCACCTCCAGTCTAGTTCATCTCTCTCTAGATA				
					CTCAACTGTATACATATCATCAAAATTTATCTTAAATCAAAACCAATGAACCTTTAA				
common_539	NC_044370.1	101113276	A	G	TGCTATGATTTCAGCTTCAGAGGGGTTTGTGTTGGCCACTCAACCAACCAACCAAC				
					CACAGTTCTGCTCTCTTCT[A/G]AACCTTCTACTTCATCAGCTCTGCACTCCGCC				
common_539	NC_044370.1	101113276	A	G	TTTCATGCTAAGGTTCTTAATCTCCATTTTCTTCTCTCTCTCTCTCTCTCTCTCTCTCT				
					ATGTTTTCATTTATTTAACTTCTCTCTCTCTCTCTCTCTCTCTCTCTCTCTCTCTCTCT				
common_539	NC_044370.1	101113276	A	G	CGGCTGCTGTCGAACACTCTCTAATGAGGAACCTTTACAT (SEQ ID NO: 10)				

TABLE 11-continued

Detailed information of each of the SNPs associated with hermaphroditic inflorescence in Cannabis as provided in Tables 1 and 3 to 10. The reference allele "Ref" is the allele that is associated with hermaphroditism, while the alternative allele "Alt" indicates the allele that when homozygous for that allele results in plants that have 0 or close to 0 likelihood of producing hermaphroditic inflorescence. The positions of the SNPs as well as context sequences are provided with reference to the CS10 reference genome as described herein. BP refers to the nucleotide position of the SNP. All of the sequences and alleles are provided with reference to the plus strand.					
SNP Marker	Chromosome	BP	Ref	Alt	Context sequence
common_544	NC_044370.1	101523781	A	G	TACAATTACATTTACAAATCCCAATGAGGCTTAACAAGGCCATGATGAACACGACGTCGTCGTCGTTGATGGGTAAACGCTTCTCGAAGCTTGTGTACTTTGTATATCAAGCTATGAATCTCCATGAGCCGTTTGGCTGGAGAAAGCGGCTGGAAAGTAGCCTTGGAAAGTTGGCTCGTCGAGTACGGAGJA/GJAAGAACCGCTAGCCCAAGCGAAATGGATCGATGTCACAAAGAGCATATGGCTCCATCAGTCCCACTAGCTCCCAAAAGGTTAGTTCATGCGCCGCAAGTGTGGCCGAGTACGAGCCAGCCCAATATAAAGTCCCAACGACCAATTCCTTGTCTTCTGTCGATCATCGATTTTGTTCCTCATTTCC (SEQ ID NO: 11)
common_545	NC_044370.1	101637128	A	G	TCTTCAACAGATTTCTCGCATAAATGAAAAATGTTTGAAGAAGAAAACTATGTTTAAATTTGTATCTTTACCAAAATGAGTAATAGATACCTGACAGTCTCTCTTTTAAATTTGGCTGCGCAAGTCTGCTAAAGCTTGTGAAGAAGACAAATCAACAACCTTTATCTCGTGGACTGGAJA/GJCTCCCAATCTTATCTGGAATAAAGTATAAATGGTCTGTGATTCATTACAAAAGAAAGAAAAAAGATCGATCGTGTAAAGCATATAAATTATAATGAACCCCGACAAAGTCAACCGTCCATTTTGGACAAATAAAGATAAGACATGAGCCAGGAATATCTCATTAATATATTATTACAGATTACAT (SEQ ID NO: 12)
common_546	NC_044370.1	101726389	C	A	ATCCAATGATTAATCCGTGCGCTTGGTAAAGAGAAGTTTCAGTAACGCCAAAGATCCTCAAGTATATACATCAGTACAGAGAGACAAAGAGTTGTCTATTTCTCACTAGAACCAGCCATACGAAACACGACTGTGATCATCAAGCTTCCCCCATAGTAGCGGAACCGCATCTTGAGCAGCTGTGAAAAC/AJGGTGTGCTGCTTCCTTGTCTGTGCTCGTTGACGCGCAATGCTGTATTTTGTGCTGTGCTGAGGCCCTTAAGCCAGCTCGCATGGCATGCATCTACTGCGGTATCTCATTTGTGTACAGTGTCTCAAGAGTCCATCAGTCCAGCAATGATGACATCCCGGAGCTACAGGTACTGTAAAAACCTGAATC (SEQ ID NO: 13)
rare_50	NC_044370.1	79534090	A	G	CATCCACTTCGAACCGTACTTTGGCAACAGTAAGCTCCGTACACCCCTTACCGTCGATGTCCTTGACCGAAACAAGATTTCTTCCCGAATGATACCGAGTTCCACTGTGCCAACAAGGCTTCAGAGTTGCTTAGCGCTGGGGCGGTAGTCGCTGCTCATGTGTGCAACC
rare_50	NC_044370.1	79534090	A	G	TCATCCATCAGTGAATCCGCCATCJA/GJAACAGAGCCTTAGTGAACACGTCGGTTA
					TAAATCTCTCTTTCATCTCGAATATTTTCAAGGCTGAGCCCGCTGACGCTACTTTTCTGTCATCCACTCTCTCTCTACACAGTAAGAGGAATATTGCGTTCCCAATGATTTTGTCCTTTTGATGTCGATCTCAAGAGTGAATACAATGTGTTCAGAGAAA (SEQ ID NO: 14)
rare_57	NC_044370.1	98263682	G	C	AAATGGAGGCTCTTTAGCAGGGGTATAGCTGGACCTTCATGCTTCTTACTGGAGTTAATACGACACATAGTTTGGCTATTTACATACCTATATCGGAGCTCGGTTTTAGCGTCGGTTGTGGGATTAAGCAACGGTTTGGGAGATATTGTAAGCCACTCACATGGCTCATGGCGACATTTTCCCTATGTGTG/CJCTCTTCTTCCCAAAATCTGTACGTATCTTTCAATTTGATTTGATTTTCCATCTAATCTTGTGTGTTTGTGTTTATTTCTCTCTAGTTTCGTTTGTATTCTTCTAATTGGGATTCGAAAATGATTTTGGTTTATGGCATTTGATTGAATTTTCGAGTTGGCTTGTTCCTCTCTTTATGGGCGTGT (SEQ ID NO: 15)

Detailed information of each of the SNPs associated with hermaphroditic inflorescence in Cannabis as provided in Tables 1 and 3 to 10. The reference allele "Ref" is the allele that is associated with hermaphroditism, while the alternative allele "Alt" indicates the allele that when homozygous for that allele results in plants that have 0 or close to 0 likelihood of producing hermaphroditic inflorescence. The positions of the SNPs as well as context sequences are provided with reference to the CS10 reference genome as described herein. BP refers to the nucleotide position of the SNP. All of the sequences and alleles are provided with reference to the plus strand.						
SNP Marker	Chromosome	BP	Ref	Alt	Context sequence	
common_472	NC_044370.1	89337710	T	C	ACAATTGTAATCCAAAATTCAAATGCAACCCCTGTTTAAAGCAAGTAAATTTCAACTGATTT AATCAAGATTTCTGAGCTATAAAGCGAGTTCCTTATTATTCAAAAGTCAAAATTTCTCTTTA ACATTTTAATTGTCTATAATCCACAGTTTCGAAATGGTCCGGAAGATCAACGAAGA ACATAGTAGAGATCATATT[T/C]CAATCAAGCTGGCTCAAAAAGGAAGCCCAATATGCA AAATCGACCGAATACTGAAAGTTCAAAATACCAGAAAACCATATCAAGTTTCGGAGG TATCGAGAAGCCATTAAAGCCAGGCCACCAAGCTCCCAAGCAAGACACCTCGATGCA TCGCCGAGCGGAGCGACTCCTCCGCTTCCATTTGCGCCACCTTCATG (SEQ ID NO: 16)	
common_474	NC_044370.1	89539255	C	T	TATATATACATATATATAATCAATTTCTATCTTTCTTCATCATCATCGTCATCAGTTTCTCTG TCTCGGAAAATCCGTTTGTGTTTTTCTGTAAATGATTAGATGGAAAGGACGACTACA ACAATTACGATGAGAGAGACAGAGGCTTTCTCTCATCTTGTTAGTCATGACTATCTTC CAGCAGCACCACTGGATC[C/T]TACCCCTCCGCCCGCGCTCTGCATATCCACCGCC GCCTCATCATCAAGCTACCCCTCTCCACCGGCTACCCATCTGCTGGATACCCCTC CTCGTCGCTCAGCGCATACCCGCTTACTCTCTCCGTCGATACCCCTCC TTCGCTTATCCTCATCCTCCTCCTGCTGCTCCATATCCATATCCTGTGA (SEQ ID NO: 17)	
common_483	NC_044370.1	90999267	G	A	TTTCACCATCTCTGCTCTGCTCAATTTCCACTCAACATCCCTCCTCCCTCATCCCGAT CATATAACACCCACATCTCATTCGTCGTAAGACGCGCTCATCAACGCTCGACATTT CAGAGCTGCTCTCTAGCCCGGACTCAATCCCTCTCTCTCTCTCTCTCTCTCTCTCTCT CCTCATTACTCAACAGTTCATT[G/A]TCAAGCTCAGCTTGACCATCATGCTCGCGTTG CCACCGGCTTCTTAAAGCTAGCTCTGTTCCCATGAAGAATATCTCTTCTTTTAG CTCAGTTGGCCACTTTGGGTGAGTAAGTAGTGTTCATTAAATTAAGAAGAAATAA ATAAAAATTTCCGACTCAATGATGATTAATAAATAAATCTCCCT (SEQ ID NO: 18)	
common_484	NC_044370.1	91453227	T	C	AATAGCTCAACTGCTATAACAGATAATACAAATTCGCCACCTTTATGAGAGTGATCCAC AGGCAACAGCTTTCTATGCTCTTTAAGGGCTACAAAATATCATTTTGATTCAAATAGTGT TTTCAAGTTTGAAGACATTGATTGTTTACCCCCCATGGTTTTTGTCTGTTTGTGA ATTTGACAGTCAAAAGCTCT[T/C]CAGAAGATGAGATGACAAAGTTAGCAGCAGA AGGAGGAATAGTTTGAGAGCTCATACCTTAAATCTCCCTTTTATAGATGAATCTTTT TTTTTTTATATCAACTTCTCTCTTTTCTTATTTCCAAAGTCCCAAGGCTTAGAGA ACGCAATTTGGCTAAATTCCTGCTGCTCCACACCCCATGTAAACAG (SEQ ID NO: 19)	
common_486	NC_044370.1	91877425	T	C	CTGTCTAGACTGACATCACCGTTGCTGCTCACTGTCTCAGCCAGGTTTATGGCAATCC ATAGCAACTCATCTCCAAGAGTTTCATCATCTACTACACTACCTTAAAGAAAACTCTGG TCAAGGGCTCTTCTATTCACCTACTTCAGCTTTTACACTACCGGTTTCTGTGACTTTGA TTGGGCATCTTGCCCTGTACT[T/C]GATGTTCAACACAGGTTTGTATCTTCTCGG AGACAGTCTCATTTCTTGGAGACGAAGAGCAGTCAATAATATCCAAAGCTCTGCAG AAGTCGAGTATGAGCCCTTGGATGCAACAAAGTAAATGACATGGCTGCAATATCTT CTCCATGACTTTCAGATACCAAGCCACATCTTGCTTTTATTTACTGTG (SEQ ID NO: 20)	

the plus strand.

SNP Marker	Chromosome	BP	Ref	Alt	Context sequence
common_514	NC_044370.1	98401235	A	G	GAAGACGATTTTCAATTTGCAAGCTCAGAGGCGGTTAGTGGTGAATTAAGGAGATG TCAGAGGTGGATGATACATAGTATGGTTCCAGAGAGATCTAAGGTGAGAGATATTC CAGCTTTAGCAGCTGGTGTAGAGCAGGTGCTGTGATGTCTGTTTCTGGGGACC CGAGGAAGAAGTCAATTACTATCCTGGJ/A/GJAGGTATCAAGATGGTGGCTTAAGCAA GTTTGGCTCATCTTGATTTCTCTTTGCGAAGTTTGGCATTCTCTCATCACCACAAAGAT CTGCTGATAGTGTCTTCTTCTGAGTTGTCAAGTCCACTGGTGCACCAACAAATAC TCCTCAACCCTATATGTCAGTATTTTCTTTCGTTATATATTTTTCACAGC (SEQ ID No: 21)
common_521	NC_044370.1	99499651	C	G	AACAAAAAGTTTATCTCTAGTCAACATAAGAAAGAGACACGACACAGCTCTTATCTTA GCATATAAGAAACATAAGCCACTACCTACTTCTTGGGCGAGTGCTTCACAGCTCGG CCATCGCGGACGACCTTGGCTATATCTTTTGGCAGTGGCATCAGGCTTTTGGTAGT GTATAGTGTGAATATCCAATGGT[A/C]/GJGATGAGTAAGCCACCAATGGCCATGAT GGCTGGGCTGTAGGAAGCTTCTTCTCTGATGGAGACACCCCCAGGATPTGGGCTT GGTGCCCTTGAGACCTTCTTACTCCAGCTGCTTTTATTTCTCTTACTGATTTTGTGG GTGTCGCCATTTTCTCCACTCTCTGCTTCTTTTCCATTTTCTTCTCTTAGA (SEQ ID No: 22)
common_527	NC_044370.1	99976278	G	A	GAATACAAGCCAAACCCCATATAAAATTTCTCAATCAATAAGTAGTTTCATCTTCATTG GGAGCGACGACGACGACAAACAGAGAAATCGTTGGGTTTGGGCAAAAGGGGTATTAAC TTGATTTCTACATCAATACAGGAAGAGACATTTGGTTTCAGCCCATGTCTTCCAAGTCAA AATTTCCATAGCCCATGATTTCTT[G/A]/CCTGAGCAACAACCGACCCACCATACTAGTCCA TTGCTCTTAAGATGAGAAATAGCAACTAAGACACCAACAGAGACCACACAGAAAGTAA CCAGTGCTCACTCTCTGCTGTCGCGCACACTACTACTACTGCTGATGCTGAGGCT GTGATGACATTTCTGTTAAAAAGAGATTAGTTTAAACAGAGCTGACCCTACA (SEQ ID No: 23)
common_532	NC_044370.1	100388338	C	G	AGAAACCTTAATCTTCAATTAAGACCATTTGTTCTTATTTATTTCTGCTTCTTCGTGAGCCCA TCCCCCTCTTGAAACACCGAATCTCCGAAATACCTCCACGACTTTCATGATCAATGTCGTT GTACGGAAGCTTTCATGCGCGCGCACCCGCCGCTGCTACGGGGGTTGCGAGGTT CGCGAAACACCAAAACCCATTCC[G/A]/CCTGACCAACGCTGGCGAATAAGCCACCG GGGTGAATAATACCTCTTGGGAATTCCTATTGAGAAGCCCTTAACCGGTGACGAGGT AAAAACCATTTTCTGCTCATCAATAGTTGAACTCTCTTACCCTCAATATCTCAAAA ATAAAGACACAAACTTTTCGACACGCTCTCTTTTATTTGCTTCTTCTTCTT (SEQ ID No: 24)
common_552	NC_044370.1	102179472	A	T	TGTCCAATCAATCTCTTCCCACTCTCTCTGTTTCCATTACAAAGAGGCGAAACCGTCG TCGTTTATGCTCCCGAGGCTGGAACGTGTGAAATCATATTTCCGTTGCCAAAGGCT GGCCTTTTGACGTCACCAATTTCTCCACAGCAACCACTTCAGACACTGCCACCATCT TATAAGAGTCTTATCCCAACCAA[A/T]/JCCAAAGCGCAATAGCTCCACCAACTTACAT AACCAAAATCAACATTCAGCTTCTCTCTCGGCCACTTCAAACTTGCACGCTT GTTAAAAAATCAAGTGGATCTCAACTCAACTCAAGTCAAGAGTATCTCAAAACGACAT GTTAAAAAATCAAGTGGATCTCAACTCAACTCAAGTCAAGAGTATCTCAAAACGACAT

TABLE 11-continued

Detailed information of each of the SNPs associated with hermaphroditic inflorescence in Cannabis as provided in Tables 1 and 3 to 10. The reference allele "Ref" is the allele that is associated with hermaphroditism, while the alternative allele "Alt" indicates the allele that when homozygous for that allele results in plants that have 0 or close to 0 likelihood of producing hermaphroditic inflorescence. The positions of the SNPs as well as context sequences are provided with reference to the CS10 reference genome as described herein. BP refers to the nucleotide position of the SNP. All of the sequences and alleles are provided with reference to the plus strand.					
SNP Marker	Chromosome	BP	Ref	Alt	Context sequence
common_553	NC_044370.1	102227038	A	G	CTTTGAAGTTATTTCGATTCTGTTTGTGCTATAAAGGGTGTAAATGCAGAA (SEQ ID NO: 25) ATATCGTGCTTAATATAAAGAACAATAAATAACATGTTATTATAGCTAATTAATATAGCACTTAATAAATCTTATACATATAGTACGTACATAATGTGGAAGCTAGCAATTAAGTGATGATGATGATTAATTAATACCTGTGTGATCATTAATATGATCAATTAGGAAATCCTCTAAGGTGGTGCAGGAGAG/G/CAGAGCCTCTGTAATGGGTGAGGAATGGTCA GATTCAAATTTGAGGAGTCTGCACAAGATTTCTGAGTGTCTCTTGACTTCGTCGG AGTCTTCCCACTGCTTTGGCCACATTTGGCATCGGTGAGGGGTGCTTGTGCAT ACACAGCTAAAGCCTTTCAAATTTGCTTTTGTCTTAGACGTCCCAAG (SEQ ID NO: 26)
GBScompat_common_91	NC_044370.1	89565596	C	T	GGTGAAGATACATGATCTTTCTGGCTCAGCTGTGTCGCGAGCGTTCAATGACAACCTCCGTTTGTGCTTTCCACAGGTTGTGATGGGTGCCAAGTCCAACCCAGGAGCTTGGTTGATTTGTCGGCTGATGCTTTCAGGCCAGAGCTGCGCAGCCATGGGGTAACTCGAAGCTTGGCGTAACCTGGTATGAGGACTC/TjGTTTGTCTAGATTTTCGCTCATGCTCGAAAGGTGATCATCAGGAGCAGGGAGCTTCTCATGCTCAGATTTATATCAATGCGCAAGGGGAGGGAGTTTTTCATAGACACGATAGCAGCGCTTGCAACGGCAGCACTCCAATCCCAAGCCGCAAGCAGCGAGACTTTTGACGAGCTCAGCCCAAGTTGTTGAGGC (SEQ ID NO: 27)
GBScompat_common_99	NC_044370.1	101752789	A	G	AGCCCCAACACCCCCACCAACACCACTGCTCCCTACCTATGCTCAGACCAAGAAGATCAGCGTATGGCGGAGTCTCGAAGCTCAACGATGGGCCATGACTAGATCTGTCAAGCCAGATTTCCACAGGCTGAAGCCCAAGCCCAAGCCAGCTGCCAAGTCGACGAACACTTTCGGCTTGATTTAGTjA/GjCTGTTCCTGCTGCTGCCAAGACTGTCTTTAACGGAGGCTGAAGATGTTGATGTTGTCAAGCAAACTGCACACAGCGAGGAATTTGAAGCGAAGCACCATTTGGGCTGCTTTTGAAGCAATGCCGTTGCAGACATTGCTGAGGCTCAGTGTGATGGGCTGAAGTTGCGGTGCTCGGCTGCTCTTGTGTTTGA (SEQ ID NO: 28)
GBScompat_rare_10	NC_044370.1	75985317	A	T	CAAGAATTTTGACGATCGATTCCGAAGGCTCTTCGTCAACCTTGAAATCGAAACTTCCATAGCTAGCAGGAGAGTGTGTTTCCGACTCCAGAAATGAACCTGGCTGGAGTCAAAAGTCAACTGGGAATCGTCAACTAGTTTGCATATGATCGGGTGAGGATCTTAGAGAGAAGAACTCGGAGTGTGCjA/TjGCGGAAGAGGAATTTGGTGGTGTAGAAATCGTCCCTAATTTGTTACTACTTAATCCAGTTATAGAGAGATCAGCTGACTTACTTGTACTTGTGAGAGTAAAGTATGATTAATGAACACAGAGCAAAATATGATCATATTTTTTGTGTGTATGATTAGCGTGTGTTGGTAAATCAGATTTTCGTGTTTGTGCAT (SEQ ID NO: 29)
common_203	NC_044370.1	20652741	A	T	AGAAATTTAGACCGTGTGAGTTTGGTTGTGTGAAATGCGCCAGGCTTCGCTTAGCAA GCACGCTGGTTGAGTGAACAGAGATCGGAATCCCAAGCCCGCTTCTCTTTTAGA CCTACAAAGTTTTTTTCCAACTTTGCCAATGAATCTTGGGGCAATTTGTTCTTGAATCC CCACAGAAATTAGCCATAATGGACjA/TjCCAGAGAGTGGCAAGTGGCCACAGGGAGA CGAAAGCAGGCCATAGTATAGTGGGGAATTCCTGTAATGACAGACTTGAGAGGATTT

the plus strand.

SNP Marker	Chromosome	BP	Ref	Alt	Context sequence
common_288	NC_044370.1	30449533	G	A	CCTTACCTCCCTTAGAGAAGACTTTATTTTTCCAAATTATGAAGTGGGCCCAAACCTTCTCATGAAGATAGTGAAAAAAATTTATTTTTTTCGTTTCGGCCAAATATATTGAGGGAG (SEQ ID NO: 30)
					CAACCCAAAGAAAGAAAAAATAATCTAAACAACCGATTATATGATGTTGTATTGCTAAAATAAATCCCAATCCAAATCCAAACCATCAATAGAAAGAAAGAGAGAAAGATGATGATAATGAGATGAGATCCCAATTCTCAAACCCCTAAGCAAAATCCAAACACATACCTCTCACTCTGTGACCCCTCAGTAGTCCCG(G/A)CTCTGCAATCACCATCAGAAATCTTCGCAATTCATTCGTGTTTCAACCGCTCCTTCGTTCTTCTCCCTCCCAATAAGACGACACGACACCAACCGCCACCGGCTCTCTGTCGGCAGCTCGACACCTTCATCATTTCTCTCCAGGATTGCCACCGCACCTTACGTACTCTCCGATGTCTTCCTCCCTCCGCT (SEQ ID NO: 31)
common_300	NC_044370.1	32721609	A	C	TGAAACTTCCTAAAGAAATGCATCTTGGTATCTCTTGAATGATCTCGGAAAAAGAAATTTTGTGAAGAGCCAAAAGTACTAATCATTTTCATTTGGTTTACCCTGTAAATTTGGTTGAGAACCAATTTGAGACTCATTTCTTTGTGCATCATGTGCATCGGTATTGAAACTACCTTCCTCAAATGCTACCTTCAATGTCAATTTGTAGCGTCA(G/A)CJAGTGAATAGCCGGCAATCGATGCATPACCACTACCCTCAATAATAGCTGATCCCTCATCAATCTAAACAAAACCTTCCTTTGCTTTGGGCTCCAGTTGTTCCTTGCTCACATAGATCGAGAAAAAGCAGATACGTAGACAGGTTTGGGCTCCAGTTGTTCGAACCATCGGCTGCAAAAAGCTTTTTTCCAGTAAGTCAAAGCCTCG (SEQ ID NO: 32)
					GGATAGAAAGACACCGTTATCATAGAGGGCCACAACCTTAGTAGTGTACAAAAATTCAGGGAATAAGTTAAATTTGTTCTTTAGAAAAAGCAATATGGGAGCCTAAATTTTGCAAAATTAGAAATCAATTAGCATTTTTCTCACATGAGAAGATTCACGCTCAAAATCAGATCAAGTGCATAGTCATAATATAGGACTTGGACT(G/A)GJCTCAACCCACAAGCTTGAATCATCTTGAATGTATCATATCGACTTCGCCAAAGAACTCAATTTCTCTCGCAATTCGCAAAACCATGAATGCCAAGAAGTGTCAATCCATCAGAGCCACTTACATCGATTTCTCACTATCGCTCTCAGATTCTCTGCAACACACGACACACACCGTCAATAAGCAAGCTTAGAGAAT (SEQ ID NO: 33)
common_375	NC_044370.1	57469730	A	G	TTTGGGACAGATAAACTATTTTGGGTACCAATCATGTCAAGTCCAAAAAGAACTTGGTTCTACTAGGACCCGTCACCAAGCTTTGTCCAGATGTGCTTGAAAGTATCAATGTGCACAATCTGTGTGGACAGAAAAATGTTAATAGAAATCAATGACGTCAAGGAGAAATACAGAAA(GCCATTCCTGACTCAGTTAAGGA(G/A)GJCTTGAATGGGAAAGGGCTGATATACATGTTGCAGCAATTGCTTCTCTTGGACAAAGGCATTCAAATGGCTTACTGTGTTGTGATTGCTGTGTGCTTTTATCAGATGTTATTTACTCTCTAGGACACGGAATTAGTCAATGCCATTTGGCTCTTGGCTGGTTGTTTATGCTGATTTCTTTTAAAGAAACA (SEQ ID NO: 34)
					TTGTCCGCTCAAAACAAATATGCAATATCAGAAACCTTATCACCCTACACACCATGTGTTTATTCAAATCAACATATCTAGATAAAAAAGCTGTAGACATGGCTGACCTGAGAACCAAGTATGAGATCCAAATGCGCGGATCTCATGGGTGACACGTGGGTGAGATAACGCAATCAATCCATCGTTTGTGAAATCG(C/T)CTTGGCTGAAAAAATCAAGGTTGTGAAAGCCAAATCAAGATTAGTCCGTAATAGATTGAGAAATTTTGAATTTTACTTTGCGAAAAATAGATC

TABLE 11-continued

Detailed information of each of the SNPs associated with hermaphroditic inflorescence in Cannabis as provided in Tables 1 and 3 to 10. The reference allele "Ref" is the allele that is associated with hermaphroditism, while the alternative allele "Alt" indicates the allele that when homozygous for that allele results in plants that have 0 or close to 0 likelihood of producing hermaphroditic inflorescence. The positions of the SNPs as well as context sequences are provided with reference to the CS10 reference genome as described herein. BP refers to the nucleotide position of the SNP. All of the sequences and alleles are provided with reference to the plus strand.

SNP Marker	Chromosome	BP	Ref	Alt	Context sequence
common_425	NC_044370.1	70407835	C	A	GGAGGATAAGTACTGTCGTGTTAAAGGATCGTATCTCTCCAGGTTGCTTGAAGT (SEQ ID NO: 35)
					TGTTCTCCTTCTCGATAATGCACACTACACTACCAATAAATACACACAT (SEQ ID NO: 35)
common_426	NC_044370.1	70445826	G	C	AAGCAATTACAGAAGCAGCAGAGGTGGTGCATAATGCAGGTGGATATGATGCCCATG ATCCGTTTCTTGGTTCGGCTGCTCAGGTAGTTTGGTTGCTTTTGTAAATACCGTTTCAA ATTACAGAAACAGCAATTATCTTGGATGTTTCTGGAATGTTCTGATTTCTTCTTACTAGTTCCAAA TCAAGCTTATTCAGCATCCG[C/A]TTCGAGTTCATTAAACCAITGCCCTCCAGGAC TGCTTCGGCGAGAGGGCTACAGGCCATTCCGAGTGAGTATCTTAGTTCCAAAA CCTTGATATTACTGTCATCATCTGTTCTCTCTGTTATACATGAACCTCCGGACCAT AAGATATGAACCTTCTGTTATCATTTTGAAGTCCCTTAATGCCACTT (SEQ ID NO: 36)
					GAGTCTTCTTTAGATTCATCAATCGATAAATTCACAAAGCTGTACTGCATGCTCTTG AATCTTTGAATCTGGATAGATAGGAGCTGTACTAATGTGGGATTCCTCCATTGTGG CAATCAAAATTCGTTCTCAGGGTCTCTTTGGAGAGCAACCGAACCCTTTTAACGCC TTTCGCTGCGTTTCTAAATGGCT[G/C]GAAGATAGTCTTCAACAGGGAACAATCTCC TCTTTGAGATCAGCTGAGAGCTTTCATGGTCTCTGATTTATCTTTGAGGGAAGTTT AAAATGTTCTTCTCACTCCACTGCAAGATTATGTTTAAAGACATAATTTGACGCAAG TGTCAGATGAGGAAGAGTTTGGCTGGTTTGGGACAAAGTCTCTGTGATTA (SEQ ID NO: 37)
common_428	NC_044370.1	70624527	G	A	AGGAAGCAAAAGATCCAAATAAAGAAATTTGGAAACAAGAGTAACAACATGTCACTT TCTTAACCGCGCTCTGGTCTCTTCAAGAAAGCTGGCGAGCTCAGCCTTCTTTGTGG GTCCAGGTGCAGTCAATCGTCTTCTCCACTGGCAACTCTCTGCTTCGGTCAATC CATCTGGAGACGCGCTTCGCTC[G/A]TATCTGATTAATAACCAACAGTAGTAG TATTACCTGGTAGTAATGATAACATTACTACTACTACTACTACTACTACTACTACTTTGT AAAAGCTATGACTGAGTTGGAAGGCCGAGAAAAGAGAGCAATTGGTGAGACAA GCGAGATTGGCTCGATTTTAGATAATAATAGTTGCGCGTGGGGAGGAA (SEQ ID NO: 38)
					ATCCCAGCAGACACCATCAAGCCGCTCAATATCTCATCATCATCATCATCATCATCT TCCATGCGATCTCATCCGTTGATTTCCAGCTCAGGACGCGGTTGCTGAGTTCACT GCTGGGTAAACCAACGGGAATTCGGACCTCGTTTGGCTGACCAACGGGTTGACCC GACTGAGACCTCCAGACTGCTC[G/C]ACGATATCTTCAGCTTCCACAGCCCAA GAATCTCCCGGATTAACCGCTCCGCTAGAGATCTCTAGAGATTTTCCCTCG CTTCGTAGCGTGTACGGTATTTTCAGACCTCGATTCTGTTAATCAAGGAACAAT CAGCGCCCAAGTGGGTTTAGAAAGAGACGAGTTAAAGGTTTCTCTTTACG (SEQ ID NO: 39)
common_432	NC_044370.1	70956672	C	A	TCAGTAACAGTCTTAAGTCTCTCCATTATTATGGGCTGCGCCAACTCGAAGCTCAAGAT CTTCGCGCGTGGCTTGTCCGGGACCGATGTAATACCTAGACGAAGCTCTCCGCC ATAGCAAGACTTCGCTGAAGCTCACGCGCTTACTCTGACTCCCTCAAGCATTTGGG TCCGCTTTTGATCGTTTCTTCAGTA[C/A]AAAAGTACCAGTCAACATGAACCTTTAAAA

TABLE 11-continued

Detailed information of each of the SNPs associated with hermaphroditic inflorescence in Cannabis as provided in Tables 1 and 3 to 10. The reference allele "Ref" is the allele that is associated with hermaphroditism, while the alternative allele "Alt" indicates the allele that when homozygous for that allele results in plants that have 0 or close to 0 likelihood of producing hermaphroditic inflorescence. The positions of the SNPs as well as context sequences are provided with reference to the CS10 reference genome as described herein. BP refers to the nucleotide position of the SNP. All of the sequences and alleles are provided with reference to the plus strand.									
SNP Marker	Chromosome	BP	Ref	Alt	Context sequence				
common_434	NC_044370.1	71490671	C	A	AAGTCCTCTCTTTATCTCCGGCGCGTGGCTTCTTCTCCCGCTGCATAAATCAAA CTCAGCAGACTCGCACCTTCGATCCCTCCGATCTTCTGAGGACGAGATAGTGATA ATCAATAAATACTCGGCATAATTGCTCAACGAGAGAAATCCCAAGGCTT (SEQ ID No: 40)				
					TCATGACCTGACTGTGACTTTTAAATTTAAAGTCGTTTGGAAAGATTAGATTGT ATATAATCTTTTCGGCTTCCTATGTTGGACCAACCATCTGCTCAGGGTTGTGA GAGTTTGGGAGATGCTGAATGAAGTAGTTGAAGGAGATAGAGTATCCCAACATTT CTTTCAGATTGGGGAGTGTTC/[A/G]ACTGCAAAATCAAGAGAGATTAACCTCTGTTCG AAATTCCTCTCAAGTCTCACCTTGGATCGCAGATATAGTCCACAGATTACAGAGA CCTCAAGGAGACCTTTGTACCAATTTCTGTTTGTTCAGTTTGTAGTAGTACTGT GGTTGATATCTCAATCTTACAAAATTAGCTGAAATCCCTCCAAAC (SEQ ID No: 41)				
common_436	NC_044370.1	71705041	T	C	CTTCTCTAATATAGTATAGCTATGCTTATTGGATAATCTTAAAGCTCCATTTAATTATGC TAGACTAAGTTCACTTTTGCCACATGCTATTTATTTGCGCAGAGTTGTAGTGCCAC TCCTTCCAGCGGACCAAGAGCCCTCCCAATCTTCTATACAAACATCTCACCAAT TACTGCAATAGAGTTAGT/[C/G]TCACTCCAGGATCTTAATGATGATCGGTGT GAATGCTCAGCGCTTTCGAATTTCAATCTGTGGAGTAAATAAACAATAAAGTTTCAC CAAGCATGAATAATTTACTTGGTACATATCAACAAAATAGTGTCTTACCGACTT CACAAATTTGATCATGAATAGAGCTTCCCAATCGTAAATGGTCAAT (SEQ ID No: 42)				
common_437	NC_044370.1	71800647	G	1	GATATCTACTTGCAGATGTACGCTAAAGATTGTTTCAGAAAGACTTCTTTAGAGGAAG AAGAAATGAAGATGAAGAGACAAGAGTGAATATTAAAGAGACCAACTCAAAA CTCAAGGATTACAAAAGAAAGAAAGTTGTCTAGTGTAAACGCCAAAGTAAAAAA AGAGTGTAGAGATAGAGTGTGT/[G/T]GGGTGTAATAATTAATTAATGAGACGACTTC TGATTAATTAATCAGATATAGATTAAAGTTTGGCAGAAAGGTAATTAATGATTGAC CGTTACATCCACGCTCAATCCCTAATTTCTTTCTCAGCTTTGTGTCAGACTAAAA TTTACTCCCTTCTTTAATTAATGCTCTCTTTTGGAACTAGCCCAT (SEQ ID No: 43)				
common_439	NC_044370.1	72435815	G	C	GGCAGACAATAACCTCTAATTTCAATTTCCCTTCTTCTTTTGGTCTCTCTCTCT CTTCCACATACAAATCCAAAGTTTCGTGAAGTACACAATAAGAAATTAACAAGATTAT AATAACGTACTAGATAGGTTTAATATATAGTTATTAATCTGATGATGCGAAAGTAT GGCTTCATCTTCTTCTTCT/[G/C]TGTGTTTATATATCGTCACATAATAACAATAAG TATGTTTATTTTATTAATCCATCGTGGGTTGATTAAGTTTACCGCAATACAGACAG TTCCGCGAGTGATCGTTCGGAGACTCGATCATGGACACAGGGAACAACAACGCTCT TAAGACTCTGTCAGTGTGATTTTCGCTCTCTTATGTTGAAAAATTT (SEQ ID No: 44)				
common_445	NC_044370.1	79634333	T	C	TTTGATATTTTAGTACGATCGGGAAGGAGAGAGTTGTTGGAGATGGCGCGAC TAGATTTCGAAAGCGATTGGGGGGTGAAGGACAGACTAGTATTGGAAATAGCTAAG GTGGTAGCAACATGGCGCGGAGCTTGAGTTGCGATTGTCAAGGACTAGTCTAC GACGATGATCCGGTAGCGAAGTACAT/[C/G]AAGAGAGATCCTGAATCTACATCTTA CTCTGAGGTTACGTTTCATGCTGTGTACGGCAATTCGAAGCGTTTGGGGAAGCG				

TABLE 11-continued

Detailed information of each of the SNPs associated with hermaphroditic inflorescence in Cannabis as provided in Tables 1 and 3 to 10. The reference allele "Ref" is the allele that is associated with hermaphroditism, while the alternative allele "Alt" indicates the allele that when homozygous for that allele results in plants that have 0 or close to 0 likelihood of producing hermaphroditic inflorescence. The positions of the SNPs as well as context sequences are provided with reference to the CS10 reference genome as described herein. BP refers to the nucleotide position of the SNP. All of the sequences and alleles are provided with reference to the plus strand.					
SNP Marker	Chromosome	BP	Ref	Alt	Context sequence
common_448	NC_044370.1	79909671	C	G	CGGACTGGATTGTGGCAGCTCAAGGCTTTGATGCTGATCCATCGTTTGCTTAATGAAGG GGATCCTTTGTTTCAGGAAGAGATCTTGTTCCGTACCAGAAAGGGACAGGCTTCT (SEQ ID NO: 45) ATCTCAATGAATTGGATTGACAAATGATGGTTATTCTCTTGTCCCAACAGTCTCAATT CGTTTCGGTTTACCACCACTAACCAAGATGTCAAAAATTTCTGCTAGTAATCGTCCA CATGGGAAGCCAAAGCTTATTGGACAGCGAATATAGACACCCCACTGTCCTCAATTC GATAAGTTTGTGTCGATGAAA[C/G]GGTAGAGTTTCTTGCAAAAAGGACAGTGTG GTTTGTCCATGATCTGGTGGCAAGGAGCTTCGCTATGAGTTGATGGATCAGG TAATTTGGTTCTGTTGGAGAGACGAGAATAGAAAATATGGAGAGTTTGGACCATC CAACCGATACCTTTTATGGGACAGGAATTTGTTGAAGGATGAGACTT (SEQ ID NO: 46)
common_449	NC_044370.1	79968518	T	C	AGTGGATCATCTACTGAACAGAGGTTCTGTATTCATAGTAAAAATTTGTTATGAGAGCA CTGAATTTCTTCGTTTCTTCAATTTGTTGATGGCGGTGACAAACATTTATTTAATG GGTACTCTGCTATCTGATGCTGTTGACATTTTTTTATGCTTTCATCAGCTACTGGA AAGTCGAGGTAGCTATGCC[T/C]GAGGACCTTACGFGAGTGTGATTTCTCCTTTCCAA TTGTTCTTATCTTGAGCTTATGATTAATTAATTTGTTCTTTCTTTTACTATAATACA ATCTTTTCTTCAAATCTTTTATAGTTAGTGGTCTAGCTATATATATCTTTCTTTATTCCTCC TTCCCTATTATGTTTGGTGGCAAAATTTGACAGATCGATT (SEQ ID NO: 47) AATTAATAATGGCATCTTCGCTCAGAACTCTCACTTTCATCTTCTTCATCTCTCTACT CCATTTGGGTTTCATCGGGAGGATCTTTCCGACGAGCTGACCAACCCACAGGCT CTTCCCTTCAATACATAAATGGCCCTCTCTGTTGGAAAAATCTCCATTAACCTTAATAT GGTACGGAACCTTCAACCA[C/T]CCACGAGCCATTTGTCCTCGACTCATTCACCTCCC TCATCTCATCTCTAATAACACACAGACCAACATCAGTCACACGCTGGTGAAGACC ATCGAAAGCTACCAATACACACAGCTAGCAACTCCGCTCTCGTTAGGATCCAGTT CATCGACGAGAACTACTCTCCCTGGGAAATCGTTGACAGTCAACAAAT (SEQ ID NO: 48) GAGTATTTTCAAGTTTAAGATCTCTATGACATATTTGCTGCCAGAGTCAATTTATTAATA ATTAGACAATGACAAAGTTGATATCAATGCAAGTTCTTTAGAGATATGTAATATCTAT CACAAAAGGATAGTTCAACAAAAATCTTACCATTTGAATGACAGTACTGACTCCTGATA TTAGTTGTTGAAGAAAAA[T/C]CTTCCCTGATTCATAGAACAAAAAGATGCTTAAGTGC TTCAATTCATTAAGATTCTTGGTGTAAAAACATAAGTTTAGTTAGAAAATTTCTCCAT TACCTCATCTTCAATAATCTACCAGCAGTGATATCTTTTCAAGAGCTCTCCACACC AGCATCTCCATCACAAATGCCAGATGAGTTGGAGTTAAAG (SEQ ID NO: 49) CTTTTATTATCTGAGATTTAGTAAGTGCCCATATGTTTGTAGATCGTGGGATTTAGTC GATTAGGGTACTAGTTGGAGGTTTTGTTTTTATCAATTAGTTGCTTCAAGAACTCACT TTCTTTTTTCTGTTTTATGCTAGTAAGAACTTGGAATGCTGAATAGGATCTCCAGCAGGCGT GTAGGAATCACTGAGAGTGT[G/C]CATGTGCTCTTGTGAGAGAGCTCAACGGGAG GTAGCCAGGCTATGGTTCTCGTTGTAATGCTCAGCAGAGGGCAACTCTAGAAAATTGTC
common_452	NC_044370.1	81689735	C	T	
common_453	NC_044370.1	81762017	T	C	
common_454	NC_044370.1	81831758	G	C	

TABLE 11-continued

Detailed information of each of the SNPs associated with hermaphroditic inflorescence in Cannabis as provided in Tables 1 and 3 to 10. The reference allele "Ref" is the allele that is associated with hermaphroditism, while the alternative allele "Alt" indicates the allele that when homozygous for that allele results in plants that have 0 or close to 0 likelihood of producing hermaphroditic inflorescence. The positions of the SNPs as well as context sequences are provided with reference to the CS10 reference genome as described herein. BP refers to the nucleotide position of the SNP. All of the sequences and alleles are provided with reference to the plus strand.						
SNP Marker	Chromosome	BP	Ref	Alt	Context sequence	
common_455	NC_044370.1	81901321	C	G	GTAGGATGAACCTACTACAGAGAGGACCCTCCAAAAGGACAAATGAAAAAAGA	No: 50)
					GGGCCAAGACACCCCGCAGCGCTTACAATGACTCCATTGGGAG (SEQ ID	
common_459	NC_044370.1	82560029	C	G	GTGGTTTACATACCCAGCACGCATCTGATTGAGCCACCATTGGTATGAAC TAGAAG	No: 51)
					GTAACTCGAGACCCGCGAGGGCTGATACACATCAAGATAATTAATACAATCACA	
common_470	NC_044370.1	89227662	T	C	TATCTGAATAGAAAATATACCAATTATGTTCTTACGTGATATAATTAGACTTTGGTTTG	No: 52)
					CACATGCGACGAATCTCGATT[C/G]GATGGAGTTTCCATAGCTTTTCCATTCCAAAA	
common_473	NC_044370.1	89460138	T	G	TTCCACTCGCTTCATCCACTGCATCTAAAGGAGTGAATGGAAAAAATGCAT	No: 53)
					GATTTGTCGATTATTACAAAAGAGGTAGTTATAATACGAATACCCCTCTCTCTTT	
common_476	NC_044370.1	89639165	C	A	TTAATCAAAATTTCCCTCTTATTATGGAAGTCTCTCCTACTAAG (SEQ ID No: 51)	No: 54)
					AGCTCACACAAATGAAGAAATCGGGACACCTTGGAATTCACCGTCGACAATATTC	
common_477	NC_044370.1	89639165	C	A	GACAAATGCTTACAAATCTCTCGAAGCGAAGATATTTCTTCCAGACCAATCT	No: 55)
					CTACTCACCCACCACTAAGCCCTTAGCTCTCAAAATTCGCTCTGCTCTTAAACAAGA	
common_478	NC_044370.1	89639165	T	C	CTTCTGATAAAGCCTTCTGTTGAGT[C/G]ATGATCAGATGAGTAGCATCCATGCTAGTG	No: 56)
					GCCAAGAAATTAGCTCAACTCGAAGTCTGTTAATTAATTAATTAATTAATTAATTAAT	
common_479	NC_044370.1	89639165	T	C	TCAATATTAATTAATTAATTAATTAATTAATTAATTAATTAATTAATTAATTAATTAAT	No: 57)
					ATAAGATGAACCAACCAATTAAGATGATGATTAATTAATTAATTAATTAATTAATTAAT	
common_480	NC_044370.1	89639165	T	C	GAACCTGTCTGAAAAATAGTTTCAATGTAATTAAGATAGGAAAAACAAGTTCCAGCA	No: 58)
					GTGAATAAGAGGAAATTAACAACTCAAAATGATCTCTTGAATACCAATCCATCTTGACC	
common_481	NC_044370.1	89639165	T	C	ACCAACAGGCTTCACACATGCACACCGAAGAAATTCGTCAAGGGAATTTTCATACC	No: 59)
					GACACATGTTGATGATAACTA[T/C]AAGCATCCGTAATAGACCTGAAATAATATTAC	
common_482	NC_044370.1	89639165	T	C	AATTAGTTTATTTTACCTGACCTTCAAAACCGCAATTTATGCTACTCAAAATGGCTTTG	No: 60)
					AGACAGCATTAAGCTCATAATAAGCAAAAGTCAAGAGTAGGCAACAAACCTTGACT	
common_483	NC_044370.1	89639165	T	C	TCTGATTTCTTCCCGTCTTTATCATAAACCTATCATCTTGTGTCAAG (SEQ ID No: 53)	No: 61)
					CTATAATATACTTTCCAAAAGGACTATTGCTTTTAATCTAATGTAATAGTACAAAAA	
common_484	NC_044370.1	89639165	T	C	TGTTGTTTCACTTTTCTTTTATTTAGAAAAAGTACCCTTTTATCCGTTGAAGCAT	No: 62)
					TGCACCAATTTGCATATCTTTATCTTCGTCTGCAATTTGTCATGCTACTTTCTGTGTTT	
common_485	NC_044370.1	89639165	T	C	TCATGTTGTTGTTGAGG[T/G]GTAAGGTAGTATATCTTCCAAATTCATATGCTTCTC	No: 63)
					TCCTTTTACTACTACTTTGTTTTGTTTATGAGAAATATTTCTACACAAC TAG	
common_486	NC_044370.1	89639165	T	C	ATAGTCATAGGAAACGTTGGAGATTCTATTGCAATAGAGTTTATGACTGACTGAC	No: 64)
					AGTFAAGTATTTCTCAGTACCACTTTTCACAATATTATAGAAA (SEQ ID No: 54)	
common_487	NC_044370.1	89639165	T	C	AATGTCATCTCTATCGAAACGCCGCTTTGATATCTTAGCCATTGCTGACACAGGCA	No: 65)
					GCGATCTGAGTGCAGTCAGTCAGCCCTTGCAAAAATGTTACAAAGAAAGTGCTCC	
common_488	NC_044370.1	89639165	T	C	TCCTTCAAAACCACTCTTCAAAACATACAGAGATGCTACCTGTGATTCCTCTGTTTG	No: 66)
					TAAAGTCGCCACCGGAGCTAAAC[C/A]TCTTGCTCTCTCTCGACATTCAATGCCAATA	
common_489	NC_044370.1	89639165	T	C	CTCGTATCTTACGGCGACCAATCTTCTCCACGGTAACTTGCTACTGCTGCTCTCA	No: 67)
					CCCTCTCTCCACCTCTGGAAGACCCGTCACCTTCCCAATTTCAATCATCGGTTGCAGC	

TABLE 11-continued

Detailed information of each of the SNPs associated with hermaphroditic inflorescence in Cannabis as provided in Tables 1 and 3 to 10. The reference allele "Ref" is the allele that is associated with hermaphroditism, while the alternative allele "Alt" indicates the allele that when homozygous for that allele results in plants that have 0 or close to 0 likelihood of producing hermaphroditic inflorescence. The positions of the SNPs as well as context sequences are provided with reference to the CS10 reference genome as described herein. BP refers to the nucleotide position of the SNP. All of the sequences and alleles are provided with reference to the plus strand.										
SNP Marker	Chromosome	BP	Ref	Alt	Context sequence					
common_478	NC_044370.1	89807220	G	T	CACAAATAGTGTATATATATAATAATAATTAATTATGGCGTGATAACCTTAT (SEQ ID NO: 55)					
					CCCTTGGTTGAAACCGCTGAGCAAGCTCTACAGTTGCATTTTGTCCACCATCAGTAGGCC					
					AGCCAACTCCACCATGATTAATGGACAAGTTTCCAAATCCATTTTCTGCGAGAGCCATA					
					CCAGTGTGTCATGGTTTGCATCCCAAGCACTTCTGGTAGATTTTCCATTTCTGCTGTTATGG					
					CAGCGAATAGCCATCAAGAA[G/T]CGGAAATCACAGGAAATGGGATCGTTGTGG					
					AGGCTGATAAAGATAGATGTTTACAGTAAGGAACCTCCATTTGCTTGAAGACTT					
					AACGATGGCCACCATGAGATCTTTTATGCTCTTTCGAGTCACCGTCGGAGGTTCT					
					CACTCGAGTGCCTATACATCAGCGTTTAAAGGATAGTGACTTAACT (SEQ ID NO: 56)					
common_479	NC_044370.1	90357946	T	C	AGTAATTTTAAAGTTCTGGTTGTTTTCAGGAAGAAAATTTGGTGTATATAAAAAGTAGGCAA					
					TGAGAGTTACAGAGAGAAATCAGATCAGATGAGTTAAATACAAAAAGCAAACTCTTT					
					CAGCTCAAGGAAGGAGGGTTCAGCTACCAATAATCCAAAATATAGAGGTAAGAAA					
					TTCCCTATTGGTGAAATTTTAAAGT/C]CAACCTTGTCGAATGAAGCCCTCCCAATACCA					
					ATGCCTTAAAGTTTCCAAATGCTTCTCCTAACACTTACAGTAGGCTTACCTTTCAAC					
					TTCAATTCAAACCCCTTTTCTTTGTAAAACTAACTTAGTATATCAAAAGTAGTTTAAAGTAG					
					AATATAATAGAACTATTTTCCCAATATTAGAAGAGTTGGGATTG (SEQ ID NO: 57)					
common_487	NC_044370.1	91953671	T	C	GTTTGAATATAAAGCTAGCAGCACCTTCAATGATTAATTAATGTTCTTAAACCAGAAAGAATA					
					GCTACAACTCTGAACCTTTTCAAGTACAATGATATGTTATGTTTGACTAATCAATCTTAC					
					CAAAATATTTTGGCAATGTGGAACGAATTCCTAGTTGACTGCTCTCTTCCCAAGTGC					
					ACGGTTAGGTTTCCCTTTCAAT/C]AATGATCAGGAGTGTCAITTCCTAAAACCTCCTGA					
					GCAITGGCGTAAGCAGTTGCGGTACTCTACTTGTGTTTCAATCCATCCATCGCATC					
					ATAGTCCTCTCTACAGGCTCAGTAAATACCCATATCCTGTAGAGAGATTTCACACA					
					GAATTACGAAAAAATTAATTTCCCATCAAGCCAGAAAAGTTTAGGTG (SEQ ID NO: 58)					
common_488	NC_044370.1	92041720	G	A	TAAGCTTTCTTACAACTGCTGTCATGAAGCGCCACACAGTAGCCACAGCAACAAAG					
					TAAGCAGCTGCAATAAATACATCACGCTTTTAGTAAAGTACCATAATTCGGTCTGCTGT					
					AAAGGTGTTCAACAGACAAATGAATATAGATATGATCAAGAGAGAGTATATGTAAG					
					GACACTAACCATAGGGAACGGG/A]AAACGTTTAGAAGTAAATATTTACACACAGACA					
					TAGAAGGAGAAACGTCATTGCAAAATGGCTGTTGCATCAGACCTACACAGAGT					
					AACCACTAAATTAATGATTAATGATTAATCAATAATAATACTTTAAAGTT					
					GAATGCTATTTTGTCTTTTACTATGCTAGGGAAAACTCAATCCCTTA (SEQ ID NO: 59)					
common_489	NC_044370.1	92434073	A	C	ATTCAGAAGCTGGGATGAAGATATTAATTCAGAGACCTTTTGGAGATAGAGATCAGACC					
					CTCATTTTCTTATTCAGCTAAGTGAATGCTTGGAGGATCATTTGGTGTCTAGAAGTTT					
					GAGNACGATGGAGTTACATGGTCAACAATGCGTACACAGATCTTCAGGTGCTCAAGG					
					GAGCTGGCGTGTGCACAAACCGA/A]AAATCTTTGGCATACTCTCTGGAGCTTTAAAG					
					TGCCCCAAAGGCTGCAATTTTGTGGAGGGCAGCGTCAGGTGTTTGCCTACTTGT					
					GTACAATTGCAGAGAGACATGTACACAGTAGTATAATCTGCCCCTGTTTGCATGTGGA					

TABLE 11-continued

Detailed information of each of the SNPs associated with hermaphroditic inflorescence in Cannabis as provided in Tables 1 and 3 to 10. The reference allele "Ref" is the allele that is associated with hermaphroditism, while the alternative allele "Alt" indicates the allele that when homozygous for that allele results in plants that have 0 or close to 0 likelihood of producing hermaphroditic inflorescence. The positions of the SNPs as well as context sequences are provided with reference to the CS10 reference genome as described herein. BP refers to the nucleotide position of the SNP. All of the sequences and alleles are provided with reference to the plus strand.					
SNP Marker	Chromosome	BP	Ref	Alt	Context sequence
common_492	NC_044370.1	94257304	G	A	TGATGAGACTATTTTCCATGCTCTTGTGTGATTGCTCTGTCGCGAGATCGTG (SEQ ID No: 60)
					CCGCACAAAAGAAAAATTATACTCATAAAGTTGATGAAAAAGAAAGGGTCCAAAGCAGTA TAATTCATGTTAAATTAGTTGTGATATATTGTTTACATTTCTCATCCACACAGCGAAAGT GTACTGATAAGCAGAACTTGTATGTCCTGACTCTGACTCCAGGATGGATTTGTTTCTT CCACCAAAATAGTGCACCT[G/A]TATGATGCACATCTTTATAATTATTATGTCATATAT ATATATAGGAGAGAAAAATTAAATTATAAACTAATTAATTAATTATTAATTAACCTCATCAATCA TTGTTGAAAATTCGCGCAGGTTGATACATAAATCAATTCCTTAATCTGATGATCAATTA AAAATATAAAATAAACATTAAATTTTAAAGTTAAGGTCAAC (SEQ ID No: 61)
common_494	NC_044370.1	94521889	C	T	ACCAACAAAGAGTGCCTCAGCATCGGGCTAAAGGATATTCAGCAATCTCGCCAAAT ATGTCATCTCTTGGCCCTTTAGAGTACCTGACTGTGATCGATGATGAAACAAGTCT GCAGTTTCAGCCATGGCCAGAAACCGGCCATCGTTAGTGAACCTAACAGCTCTTATTTG CCCCATCTTCCCTTCAGACGGC[C/T]AAGGACTCTGACAGTTCCCTTATGTCCTAC AATCTCGAGTGTGTCTTGGTTCCTCCGATAGCCAAATACGTCATCTGGATGCCAAG CAGAGCAAAAGAGTAGTCCCAAGTCCCTTTGAGGCTCCCAAGTATCTGAGAGGAAGG AAATTGAATCAGGAACAGAGAGGAATGGAGACCAACAAAAATCAAGGATAAAGT (SEQ ID No: 62)
					GCCAAAACTCGCAACAAAGTCAAGTTTGTGTACTCGGATGACATTAACACGAAAGAAA TAATGGAGGATCACTTTGATATGGATCAACTGGAGTCTGCATTTGGTGAATGATACT ACAGTTTCGATATTAATAAATATGACAGAGAGATGAAGAGGATGACAAAGATGATGCC TGCTTTCGGACTAAAGGAAATCC[C/T]CCAGCTCCAGCAACCTCAGAACCTGTCCCGA ACATGACACTCCATCTTCGACTCTACCATCAAGTTAGAGATGATGACAGCTGATTC ATCGAGAGAGAAATGGATCGAAGGGGTGCTCCAGCCCCCAATCACACTATGCTCA CTGTTGACAGTAGCAGAAATCCTACTAAAGAGGTTCAAGTAATTAAGATCCTT (SEQ ID No: 63)
common_497	NC_044370.1	95062684	T	C	AGTTTCGCCCTCCAGGTAAGTAGGAGTGGTGAATTCGTATCTCATAGCCCATCTAC CCGTTATCACCAATTAATPACCCGACAGACAGCTGCTCCTCTCTTCCAAAGGCA TCAGTCAATCAAGACAATCGAGCGGTGCTGACTCTCCGACAAGGATTAAGGACAGTCG AACAGGAGGATCTGCAGCAGAGGC[T/C]GAAATAAGAGAGTCCCTCCATGACAGCA ACCTGTTGGAGGACAAAGGAGGAATACCTTCTCTTTGATATACAACTGTTCCT TTGAGCCCAACTGCACTAAAAGATAGCCATGTAGTTGATGACAGAGTTCGGTCAG GAGTGGGAAAGATGGATACTAAATCTCCACATACTCCTGCATTGCTCACCA (SEQ ID No: 64)
					TCCTTGCCTGTGGGTAATGCCATCAATCAGATCAATCTCCCATATTAATAATGGGTATG GGGAGGTCACTATCTCAACTCGGTGGTGTATGCAAGGTATATGATGAAAACTTTTGG CATATCGCATTTCTCAACAAATTCGTGTGTCTTTGTTGATTATTCGGTAGTAATATCTT TAACGGATAATCTTCTTCCATA[A/G]ACTCTGCCAACCTACATGATCCCCACAGAGTCT
common_500	NC_044370.1	95278820	A	G	

Context sequence

[illegible]

the plus strand.

[illegible]

TABLE 11-continued

Detailed information of each of the SNPs associated with hermaphroditic inflorescence in Cannabis as provided in Tables 1 and 3 to 10. The reference allele "Ref" is the allele that is associated with hermaphroditism, while the alternative allele "Alt" indicates the allele that when homozygous for that allele results in plants that have 0 or close to 0 likelihood of producing hermaphroditic inflorescence. The positions of the SNPs as well as context sequences are provided with reference to the CS10 reference genome as described herein. BP refers to the nucleotide position of the SNP. All of the sequences and alleles are provided with reference to the plus strand.					
SNP Marker	Chromosome	BP	Ref	Alt	Context sequence
GBScompat_ common_79	NC_044370.1	70516405	T	G	GGCACAGAAACACAAAGTTGTTCTCCTTGGTGACAGTAAATGATAAACCACCAATTTTAC ATATATTAAAGATAGACATAAATTGTTACTTTTACATATCTTAAATTAGTAGTAC (SEQ ID No: 75)
					ATATCATAAATGCACATTACTAATATTTTGTGACTAATTTATAGATACAAAGTAAGTGA ATTGATTTATTGATTCAGAAATCTAGCTGTGTACAGTAACTATATGGAATTCAC CGGATATAAATGAAATATGAGAGGGTCTGATTTGATGGCTTTGAGGTACTTAA TTGCTGCTTTCTGTTGTTA[T/G]TGTTACTCAATGAATTCCTCTTTAATGGAATCTCTGA CCATTAGTCAAAATACAGGTTCACTACAGTTTGAATTAATGGAATGAGACGTTGATC TCATGTCAACTTGAATGATAGATCTTAGCTGTTCAATCTGTGCCAAGGAGAACTTC AGTTGGTTGGGTTGCTGCCCTACTTCTAGCTTGCAAAATACGAA (SEQ ID No: 76)
GBScompat_ common_81	NC_044370.1	71626242	A	T	TAGACGAAAGGAGTAGTAGTACAAATTTATAGAAACAAAAAAGAAACATAATGAT GCGAATGATAGTTATTAATGTTTCAATAACTAGAAAAATACAAATCTACTCTGATGAAGC CAGTAAACCTTCCTCCTTGTGACAAACCAATATCTGAAGTCCACGCTGGCCTGA ATTTTCTTTTGGACAAAT[A/T]ATTATGCAACCAATCAGCTGCGATTTGAACTTAT GCACACATTTAATTAGTTGTTTAAAGGTGATATAGACTAACCTATGGTAACTGTG GAATCGCAGCATGTCATTACACCAACATGAGCTAGTGACAGCCAAAAAGATCAGGT CTCTGTAAACAAATGAAAAAATTTGTTTGTAAATCAACACGCTAAG (SEQ ID No: 77)
					CAATATTGGGTGTGATCTATCTTTTGGATTGGGTCTTATGAGTGTGAGAACAGCAAC CCAGGCTATTGACAGCTTAATCCCGAGAGGAGGCAAGTTTGTGACCAATACC ATTATTGATAAAGGTGCTCCATGTTTCTCCTAGAGGCAATGCTTGCCAGGCATTAA GTTCAACAAAGGTGTAGAGCAGC[C/T]GTGCGAAACGAAAGCCATCTTTGATTAATG GAATCGACGATCAGAGTTTGTAAATTTGAAGCAGCTCCTCTTTCTCCACTTAAATG GAAAAATGAGAAATGCTGGATCATAGTCTGATTTTCTTCTTGGTTAACTTTGTGG TGCAATTTTGTGTACTCCCGGATGTTTATGACCCAGAGCTACCCAGA (SEQ ID No: 78)
GBScompat_ common_94	NC_044370.1	92378793	A	G	GCTACATAAGACAGAAACACCTTTTTTTTTTGTGAAACTACACAAAGTAAAAAATTTG TAGAAAAACAAATCCATAAAAAAGAAAAAATATATATATTATAGAAAAAATACAAT TTAAGTAGTGTAAAGATATTATAGTAACCTGGGTTTGAATATATGCAATCGCAATTA AATGAGATATGGCAGCAG[A/G]GCGAATGATTGACTCAGGTATGTCATCTTTCAAAAT TCAATACTACTGATTTCTTTAAATGAACCAACATCCAACTCAATTAACAAATAAA AAATAACATATAATGTAATGTTCACTCACTTACAGTATCTAGAAAAATGATACATCAA ACATGGCTGCATCTCCAGATCAAGAGTGTCTATCCTCCAGC (SEQ ID No: 79)
					CGAATCCTCTTCTTCTCAACGCTCTAGAGTTTCAACAATGGGTTTCCCACTACCAA ACTGTCCAGTTCTGTAATATCAATATACGATCCGAATCCGATTTAAACACCGAGGAAAGT GATCAGACTGTGTCACCATGAGCCCGCCAGCTATTCTCCACGACCTCCCAAGAG TGTTCCACGAGCTGAGCTCCAAT[C/A]TAGCGTCGGGTGCGATTCGGATCAGTTATCT GCCCTACAACTTCTAGAGATCTCTGGTTCCCAACCCGCGCCGATCCTGATCGCTG

TABLE 11-continued

Detailed information of each of the SNPs associated with hermaphroditic inflorescence in Cannabis as provided in Tables 1 and 3 to 10. The reference allele "Ref" is the allele that is associated with hermaphroditism, while the alternative allele "Alt" indicates the allele that when homozygous for that allele results in plants that have 0 or close to 0 likelihood of producing hermaphroditic inflorescence. The positions of the SNPs as well as context sequences are provided with reference to the CS10 reference genome as described herein. BP refers to the nucleotide position of the SNP. All of the sequences and alleles are provided with reference to the plus strand.					
SNP Marker	Chromosome	BP	Ref	Alt	Context sequence
GBScompat_ common_97	NC_044370.1	100988215	G	A	CGCGGAGTTGATGCGCGGTGTAGTGGGGAGTTTAAATTTCTCAGGTTGGGGCGAAACC TGCGTCGAAGATTTTGTAGCTCTCTGCTTGGCGGAGCGAAACCGGAGAGTTGCC (SEQ ID NO: 80)
					TGATCTTCAGAAGCAGGTGCCCGAGATGGCTGGAATGAGAAACAGATTCCTTAGTATTG AAGAAGAAATAAGAAAGATTACAGAATTGGTTGGTGTGATGTGCTACAGATCCTGGACTA GAAGTCAAAATTTCTTCGCGAGTAAACATTGAGTGTCTAGAAATGTTGCTGAGGAAGCT TCTAGAAAATTATGCAAAATTTTC(G/A)ACCATAGAACCCTGTACTTGGTGGTGAATTGAT GAGCTGCAGACTGACGTGATGACTGTAGAGGCAGCTAAGAACTTAAGCAAAACCCCATG CTGGGAGTCCGATGAAGTTATCATGAAGAAAGAGCTTGAGGAAGCTTTTCACGAGTT GATTCCTGTGAAGGAGGAGAGAGGTGTATTTGTTGAGAACAACATCTTTG (SEQ ID NO: 81)
GBScompat_ common_98	NC_044370.1	101069452	G	A	TGTGCACCAACCAAGGAAATGGAGTTGAAGAAAGACAAGTTTGTCAAGTGAACCTCCTT TACTCTAATATCTTTCTTTTCAATCATGATGCTTAACCTGAGGTTTTTCATTTCTCCAA TTGTAGATTTTACCATGAGAAACAATAAGATAGTAACCTTAGAAGAGCCGTTACCAGT GTTCAAAGTAGCAGCAGAG(G/A)CCTTTGTTTGGCGCAGAGAGGAGGTAGAGTAGTA GTAGTAGAAACAGAAATTTATATCCAAAATTTGGAAGGTTTAAAGTTTTCCCGAGAAAATG AGAATCCATGGAGGAGAGATTTCTGACCTGGAAGTGATATTTTCTGCAATGGAAC AGATTTTCTCTCTTTTGTGTTGGTAGCATTTTTTGTGATCCACTC (SEQ ID NO: 82)
					AAAGCTCAAGAAATTTCCAATCCTGGCGTGGAACCTCTGGTATCTCTCCTACAAACCGGT TACTCTCAACGTTTAACTCCAAAGCTTAGTCAACCTCACAATGGAGTTGGCAATCTC CCATCAAACTCATTTCTGTCCAAGTAAACTTTCTTCAAGTAAACCATTTCTTCAAACTAT CGTCCGGTATGTCCTGAAAA(A/T)TCATTTGTACGACAGTACAGGCTCTCAACCTC GAAGCACTTTGAATTCAAGTATTCACCCCAAGTGGTTGCCCTATCACGCTGAGGCTG CGCAGAGTGGTTGGGAGCTTGGACAATGTGTAGACATCGATGGTGCACCTAGTCCCA TGTTTTCAGTCTTAATCCGTAAGGCTTCCATTTCACGATGTCACCT (SEQ ID NO: 83)
rare_63	NC_044370.1	101144766	A	T	

TABLE 11-continued

Detailed information of each of the SNPs associated with hermaphroditic inflorescence in Cannabis as provided in Tables 1 and 3 to 10. The reference allele "Ref" is the allele that is associated with hermaphroditism, while the alternative allele "Alt" indicates the allele that when homozygous for that allele results in plants that have 0 or close to 0 likelihood of producing hermaphroditic inflorescence. The positions of the SNPs as well as context sequences are provided with reference to the CS10 reference genome as described herein. BP refers to the nucleotide position of the SNP. All of the sequences and alleles are provided with reference to the plus strand.					
SNP Marker	Chromosome	BP	Ref	Alt	Context sequence
rare_66	NC_044370.1	102037098	A	T	ATCAAGTAACTGCAAGTTGGGAATAGACACGAGTGAACACGAGGAAGCACCATCGAGT TTATTGTCACTGAAACCAACCCCGCAGCTGAGAAAGGCTTGAAGAAACAGCCCTTTGG CAATGGACCCCTTAGACCATTAGCACTAATCAATAAACCTCAAGCTCCTCAAGCCCG TGAATCTTTAGGAATGACCCAGT[A/T]AAAAATTTCTTGCTACATTTAGCTCAACAG TCGAGAAAGTTTATTCAACTGAACAGGTATAGAAGCGGTAAAGACTATTATCAGAGTAC TTAAAACTCAAGTCTCGAGAGGTTACCTAACTGATGAGATCGACCCAGAAAAAGAA TTTGAGGAGAGTCAAGAGTAGTAAGATTTTGAAGTGAGGTAAACTCGAAC (SEQ ID NO: 84)
common_1527	NC_044372.1	61687722	G	A	TGAAGATAAACTAGCATTAATAAAATAAAAATTAATAATTTCTTATGATATCGAACTG CAGCGAGAATGCTGTGGTGAAGATCGATATCCATTGCCGTATCAAGGGGATGTTGT TCTTGAGTGATCCACTTAGACGAAGATCTTGACGTGAGGAGATGATGTTAGAGTTA TGTTCCATACAGCATTTGTGG[G/A]TCAATATTTTGTGCTAAATCGTGACAAATTGA TGTTTGTGGGATGCCAAGACCACTTTCCAGGAGCTTTAAGCGAGGAGTAGTATTA CTATTCTTGTGTTTCTTTTGTGAAGCTGTTGGAGTTGAATCATCTCATATCTGTAT CAGGTACTTTTTTTTGGATGCTGATGCCGTTGTCCTGATATCACA (SEQ ID NO: 85)
common_524	NC_044370.1	99761993	T	C	GGTACGTGCAATAAAATATTTAAACTTATTTTCAGCACGATAAAATTTTAAATTTTATA AAATTTAACATAAATTCACAAAATGATAAATAACAGTGATATAATTTTGTATACATAAT TTTAGAACACACTTTATTGAAAGCTGAAAAATTAATTATTAATCAGTACTTAAGATAGTTA GAGAACCAATCCAAAT[T/C]TTTGAATGGCCCTCAATTGCGAGATTTTGATGCTGTCA TCATCCAGTTGTACCTCACTGTCCACCATGTTCAACCTTTGGGATAATCTCAACATGG CTCTAACCTATACCATGCCAACCATTCATTAATTTATGCAATACAAATTAATTAATAAT TAAACTTATTTCATTTATACATTCATATCATATTATATCA (SEQ ID NO: 86)
rare_30	NC_044370.1	28544332	G	T	AAATCATCTTCTGCTTCTTCTTCAGCTGTTAGAAGCCCAAAATATTCGATCTCTGATCA ACATCTAAAAATTAACGGTGGTGATCGCACTGTTAAATCCTCTTTGGACTCGATGACCG TCGATGGCTACTTCGCACCTGAAGCCCTCAGCTATCACTGCTGAGTCTACTCTTTTAGAT GCACAAATTAATCTGATCGACCC[G/T]ACTCCGACAAACCTCCATTTCTGCTGCGGT GGCAGTGGCTCTTGGAGATTGAATCCGCGACATTTGGCTCATCTAGTCTCTCTAAG ACTGTTGATGAGTGTGGCGGAGATCGTCTCTGGAGATAGGAAGGAGTGTAAAGAGG AGGAGCAGGATATGGTGATACCGCTTGAAGACTTTTACTTTGCCAAAACCTGGG (SEQ ID

Detailed information of each of the SNPs associated with hermaphroditic inflorescence in Cannabis as provided in Tables 1 and 3 to 10. The reference allele "Ref" is the allele that is associated with hermaphroditism, while the alternative allele "Alt" indicates the allele that when homozygous for that allele results in plants that have 0 or close to 0 likelihood of producing hermaphroditic inflorescence. The positions of the SNPs as well as context sequences are provided with reference to the CS10 reference genome as described herein. BP refers to the nucleotide position of the SNP. All of the sequences and alleles are provided with reference to the plus strand.

SNP Marker	Chromosome	BP	Ref	Alt	Context sequence
common_10	NC_044370.1	889040	T	G	No: 87)
					GAGCACACTATTGTTATGTACTATATTTATGACATTTTAAATGAATGTCGATAAGTATTATT CAAAATCAATCCGATGAATAGCTTCTGAATCTCATGCCGATGATTTTCATCGAGGTTAG GATTAATTGCTGATTTGGCGGGTTGCAGTAGTTCATCCATTCGAGATGTTTCT ACGGAAGCCTGCTGTTGTTT T /G CTGAGAGCGTCGTTGTTCTTCTCCATTTTGAATC TCGACAGTATAGAAATTCGCCACATAGTAACCCGGAATCCGGTAATATGTAGTTCAACG AACAGATAGAACTCTTTCGATATGAAGCTGTAGATTACATGATGACCAATCTTTAAACG TCTATCTGAAGATCTTAATTGAAATAGTTTTCTTTGTAAACCAA (SEQ ID NO: 88)
GBScompat_54 common_54	NC_044370.1	32763867	T	G	CAAAGCTAGCTGTAAAGATAGAACTACTTCTGAAATTTTGATTCAACTTCGTGTAGTTCGTA CTATATCTTAAACACTGGCAGTGAACGTTTCAGATGGGTATAGCAACAGAGCTGTGGG AGAGACCGTATCCAAATTAACTTTACAAATATAGTACATGACAGTAGATTCTCTGTATCT CTTTGGGTACTCTAGATTTAT T /G TGAAGACCATATTTTCTCTGTTTCGACCTTATAT AAATAGAAATGAACATTACCAGCAGCAAGTCAATGTGTTTACATATTCACCTCCACG AAGAAATGAGAAGATAATCGGTACAACTATTGTTGTTGTATCGAGTTTTCACAT AACTAAATATTGAACTTTGAAATCTACTACAGGTCGAGAACTA (SEQ ID NO: 89)
					GTGTGATCGAAACCTATTCGCCCGAGCTTTGGCAAATCTCAAGGGAAACCTTTTCTTTGTA GCAACCATCTTGTGTAGAAAAGCAGCTTGGCAATTAACACTATTCAACAGCTTTGTTT ACAAGTATCTTGTCAACCGATGAGGGAATTCGTAAGTGTCCGGTTCCTCATCAAC CGTCTAATTGAGCTATAGAAAC A /G CCTTGTGCACAGCCTTCTATGGTTGAATTTCTGTT GCAGCCTAAGCTCTTCGTTTATCAATGTAATCTAAACCCAGGAAGACAAGACAAC AGTTTCTTGATTTATAAGCTGACAAATATGAATTGATGCCACATAATCCCAATGGAGACA AAGATTGTTAGTTCGAGACCACCTCAACAAACCAGCTGCTGTTCTGAA (SEQ ID NO: 90)
GBScompat_56 common_56	NC_044370.1	35677966	A	G	

TABLE 12

Targeted sequencing primers (5' to 3') for the SNPs identified in Tables 1 and 3-10, as described in Examples 1 to 4.		
SNP Name	Forward Primer	Reverse Primer
common_491	TGGCTAAAGTCATGCTCCATGT (SEQ ID NO: 91)	CACACCTAGTGGGATGTATCAGT (SEQ ID NO: 92)
common_512	CTGTCGAGGCCCATCTCAAA (SEQ ID NO: 93)	AGCGAGGCAATACCATCAAGA (SEQ ID NO: 94)
common_517	GGATTGCTTCACGACAGCC (SEQ ID NO: 95)	CCCAAACCACTTTCGCCAG (SEQ ID NO: 96)
common_525	AGCTTGGGTGATACAGCTGC (SEQ ID NO: 97)	ACCCGATTGCTTACTAGGTCC (SEQ ID NO: 98)
common_511	TGCTTGATTCCGAGGATCCT (SEQ ID NO: 99)	GACTTTGGCCTATGCAGAGGA (SEQ ID NO: 100)
common_518	ACACTTACCCGCTCTTCAGG (SEQ ID NO: 101)	AGAGGTGGAATTGGAATGCCA (SEQ ID NO: 102)
common_522	TATTAGAGCTGCCTTCGCCG (SEQ ID NO: 103)	AACCAATCCGTGAACACTCT (SEQ ID NO: 104)
common_533	AGTCCTCTCTTGATAGCCATCT (SEQ ID NO: 105)	CCAAAGCGGAATGTGACAA (SEQ ID NO: 106)
common_534	TGGTGTGGTTTCTTTCTGTCCT (SEQ ID NO: 107)	TGATGGTGATGGTGAGGCTG (SEQ ID NO: 108)
common_539	AGCTTCAGAGGGTTTGTGT (SEQ ID NO: 109)	AGGAGTGTTACGACAGCAGC (SEQ ID NO: 110)
common_544	AGGCCATGATGAAACGACGT (SEQ ID NO: 111)	TATTGGGCTGGGCTTCGTAC (SEQ ID NO: 112)
common_545	TTGCTGGCCAAGTGAGTTCT (SEQ ID NO: 113)	CCTTGTCGGGTGGTTCAAT (SEQ ID NO: 114)
common_546	TCCGTGCCCTTGGTAAAGAG (SEQ ID NO: 115)	ACGGCAGTAGTAGTCATGC (SEQ ID NO: 116)
rare_50	TCGAACCGTACTTTGCCACA (SEQ ID NO: 117)	GAAAAGTAGCGTCACGGTGG (SEQ ID NO: 118)
rare_57	GGTTTTAGCGTCGCGTTGTG (SEQ ID NO: 119)	ACACGCCCATAAAGACAGGG (SEQ ID NO: 120)
common_472	TGGGTCCGGAAGATACAAGC (SEQ ID NO: 121)	ATGAAGGTGGCGCAATGGAA (SEQ ID NO: 122)
common_474	TCAGTTTCTCTGTCTGCCGA (SEQ ID NO: 123)	GGAGTATAGGGCGGTGGGTA (SEQ ID NO: 124)
common_483	TCCTCCCTCCATTCCGATCA (SEQ ID NO: 125)	ACTCACCCAAAAGTGGCCAA (SEQ ID NO: 126)
common_484	CCACAGGCAGCACGTTTCTA (SEQ ID NO: 127)	CTAAGGCCGTTTTGGGACCT (SEQ ID NO: 128)
common_486	GACCTGACATCACCGTTGCT (SEQ ID NO: 129)	ACTCGACTTCTGCAGAGCTT (SEQ ID NO: 130)
common_514	AGGAGATGTCAGGAGGTGGA (SEQ ID NO: 131)	TTGTTTGGCACCAGTGGACT (SEQ ID NO: 132)
common_521	TACCCTACTTCCTTGGGCGA (SEQ ID NO: 133)	AGGAAAATGGCAGACACCCA (SEQ ID NO: 134)
common_527	TCTTCATTGGGAGGCAGCAG (SEQ ID NO: 135)	AGGCTCCACCACTGAGTGTA (SEQ ID NO: 136)
common_532	TCTCCGAATACCCTCCACGT (SEQ ID NO: 137)	GCGTGTCCGAAAGTTTGTGT (SEQ ID NO: 138)

TABLE 12-continued

Targeted sequencing primers (5' to 3') for the SNPs identified in Tables 1 and 3-10, as described in Examples 1 to 4.		
SNP Name	Forward Primer	Reverse Primer
common_552	TCTTCCCAACTCCTCCTGGT (SEQ ID NO: 139)	TGCAAGTTTTGAAGTGGCCG (SEQ ID NO: 140)
common_553	TCTAAGGTGGTTGCAGGCAG (SEQ ID NO: 141)	TGGACGTCTAAGCAAAACAAG (SEQ ID NO: 142)
GBScompat_ common_91	CAAGTCCAACCCAGGAGCTT (SEQ ID NO: 143)	GGCTGAGTCCTGCAAAGTCT (SEQ ID NO: 144)
GBScompat_ common_99	CACCACTGCCTCCACCTATG (SEQ ID NO: 145)	CAATGGGTGCTTCGGCTTC (SEQ ID NO: 146)
GBScompat_ rare_10	TTTTGCAGCGATCGATTCCG (SEQ ID NO: 147)	GGCACGATTCTACACCACCA (SEQ ID NO: 148)
common_203	ACGCCTGGTTGAAGTGAACA (SEQ ID NO: 149)	ACAAAGTTTGGGCCACCTT (SEQ ID NO: 150)
common_288	CCCAATCCCAAATCCAAACCA (SEQ ID NO: 151)	GGTGGCAATCGTGGAGAGAA (SEQ ID NO: 152)
common_300	TGGTTACCCTGTAATTGGTTGAGA (SEQ ID NO: 153)	CTTTGCAGCCGATGGTTCG (SEQ ID NO: 154)
common_313	TGAGGGCCACAACCTTAGTAGTG (SEQ ID NO: 155)	TGGCTCTGATGGAGATGACAC (SEQ ID NO: 156)
common_375	CAGGCTTTGTCCAGATGTGC (SEQ ID NO: 157)	ACAACCAGCCAAGAGACCAA (SEQ ID NO: 158)
common_424	TGGACTGACCTGAGACCAGT (SEQ ID NO: 159)	GCAACCTGGAGAGAGATACCG (SEQ ID NO: 160)
common_425	ATCCGGTTTCTTGTTCGGCT (SEQ ID NO: 161)	ATGGTCCGAGAGTTCATGT (SEQ ID NO: 162)
common_426	GGATTCTCCATTGTGGGCA (SEQ ID NO: 163)	ACTTGTCCTCCAAACAGGCA (SEQ ID NO: 164)
common_428	GCGCTCTGGTCTCTTCAAGA (SEQ ID NO: 165)	AGGCCAATCTCGCTTGTCTC (SEQ ID NO: 166)
common_431	GCAGAACACCATCAAAGCCG (SEQ ID NO: 167)	CGTCTACGAAGCGGAGGAAA (SEQ ID NO: 168)
common_432	CTCTCCGCATAGCCAAGAC (SEQ ID NO: 169)	GCCTTGGGATTCCTCTTCGT (SEQ ID NO: 170)
common_434	CTGCAGGGTTGTGGAGAGTG (SEQ ID NO: 171)	GTTTGGAGGGATTCAGAGCT (SEQ ID NO: 172)
common_436	GTTCACTTTTGCCACACTGCA (SEQ ID NO: 173)	TGTGAAGCTGGTAGGACACT (SEQ ID NO: 174)
common_437	GCAGATGTACGCTAAAGATTTGTT (SEQ ID NO: 175)	GCGTGGGATGTAACGGTCAA (SEQ ID NO: 176)
common_439	TGGTCTCTCTCTCTTCCACA (SEQ ID NO: 177)	TGATCGAGTCTCCGAACACG (SEQ ID NO: 178)
common_445	CGGGGAAGGAGAGGAAGTTG (SEQ ID NO: 179)	GCATCAAAGCCTTGAGTGCC (SEQ ID NO: 180)
common_448	CGGTTTCACCACCACTAACC (SEQ ID NO: 181)	GGGTATCGGTTGGATGGTCA (SEQ ID NO: 182)
common_449	TGTTGATGTGGCGGTGACAA (SEQ ID NO: 183)	TGCCACCAAACCATAATAGGGA (SEQ ID NO: 184)
common_452	TCCGACGAGTCTGACCAAAC (SEQ ID NO: 185)	CAACGATTTCCCCAGGGAGT (SEQ ID NO: 186)

TABLE 12-continued

Targeted sequencing primers (5' to 3') for the SNPs identified in Tables 1 and 3-10, as described in Examples 1 to 4.		
SNP Name	Forward Primer	Reverse Primer
common_453	TGACAAAGTTGATAATCAATGCAGT (SEQ ID NO: 187)	TATGCTGGTGGTGGAGAGCT (SEQ ID NO: 188)
common_454	CTAGGATCTCCAAGCAGGCG (SEQ ID NO: 189)	GTCATTGTAAGCCTGTGCGG (SEQ ID NO: 190)
common_455	TTTACATACCCAGCACGCA (SEQ ID NO: 191)	CGTGACAGTTGGATGAAGCG (SEQ ID NO: 192)
common_459	ACCACCTTGGATTCCACGTC (SEQ ID NO: 193)	CGACCTTGCAAGTTGAGCCTA (SEQ ID NO: 194)
common_470	CTTCTGACCACCAACAGGCT (SEQ ID NO: 195)	CAGAAGTCAAGTTTGTTCGCT (SEQ ID NO: 196)
common_473	CCGTTGAAGCATTGCACCAT (SEQ ID NO: 197)	TCCCAACGTTTCCATTGACG (SEQ ID NO: 198)
common_476	ATCTGACGTGGACTCAGTGC (SEQ ID NO: 199)	CACTATTGTGGCTGCAACCG (SEQ ID NO: 200)
common_478	CCAGCCAATCTACCGATGA (SEQ ID NO: 201)	GTGAGAAACCTCCGACGGT (SEQ ID NO: 202)
common_479	AGTTCTGGTTTGTTCAGGAAGA (SEQ ID NO: 203)	TGAGGAGGGCTTCAATTGCA (SEQ ID NO: 204)
common_487	AGCTAGCAGCACCTTCAATGA (SEQ ID NO: 205)	AGGAGCTATGATGCGATTGGA (SEQ ID NO: 206)
common_488	GTCCAATGAAAGCCGCAACA (SEQ ID NO: 207)	TAGGTCTCGAGTGCAACAGC (SEQ ID NO: 208)
common_489	CCTTGGAGGATCATTGGTGCT (SEQ ID NO: 209)	CCACATTGCAAACAGGGCAG (SEQ ID NO: 210)
common_492	AAGAAAGGTTCCAAAGCAGT (SEQ ID NO: 211)	ATGTATCAACCTGCGGCAGA (SEQ ID NO: 212)
common_494	CCAACAAAGAGTGCCTCAGC (SEQ ID NO: 213)	TCTGCTTGGCATCCAGATGG (SEQ ID NO: 214)
common_496	TGGAGTCTGCATTGGTGGA (SEQ ID NO: 215)	GCATAGTGTGATTGGGGCT (SEQ ID NO: 216)
common_497	CCTCCAGGTTAGAGTAGGCA (SEQ ID NO: 217)	TGCAGTTTGGGCTGAAAGGA (SEQ ID NO: 218)
common_500	AAC TCGGTTGGTGGTATGCA (SEQ ID NO: 219)	ACCGAGGCACACAACTCTT (SEQ ID NO: 220)
common_510	CGGTACTACCCATCGTCGTC (SEQ ID NO: 221)	TGGCAGAGTCCACAGCATTT (SEQ ID NO: 222)
common_515	TGCTACCCGAGAATCATCACA (SEQ ID NO: 223)	AGAGCCAAATGGAAGGACTGG (SEQ ID NO: 224)
common_520	ATAGAGCCAGACGTGGTGGA (SEQ ID NO: 225)	CTGTTTCATAGGTCAACCCCA (SEQ ID NO: 226)
common_523	ATTCATAGCATGCGCGCAAC (SEQ ID NO: 227)	TCAGCTCCAACCTCATTGCAG (SEQ ID NO: 228)
common_526	AGGTTTGGCCGCCTTCAATA (SEQ ID NO: 229)	CGAGACGGAGAGAACCTGTT (SEQ ID NO: 230)
common_528	TGCAAGCCCATCGAAGTGAA (SEQ ID NO: 231)	GGTGAGGCTGAACAAGAACC (SEQ ID NO: 232)
common_531	GGCTCTTATTCTAGGGGCAC (SEQ ID NO: 233)	GAAAGGTTGAGGAGGAGCG (SEQ ID NO: 234)

TABLE 12-continued

Targeted sequencing primers (5' to 3') for the SNPs identified in Tables 1 and 3-10, as described in Examples 1 to 4.		
SNP Name	Forward Primer	Reverse Primer
common_541	GTGCTGGTTGCCAAAGTGAG (SEQ ID NO: 235)	ACATCACATTTCCGGATGATAGTTCT (SEQ ID NO: 236)
common_549	TCTTCTTCTTTCTTCACCCCGA (SEQ ID NO: 237)	GCCTACCTCTCTTTCTCGGC (SEQ ID NO: 238)
common_550	GGCAACTGCATGCTACCATC (SEQ ID NO: 239)	ACCTGCACCAAGGAGAACAA (SEQ ID NO: 240)
GBScompat_ common_79	AGGGGTTCTGATTGATTGGCT (SEQ ID NO: 241)	GCAGGCTAGAAGTAGGGCAG (SEQ ID NO: 242)
GBScompat_ common_81	CACGCGTGGCCTGAATTTTT (SEQ ID NO: 243)	TTTGGCTGTGCACTAGCTCA (SEQ ID NO: 244)
GBScompat_ common_84	GGGAGGCAAGTTTTGTGACC (SEQ ID NO: 245)	TGCATAACACATCCGGGAGT (SEQ ID NO: 246)
GBScompat_ common_94	ATGGCAGCAGAGCAGATTGA (SEQ ID NO: 247)	GCTGGGAGGATGACCACTCT (SEQ ID NO: 248)
GBScompat_ common_96	AGAGCTTCACACAATGGCGT (SEQ ID NO: 249)	TAAACTCCCCACTACACCGC (SEQ ID NO: 250)
GBScompat_ common_97	CCGAGATGGCTGGAAATGAGA (SEQ ID NO: 251)	AACTTCATCGGACTCCCCAG (SEQ ID NO: 252)
GBScompat_ common_98	TCATGCATGCTTAACTGGGGT (SEQ ID NO: 253)	ACTCTGTTCCATTGCAGGAA (SEQ ID NO: 254)
rare_63	ATCCTGGCGTGAAACTCTG (SEQ ID NO: 255)	CCAAGCTCCCAACCACTCTG (SEQ ID NO: 256)
rare_66	TCACTCAGAACCAACCCG (SEQ ID NO: 257)	CTGGGTCGATTCCATCAGGT (SEQ ID NO: 258)
common_1527	TGCCGTATTCAAGGGGATGT (SEQ ID NO: 259)	ATCAGGGACAACGGCATCAG (SEQ ID NO: 260)
common_524	TGGAATGGCCTCATTTGCA (SEQ ID NO: 261)	AGGTTGAGAGCCATGTTGAGA (SEQ ID NO: 262)
rare_30	CGGTGGTGATCGCACTGTTA (SEQ ID NO: 263)	GCGTCATCACCATATCCTGC (SEQ ID NO: 264)
common_10	GCTTCTGAATCTCCATGCCG (SEQ ID NO: 265)	ACAGCTTCATATGCAAGGAGT (SEQ ID NO: 266)
GBScompat_ common_54	GCAACAGAGCTGTGGGAGA (SEQ ID NO: 267)	AGTTCTCGACCTGTCAGTAGA (SEQ ID NO: 268)
GBScompat_ common_56	TTCGTAAGTGTCGGTTCCC (SEQ ID NO: 269)	GCAGCTGGTTTGTGAGTGG (SEQ ID NO: 270)

Example 5

Gene Identification

[0187] There are presently no known genes identified in Cannabis that have been shown to regulate hermaphroditism or sex determination in Cannabis. Several molecular markers have been proposed for sex determination, but these regions are distinct from the QTLs identified herein. Genes that control sex determination have been described and characterized in several plant species, however the multitude of different genes involved in this process does not easily allow the identification of sex determination genes in Cannabis. The inventors considered genes that may stimulate

homeotic transformation or those that are involved in the interplay between flowering and the stress response to be potentially involved in enhancing or eliminating the likelihood of the emergence of hermaphroditic inflorescence in Cannabis. They next sought to identify putative genes that could encode proteins that may be responsible for both the emergence of anthers in pistillate flowers or those that could stimulate male flower development on female plants. Using the findings of the association studies they identified candidate genes at the QTLs identified.

[0188] Based on the results of the association studies for hermaphroditic inflorescence from a QTL found on NC_044370.1 based on the SNP marker “common_10” at position 889040 in Example 1, the inventors searched for

genes that may encode proteins involved in the stress response or those that could play a role in floral development from an annotated gene list for this region of NC_044370.1 from the *Cannabis sativa* CS10 genome. Upon inspection of this genomic region and BLAST analysis of putative candidates they identified a single candidate gene LOC115715793, Table 13. LOC115715793 encodes a protein with homology to *Arabidopsis thaliana* DELLA protein RGL2. DELLA proteins are repressors of Gibberellin signalling. This candidate makes sense in light of Gibberellin being able to induce hermaphroditic flowering.

[0189] Based on the results of the association studies for hermaphroditic inflorescence, a QTL demarcated by position 28544332 to 35677966 on chromosome NC_044370.1 was identified from Examples 1, 2 and 3. The inventors searched for genes that may encode proteins involved in the stress response or those that could play a role in floral development from an annotated gene list for this region of NC_044370.1 from the *Cannabis sativa* CS10 genome. Upon inspection of this genomic region and BLAST analysis of putative candidates they identified a single candidate gene LOC115702418 (Table 13). LOC115702418 encodes a protein with homology to nodulation-signaling pathway 2 protein in *Arabidopsis*. This protein contains a GRAS domain and is likely part of the GRAS domain family, member of this family play roles in Gibberellin signalling. This candidate makes sense in light of Gibberellin being able to induce hermaphroditic flowering.

[0190] Based on the results of the association studies for hermaphroditic inflorescence a QTL demarcated by position 70255045 to 79818534 on chromosome NC_044370.1 was identified from Examples 1, 2 and 3. The inventors searched for genes that may encode proteins involved in the stress response or those that could play a role in floral development from an annotated gene list for this region of NC_044370.1 from the *Cannabis sativa* CS10 genome. Upon inspection of this genomic region and BLAST analysis of putative candidates they identified a single candidate gene LOC115719981 (Table 13). LOC115719981 encodes a protein with homology to FT-interacting protein 7 in *Arabidopsis*. This protein regulates the movement of FT a major regulator of flowering in plants. Mis regulation of FT-interacting protein 7 could potentially lead to the aberrant induction of flowering and could potentially explain parts of the emergence of hermaphroditic inflorescence.

[0191] Based on the results of the association studies for hermaphroditic inflorescence from the QTL found on NC_044370.1 approximately between positions 94,000,000 and 102,000,000, the inventors searched for genes that may

encode proteins involved in the stress response or those that could play a role in floral development from an annotated gene list for this region of NC_044370.1 from the *Cannabis sativa* CS10 genome. Upon inspection of this genomic region and BLAST analysis of putative candidates they identified ten candidate genes LOC15698113, LOC115698183, LOC115698538, LOC115695629, LOC115699142, LOC115699627, LOC115699728, LOC115703125, LOC115700107, LOC115700622 listed in Table 13.

[0192] The gene IDs LOC115698113, LOC115698183, LOC115699627, LOC115703125 encode putative ethylene responsive transcription factors. From research on likely homologs in *Arabidopsis*, these proteins may have a role in mediating ethylene signaling and may be involved in the regulation of gene expression by stress factors and by components of stress signal transduction pathways. Ethylene signalling has been shown to be involved in some cases in plant floral development, the stress response is a known trigger of hermaphroditism in many plant species.

[0193] The inventors further identified LOC115698538 that encodes for a Clavata1-like receptor kinase. Clavata1-like receptor kinases have been implicated in playing central roles in meristem and anther development in *Arabidopsis*. LOC115695629 is a gene that may encode a protein with a Myb/SANT-like DNA-binding domain. In *Arabidopsis* proteins containing similar domains play roles in organ morphogenesis and floral development. LOC115699728 is a gene that encodes a protein with homology to *Arabidopsis* agamous-like MADS-box protein AGL12. In *Arabidopsis*, MADS-box proteins have been shown to be involved in the flowering transition. LOC115700107 is a gene that encodes a protein with homology to *Arabidopsis* E3 ubiquitin-protein ligase MBR2. MBR2 in *Arabidopsis* encodes an E3 ubiquitin-protein ligase that functions as a positive regulator of FLOWERING LOCUS T (FT) and is important to induce the expression of FT and consequently to promote flowering.

[0194] The inventors also identified LOC115700622, a gene that encodes a protein with homology to *Arabidopsis* transcription repressor OFP7. In *Arabidopsis* OFP7 acts in the regulation of homeodomain transcription factors that could play roles in floral development.

[0195] Finally, the inventors identified LOC115699142, a gene that encodes a protein with homology to *Arabidopsis* transcription repressor TIFY8. In *Arabidopsis* likely acts as a negative regulator of jasmonate signalling. Jasmonate signalling has been connected to stress induced flowering in a number of species including *Arabidopsis* and Tomato.

TABLE 13

Gene list of candidate genes identified. The gene ID is provided with reference to the publicly available CS10 genome as updated in Apr. 2020 and accessed in Feb. 2022.				
Start Position	End Position	Gene ID	Protein ID	Description
9247972	9249855	LOC115715793	XP_030500325.1	DELLA protein RGL2
31248309	31250365	LOC115702418	XP_030485743.1	nodulation-signaling pathway 2 protein
79532538	79536136	LOC115719981	XP_030505012.1	FT-interacting protein 7
94086108	94088167	LOC115698113	XP_030481151.1	ethylene-responsive transcription factor RAP2-4-like
94441292	94442472	LOC115698183	XP_030481233.1	ethylene-responsive transcription factor RAP2-4

TABLE 13-continued

Gene list of candidate genes identified. The gene ID is provided with reference to the publicly available CS10 genome as updated in Apr. 2020 and accessed in Feb. 2022.				
Start Position	End Position	Gene ID	Protein ID	Description
95149366	95152936	LOC115698538	XP_030481483.1	Clavata 1 Receptor like kinase
95437048	95447718	LOC115695629	XP_030478545.1	Myb/SANT-like DNA-binding domain
97492172	97496609	LOC115699142	XP_030482249.1	TIFY domain
99557456	99558278	LOC115699627	XP_030482993.1	AP2 domain
99684325	99689041	LOC115699728	XP_030483127.1	MADS box AGL12
99933474	99934499	LOC115703125	XP_030486484.1	AP2 domain
100782096	100786765	LOC115700107	XP_030483528.1	E3 ubiquitin-protein ligase MBR2
101822631	101823768	LOC115700622	XP_030484093.1	Ovate-Transcriptional repressor

Example 6

Genome-Wide Association Studies (GWAS) of an Expanded F2 Population

[0196] The inventors engaged in speculation regarding the validity of certain QTLs identified in the GWA of mixed populations conducted in 2020 and from a corresponding GWA conducted on a mixed F2 population. To address this disparity the inventors expanded the diversity of the mixed F2 population experiment by including additional F2 populations, including several populations that were crossed with high THC varieties.

[0197] The inventors sought to understand the genetic basis for hermaphroditism by analysing the hermaphroditism trait in 24 designed F2 populations, totaling 2758 individuals. This was accomplished by visually examining female plants for the presence of stamens or the growth of staminate flowers alongside pistillate flowers until the time of harvest during the 2021 field trial season. Hermaphroditic plants were scored as 1, while non-hermaphroditic plants were scored as 0. The percentage of hermaphroditic flowers was calculated for each of the populations. Table 14 presents an overview of the distribution of hermaphroditism in the 24 F2 populations.

TABLE 14

An overview of the F2 populations used in the GWAS study described in Example 6, including the sample size denoted by the number of plants, and the percentage of hermaphroditic flowers present.		
F2 Populations	Sample Size	Percent presence
21 002 001	229	37
21 002 002	144	21
21 002 003	145	28
21 002 004	133	20
21 002 007	79	18
21 002 012	153	12
21 002 014	96	21
21 002 016	75	15
21 002 025	120	34
21 002 026	108	24
21 002 027	112	24
21 002 028	123	16
21 002 029	89	12
21 002 031	87	13
21 002 032	81	10
21 002 035	158	16
21 002 036	128	12
21 002 037	80	0
21 002 038	116	21
21 002 039	117	9

TABLE 14-continued

An overview of the F2 populations used in the GWAS study described in Example 6, including the sample size denoted by the number of plants, and the percentage of hermaphroditic flowers present.		
F2 Populations	Sample Size	Percent presence
21 002 040	82	11
21 002 041	99	10
21 002 046	113	0
21 002 057	90	0

[0198] Following hermaphroditism scoring detailed in Table 14, DNA was extracted from each of the 2758 F2 plants using approximately 70 mg leaf discs using an adapted kit with “sbeadex” magnetic beads by LGC Genomics, which was automated on a KingFisher Flex with 96 Deep-Well Head by Thermo Fisher Scientific. The extracted DNA served as a template for the library preparation for sequencing with the library pools prepared according to the manufacturer’s instructions for the AgriSeq™ HTS Library Kit-96 sample procedure from Thermo Fisher Scientific. The subsequent targeted sequencing was conducted using a custom SNP marker panel based on the Cannabis Sativa CS10 reference genome on the Ion Torrent system by Thermo Fisher Scientific. The primers for the identified SNPs are provided in Table 17. The library pool was loaded onto Ion 550 chips with Ion Chef and sequenced with Ion GeneStudio S5 Plus according to the manufacturer’s instructions for the Ion 550™ Kit from Thermo Fisher Scientific.

[0199] A genome-wide association study (GWAS) was performed using the data from the 24 F2 populations, totaling 2758 individuals. The study aimed to detect significant associations between genotypic information derived from targeted resequencing of the custom SNP marker panel described above and the binary trait of hermaphroditism—with 1 indicating hermaphroditism and 0 indicating the lack of hermaphroditism.

[0200] The genotypic matrix was filtered for SNPs with more than 30% missing values within the population and a minor allele frequency below 5%. This resulted in 3627 SNP markers after filtering.

[0201] To improve the estimation of population structure and kinship among individuals the inventors recognised that a high rate of missing values may be impacting the modeling. To address this, they incorporated an additional step. The GWA was performed again on the combined F2 populations using instead a SNP matrix that underwent a round of imputation to reduce the number of missing values. The

imputation was performed using the HapMap_imputation software (GitHub-mwylerCH/HapMap_Imputation). Briefly, the genotype file was converted to a hapmap format (comma separated, <http://augustogarcia.me/statgen-esal/Hapmap-and-VCF-formats-and-its-integration-with-one-map/#hapmap>).

[0202] In a first step, HapMap_Imputation counts the occurrence of each nucleotide at every genotyped position. The most common nucleotide is defined as major allele, the second is defined as minor allele. Missing genotyping information is excluded. In the case major and minor alleles occur at the same number, the nucleotide of the reference cs10 genome (available as GCF_900626175.1 on NCBI) is chosen as major allele. In a second step, HapMap_Imputation sorts markers by position and parses the hapmap into the required fastPHASE (Scheet & Stephens, 2006) input format. Briefly, HapMap_Imputation splits the haplotypes into two separate rows, converts major and minor alleles into 0 and 1, respectively, and produces temporary files for each chromosome.

[0203] During the third step, HapMap_Imputation downloads the latest fastPHASE version and runs the imputation using 8 cores in parallel. fastPHASE is run with ten random starts of the imputation algorithm. After imputation, HapMap_Imputation reverses the 0 and 1 coding into the major and minor nucleotide, respectively. Subsequently, the two haplotypes are combined, and the separate chromosomes are merged into a single file.

[0204] The imputed genotypic matrix was filtered for SNPs with more than 30% missing values within the population and a minor allele frequency below 5%, resulting in 5077 SNP markers for association analysis, a significant increase compared to the non-imputed dataset. The GWAS was performed using GAPIT version 3 (Wang & Zhang, 2021) with four statistical models: General Linear Model (GLM), Mixed Linear Model (MLM), FarmCPU and Blink. A quantile-quantile plot (QQ plot) was used to assess the performance of the models. The Blink model was chosen by the inventors and was used for the analysis. SNPs with a LOD (−log₁₀ (p-value)) value greater than 5 were considered to be significantly associated with trait variation.

[0205] The inventors identified two major QTLs from the GWA on Chromosome NC_044370.1, based on the SNPs found to be significantly associated with the presence or absence of hermaphroditic flowers. The first QTL is found in the region of 18371013-35677966, where the most significantly associated SNP, “common_294” is at position 30974345, with reference to the CS10 genome. This QTL

was identified by the inventors in the 2020 mixed population experiment of Example 1, where it was based on SNPs “rare_30” and “GBScompat_common_54” from position 28544332-32763867, respectively. The QTL was further validated in the expanded mixed F2 population as a determining factor of hermaphroditism. The SNPs identified in this experiment, as listed in Table 15, can be used as markers for the selection for or against the hermaphroditism trait.

[0206] The second QTL is found in the region of 90451268-102978171, where the most significantly associated SNP, “common_517”, is at position 99121582, with reference to the CS10 genome. This QTL was also identified in the 2020 mixed population experiment described in Example 1, where it was defined by the SNP “common_524” at position 99761993. The QTL was further validated in the expanded mixed F2 population described in Example 2, where it was defined by SNPs “common_491” and “common_546” in the region of 94129798-101726389, respectively. The additional SNPs identified in this experiment can be used as markers for the selection for or against hermaphroditism.

[0207] The inventors demonstrated the use of a genomic selection model and tested this on a training population of mixed lineage diverse cannabis plants where ~36% of the plants displayed hermaphroditism. Twenty-five markers were selected in the region of the QTL at position 90451268-102978171 to test the prediction power for the selection of hermaphroditic plants compared to 25 randomly selected markers. The inventors performed a multiple regression analysis with the allele as variable and hermaphroditism as target using the random forest algorithm implemented in the ranger package (v 0.12.1, Wright and Zieger 2017). The resulting R squares are derived from the comparison of the predictions from the developed model with the measured phenotype of the training population.

[0208] The approach tested 100 permutations for the specific markers and 100 permutations of the 25 random markers resampled for each permutation. The results of the genomic selection model demonstrate that the specific markers in this region greatly improve the accuracy of selecting for hermaphroditism, with R-squared of ~0.25, in comparison with the use of random markers, with R-squared of ~0.0125.

[0209] Table 15 lists additional SNPs significantly associated with variations in the hermaphroditism trait. These SNPs, representing QTLs, can serve as markers in marker-based selection or to improve genomic selection models for the hermaphroditism trait.

TABLE 15

Additional SNPs associated with variations in hermaphroditism in the F2 populations. The table presents the positions and chromosomes of the SNPs with reference to the CS10 reference genome. The BLINK model's LOD score is included as LOD_BLINK, and the mean phenotypic values for Allele 1, 2, and 3, are presented as “Mean 1”, “Mean 2”, and “Mean 3” respectively. The number of plants having Allele 1, 2, and 3 are presented as “Count 1”, “Count 2”, and “Count 3”, respectively.						
SNP	Chromosome	Position	Alleles_1	Alleles_2	Alleles_3	Mean_1
common_517	NC_044370.1	99121582	AA	AG	GG	0.06871609
common_507	NC_044370.1	97056996	AA	AT	TT	0.25238898
common_294	NC_044370.1	30974345	AA	AT	TT	0.93333333
GBScompat_	NC_044370.1	35677966	AA	AG	GG	0.08256881
common_56						
GBScompat_	NC_044370.1	18371013	AA	AG	GG	0.18352223
common_36						

TABLE 15-continued

Additional SNPs associated with variations in hermaphroditism in the F2 populations. The table presents the positions and chromosomes of the SNPs with reference to the CS10 reference genome. The BLINK model's LOD score is included as LOD_BLINK, and the mean phenotypic values for Allele 1, 2, and 3, are presented as "Mean 1", "Mean 2", and "Mean 3" respectively. The number of plants having Allele 1, 2, and 3 are presented as "Count 1", "Count 2", and "Count 3", respectively.

common_480	NC_044370.1	90451268	CC	CG	GG	0.24893617
common_561	NC_044370.1	102978171	AA	AG	GG	0.06020558
common_685	NC_044371.1	14838226	AA	AG	GG	0.14431984
common_1025	NC_044371.1	86862040	AA	AT	TT	0.19169719
common_2949	NC_044375.1	13190709	AA	AG	GG	0.16041979
rare_7	NC_044370.1	6234467	AA	AC	CC	0.36030534
common_119	NC_044370.1	11475121	AA	AG	GG	0.17193897
common_528	NC_044370.1	100088557	AA	AG	GG	0.20042531
GBScompat_	NC_044371.1	25913418	AA	AT	TT	0.13581184
common_139						
GBScompat_	NC_044378.1	37840619	AA	AG	GG	0.29737609
rare_186						
common_489	NC_044370.1	92434073	AA	AC	CC	0.28432732
GBScompat_	NC_044378.1	100163	CC	CG	GG	0.1152815
common_883						
common_1524	NC_044372.1	60699277	AA	AG	GG	0.10067114
GBScompat_	NC_044374.1	44536110	AA	AG	GG	0.15967197
common_504						

SNP	Mean_2	Mean_3	Count_1	Count_2	Count_3	LOD_BLINK
common_517	0.09589041	0.30748422	553	1095	1109	37.53816716
common_507	0.04481793	0.01136364	1779	714	264	27.53549948
common_294	0.45652174	0.16654303	15	46	2696	17.0615317
GBScompat_	0.13923077	0.23002421	218	1300	1239	13.47064117
common_56						
GBScompat_	0.12112676	0.18518519	2294	355	108	10.96401867
common_36						
common_480	0.11869436	0.17792932	470	674	1613	9.684858537
common_561	0.15433213	0.28099174	681	1108	968	9.440837483
common_685	0.25813449	0.28163265	2051	461	245	8.498430839
common_1025	0.19555556	0.13161132	819	1125	813	8.328623727
common_2949	0.238921	0.16455696	2001	519	237	8.14447613
rare_7	0.15095677	0.05065123	655	1411	691	7.662803717
common_119	0.35849057	0.17647059	2687	53	17	7.510824509
common_528	0.22461538	0.06170599	1881	325	551	7.494616121
GBScompat_	0.20283976	0.23178808	1318	986	453	6.182667096
common_139						
GBScompat_	0.23658537	0.11794228	343	820	1594	5.880347694
rare_186						
common_489	0.06272401	0.0201005	1442	1116	199	5.521817722
GBScompat_	0.15242019	0.20736023	373	971	1413	5.299066177
common_883						
common_1524	0.19734904	0.18700184	447	679	1631	5.1934005
GBScompat_	0.19607843	0.25382263	2073	357	327	5.053464113
common_504						

[0210] The underlying mechanism and gene(s) involved in the initiation of hermaphroditism in cannabis are unknown to date. Based on a candidate gene approach the inventors were unable to identify the causative SNP responsible for the trait under investigation. Therefore, the inventors used the results of the association studies to identify candidate genes at the QTLs identified by comparing a collection of cannabis genomic sequences to identify putative causative SNPs that appeared in a similar pattern as one of the most significantly associated polymorphisms listed in Table 3, "common_517".

[0211] The inventors used SNP differences between cannabis genomes to identify candidate genes associated with the trait of interest. To achieve this, short reads from sequenced lines were dereplicated with NGSReads Treatment (version 1.3, Gaia et al. (2019)) and pre-processed with fastp (version 0.23.2, S. Chen et al. (2018)) before being aligned to the CS10 reference genome with Bowtie2 (ver-

sion 2.3.5.1, with options --rg and --rg-id to add read-group identifiers, Langmead and Salzberg (2012)). Only unique alignments with a mapping quality of at least 10 were kept. SNPs were called with freebayes and filtered for a minimal quality of 20 (version v1.3.2-40-gccee27fc, parameters -p 2 --min-coverage 20-g 30000 --min-alternate-count 4 --min-alternate-fraction 0.1 --min-mapping-quality 10 --max-complex-gap-1, Garrison and Marth (2012)). SNPs were finally filtered for a coverage between 5 and 10,000 within each line and annotated with snpEff (version 4_3t, Cingolani et al. (2012)).

[0212] For each line, the inventors constructed a pseudogenome by incorporating its variants into the CS10 reference genome with vcf-consensus (Danecek et al. (2011)). To align genes from a reference genome to a target genome, CS10 annotation was lifted over with liftoff (version 1.6.3, Shumate and Salzberg (2021)). Protein and cDNA

sequences were extracted with custom scripts and aligned with muscle (v3.8.31, Edgar (2004)).

[0213] The inventors extracted proteins located between 90451268-102978171 bps on chromosome NC_044370.1 and performed multiple sequence alignments which to generate tables representing the variant positions and protein variants. These were tested for correlation with the significant SNPs from the GWAS marker panel, “common_517”. Proteins with significant associations were kept and used to extract the SNPs within the associated genes. SNPs were tested for association with the significant SNP “common_517” from the GWAS marker panel and only significant SNPs were used as candidates to identify genes potentially associated with hermaphroditism. The remaining 117 SNPs were further filtered by examining the gene expression in a published study of masculinization in cannabis where female flowers were treated with colloidal silver to induce male flower appearance (Adal A M, Doshi K, Holbrook L, Mahmoud S S. Comparative RNA-Seq analysis reveals genes associated with masculinization in female *Cannabis sativa*. *Planta*. 2021 Jan. 4; 253 (1): 17). Using transcript data from this study the inventors identified a candidate gene, LOC115696400, that is strongly regulated in response to colloidal silver and shows differential expression between male and female flowers. This gene encodes the protein, XP_030479162.1 predicted to be a 1-aminocyclopropane-1-carboxylate synthase (ACC synthase), which produces ACC, a precursor to the gaseous hormone ethylene, known to be involved in sex determination and flowering in many plant species. The inventors identified polymorphisms present in their collection of sequence cannabis lines. A number of SNP differences were identified in LOC115696400 (NCBI cs10 reference genome gene ID) based on the genome collection, some of which impacted the amino acid sequence; 99811627 (A>G, loss of start), 99811644 (G>A, synonymous), 99811652 (C>A, Pro>His), 99811653 (C>T, synonymous), 99811703 (T>A, Val>Glu), 99811709 (C>T, Thr>Ile), 99811742 (A>G, Lys>Arg), 99814345 (A>G, Gln>Arg), 99814346 (A>T, Gln>His), 99814409 (G>A, synonymous), and 99814578 (G>C, Arg>Thr) (where the first allele given is the reference allele based on the CS10 reference genome, and the effect of the SNP on the amino acid sequence follows).

[0214] Of most interest to the inventors was the surprising discovery of a SNP in LOC115696400 at 99811627 that

changed the start codon to Val. Based on the alignments from the genome collection, the inventors identified several genomes where LOC115696400 lacked a start codon as a result of a polymorphism A>G at position 99811627 on NC_044370.1 (position 1 of SEQ ID NO: 343).

[0215] The inventors found that the polymorphism leading to the stop codon was associated with an increased propensity for hermaphroditism. It was present in 16 genotypes that were all predicted to have an increased likelihood of hermaphroditism. The inventors believe that the loss of a functional ACC synthase in the specific tissue type where it is expressed results in an increased propensity to hermaphroditism while a functional ACC synthase can prevent the emergence of hermaphroditic flowers. The inventors recognize that the polymorphism that confers the loss of the start codon in LOC115696400 at position 99811627 (A/G) can be used as a diagnostic tool to identify plants with or without the propensity for hermaphroditism. Based on the findings from Examples 1-3, the inventors point out that the heterozygous allele state for this polymorphism likely results in an intermediate propensity for hermaphroditism compared to the homozygous case. The identification of this polymorphism as a causative factor in hermaphroditism in cannabis can be used to manipulate this trait through a number of genetic modification techniques known in the art, including targeted mutagenesis, gene silencing, genetic modification, etc. The inventors note that in monocious cannabis the ratio of male to female flowers can determine seed and grain yield. The number of female to male flowers could be influenced by controlling the dose of the ACC synthase XP_030479162.1 at the specific tissue type where it is expressed, for example through providing one copy of the gene or two.

[0216] Context sequences for each of the markers identified in Table 15, as well as for the polymorphism at position 99811627 are provided in Table 16 below. The context sequence is based on the CS10 reference genome. The reference (context) sequence can be used as the basis of a KASP assay to genotype at this position. Targeted sequencing primers for each of the markers identified in Table 15 and 16 are provided in Table 17. The nucleic acid sequence from position 99811627 to 99817925 on chromosome NC_044370.1 is also provided below as SEQ ID NO: 343 with the polymorphism at position 99811627 in square brackets.

TABLE 16

Detailed information of each of the additional SNPs associated with hermaphroditism in Cannabis as provided in Table 15, and for the SNP at position 99811627. The “context sequence” is provided with the SNP given in brackets. All of the sequences and alleles are provided with reference to the plus strand.	
SNP	Context sequence
common_507	TTAATCTGAACACACTCATTCTTCAGGAGCAGATTCTCAGGTTAACCTTTTCAAACCTTCC TTGTTCAATTCAACATCTTAACATAGTAAACATAATATACATGCATCTCTTAATTATC TTGAGGATGTGACATATTCTAGTATTCCATTTTCTCAGATTATCGCATGGAGCTCTGATAAT TGGGAAAGACAAA[A/T]AGCACGTTCTTCAGACTCCAGCTGGAAGGACAGGAATGTCA GACACACAAGTGCAAGTTTCATCAGGACCAGACACACTTCCTCGTCGTACATGAGACTCAG CTATCTATATACGAAGCCTCGAACTCGAACGCATATTGCAGGTAGAACAAATCAAACATTG CTTCAAAACAGTCATGAATCTAAAAATTTATTACAAT (SEQ ID NO: 271)
common_294	GGCTACTTACGGTGAGAAAGAGTGGTATTTCTTTTCGCCGCGTGATCGGAAATACCCGAA CGGTTTCGAGACCGAATCGGGCAGCTGGAACCGGGTACTGGAAGGCAACCGGAGCGGAT AAACCGATCGGGAATCCGAAACCGGTTGGGATTAAGAAGGCTCTGGTTTTTACGCCGGA AAAGCTCCGAAAGGAGAGAAAAC[A/T]AATTGGATTATGCATGAGTACCGGCTTGCTGACG TGGACAGATCAGCTCGTAAAAAACAGCTTAAGGGTACATTTTCTAATTAACCTCCATGA

TABLE 16-continued

Detailed information of each of the additional SNPs associated with hermaphroditism in Cannabis as provided in Table 15, and for the SNP at position 99811627. The "context sequence" is provided with the SNP given in brackets. All of the sequences and alleles are provided with reference to the plus strand.	
SNP	Context sequence
	TATTAGCAGGGGTAGTTTCGTCATTTACAGCATGTCCAAGCTGTGGTGGTAGTAAATGG GTCCACAGTGTGCTATCTGACACGTGGCGGCTTATAGCGCTAGATT (SEQ ID NO: 272)
GBScompat_ common_36	GGTAGAGTTAAGATAATTAATTTAGTTTATACAGAACTGCATACCGACGTATTTTATAGTGTA TACCTTTTCTCGTACGCTTTAGAAGTTCTCCTCCTCTGCAACAAAGTTCAATGCTGTGAA AGCTTTCGATTTCATGGCCTTCGCCTTTTCGTCAGCCAATGCAATGTAGTACTTCCACAG CCCCTTTGAAGCCTTG[T/C]CTTAGAATAGCAGCTCCTCTTAAGGCTGTAGTATACACAAA GAAAACTCTCCCTCATAATTTTATTCAAAGAAGCTATTGTTTAATTAGTTTTTACATGCTAA TCTATTGTTATGAGTAAAAATAATAGGATTGCTCCTTTAACTATGACTATTGTAGTAAAT AATATTTTGTATGTGGCTAAGCTGACGTTAATC (SEQ ID NO: 273)
common_480	CAAGGAGTTATCTCACTAGAGCTACATCACTAACAAAAATTGATTCCCCTCTCATAAATAA AACTAAAAGACTGTAACCAACACATATTATTGTACTCTTGTTTGTGTTTGGTGAGAATAAAG AGATGAGAATGAACATACCTTATTGCCACTTGATCTGCAGAACTTTTCTTCAACACCTT CTAGTGCATAACTCA[C/G]TTAACTCACTCAAGCCTCAGTTGCAAACTCTCATAAGCTAG ACCTTCAATGAGAATGGTATATGTCACCTCAGTGGGTTTACATCCTTGGATACCATATGAT CCAGAAAATCGATGGCAGCTGGTTTGCCTAGTCTTACAAAGTCCCAACATGATAGAGTT GAAGGTGATTGCATTAGGCTTTATTCGAAGTCCTT (SEQ ID NO: 274)
common_561	ATGGTTCGTACATTGTTGCTATATTTATCTCTTTTAACTTAAATCTTTCTTGTGAAAACT TTCCGATTCCGGTTTGGGTGCGGTGAGTCTTGCTTGTGACATGTTTGAGAGTTAGTATTT GTGTTTGGCGCTCCGATTACGGTTGTTTATGATCTAATGGCGTCTAGTAGTAGAATATGG AAGATTTGGGACATCA[A/G]TGGGCTGATCTACAAGTGAAGATGAGAATGAAGTGGGGC TTTATTGTTGATCAGGAGGAGGTGTTGGTTGATGAGTTTGATGGGAGGTGGTGTAGTCG GGAAATTAATCTGATAGGCCGGCTGATTTTGAATCTCTAAGGAATGTAATGGCGTCTTT GTGGCAGCCGGGAAAGGGATGTACGTCAAAGAGTTGGAG (SEQ ID NO: 275)
common_685	TACACCAGAAGCAAACTTCCCAACAAGATCATCTTCAGTCATATCAAGAACCTCAGGGCTG AAGACTGAGCCATTGTCATAAACAGATTGGACAACAAGACCATATGAAAAATGGCCTGATG CCAAGCTTGGCAAGAAGGGCAGCCTCAGAGGAACCAACCTTGTCACCTCTCTTATGAGC TCAACTGGGGTGATAATTTT[C/G]ACAGTACCCTTGTTAATCTTAGTTGGGATATTAGCAC CTGTTGAGCCCAACGGTGTAGTCATAAATCTAAGGAAATACAATGAATGAAAAATATA GGTTAGTTAAGTGTGACAGCTTGAAGAAAGAGTCTGAGATGGATCGAGACCAGTGT GCCAGGAGGAACAACCACTCAATCGGGCAATCAACCCACA (SEQ ID NO: 276)
common_1025	TGTAGTTTCCGAGTGGATGCATGTGCTTTTACCTATAGCTATGGAAAGAGGAGTTGCATA ATCACAACATGGGTGCAAGTAAGATTACTAAATAAAATTTTATGCAATCAACAACTTTTT TTTTTCTTAAAAAGAAATAGAGTTTAGATAAATGTTCTTATTATTGTTTCTTTTCTGGGCAGT GGATCCATGTGGTG[C/T/A]CAGGAGAAAGTTGTAGAAATAGCCAGCAGCCTGGGACTGAA CGTATCTGTGGGTGTTGCTTATGAGGTTTTTGTGAGCAAAACAGGTATTGTAGTCACTTT TAGGAGGATGAATTGGAGATGATACGTTTGAGATTTTATAAACTAATAGCTTTCACAACA TTCAATTAAATCAAGGATTGTTTTATGATTTAATCATG (SEQ ID NO: 277)
common_2949	GAACTCTTCTTCAATCATCATATTGCTTCTTCTATACTACTAGTACTACTACTTGTACTTGT AATGAGTTTTGGGCAACAACAAGCTGAGGCAAGAGCTTCTTTGTGTTTGGAGATTCTCTT GTGGACAATGGCAATAACAATTTTAGCCACCACAGCAAGAGCTGACTCTCCTCCTTATG GGATTGACTACCTTAC[T/C]CATAGACCCACTGGCCGTTTCTCTAATGGCCTCAACATGCC TGACCTTATCAGTAAATCTCTCATTAATTTTCTCTTTAATTATCTAAGTATACATATATATGTA TGATCTCTGATTTGTGTATATCTATTAACTTGAATAAATAGGTGAGCAAAATGGGTCAG AGCCACATTGCCATACTTGAGTCCGGAGCTCAC (SEQ ID NO: 278)
rare_7	TAGTCAACTTGGATGGCCAAGAGTGTGTATTGTATGGAGTTAGAAGATTATGCATTAAGT GTATTGAACTTAAATAGCTGACTGGTTTTCAATCTATATAATCAACATAGAATCTCACTAT CATGCGGGGATCTCTGATCACTGCTTATGATTATACTCTCAGATGTTTTAATCCATTATTG ACTTAAATGGGTTTTT[G/T]GATATAGGTATTCTTGAAGGGCCCATCCTGTATGCTCTTAG GGGCTTGGCTGGCCGGTTTGCTCCTATTGGAGTTTATATAGCAATGCTACTAATAATGGC AGGTGGAATCTAAGTGCAGCTGGGAGCTTCAAAGGAACAGTCACAGTGCCGAGGGC TGAATTTTGTGTCGAGATGTACTAGGCCCACTGGGTT (SEQ ID NO: 279)
common_119	AAAAAGCACAGGAAGAGATTTTGAGATCCATGTTTGGACCAAGGAAAGGAGTACTCTCT CTCTCTCTCTGAACTGTCTGTTTGAGCTTTGGTGAGCTAAACCAACGACGAGCGAGC TTCAAGCTCAACAGGCCCTTCTTTGAGTATGTATGTAATGTACCCGCTCTCACCTTCTTCTCA TGAAACGGCTTATGAAATG[T/C]CATCCAATCTACTACCCTGTCTTACCATAGGATAT CAACCTCTTTCCCGCTCTCTCACTTCTCAACCTCCACATATCAACGTAATGGAAACCA AGACCCTCTCATATTACCAACCAACTTTATTACTCATGGAGTCTTGCTCCACCATGCT CCAGATGAAGCAATTCAGCTCAAATGACTCTCACTGGT (SEQ ID NO: 280)
GBScompat_ common_139	TCATCTGTTTGTGACATTATCTTTGCTTATTTTGTCTTAAAGAACTTCGTTTCGATTATTA AATATATGAAAGTTCCGTCCTTTATGTGGAATGTATATTTTCTTTCTGAGACAAATTT GCAATTGTTAGCCAGCTGCTACTTGCGGAACAGTCAGGCATACTCTGCGTACCATATTTG

TABLE 16-continued

Detailed information of each of the additional SNPs associated with hermaphroditism in Cannabis as provided in Table 15, and for the SNP at position 99811627. The "context sequence" is provided with the SNP given in brackets. All of the sequences and alleles are provided with reference to the plus strand.	
SNP	Context sequence
	AAAGGTACATATGTC[A/T]TAAAAATATATCTTTAGCTTACTATGGTTACGGTTTACTATTTT TTCATTTTGGCTTGAGTTTGATTATTTTTTTCCTTTACACTATTATCTGAAGGAACACAATTGG CTCAATCCCGCTACTTGTTCGAATGCTTGTCTTGAATGGATCTTCTTAATGAAGCTGAA ACAGCATGTGTCCTCTAATGAACCCAGTG (SEQ ID NO: 281)
GBScompat_ rare_186	TCGATCTGCTACCCTAGTATGATGAGTGATAAAAGTACAGTATAGAGATTTTCATACGTAT AAATAATATCGTGGTCTGACAGTACATAATATATTTTTTATATATATACAAATACAATTGAG AGTACATAATATTTAACGTATGTCCAAGTTAGTTTGATCGAGCCCTTTTCGAGCTTCGAAG CGCGTGACGCGTGT[G/A]TGTCAGTCTTGCTAGCCGACAAAGGTCGTCTTGTGAGTTT ATGAGGACTTTACTTGGCAGTTGCCACCTTCATCTACGAGCTCAATAAATCTTACTTTGCA CGCGCTTCAACGTGGTGCGTGAAATCTTTCTTATTTTGCCAAGTGGACTTAGTCAAAAGCT GCCCTACCTTCGTTTATTATTTAGAGAAATTACAT (SEQ ID NO: 282)
GBScompat_ common_883	AAATGAGGTTAACCCAGCACTAAACAGTCTAACTTAAAAATAAGAACTACCTATCACCAT CCCCACAACTTACAAGTTTAATTATGCCACTCTAGTATAAAACGAAAAAGGCAATCATTG TCATAGACAAACACAAGGCATTTAATTTTAAATAAGTAGCAACTGCAGCACCAGGACAAT ACAAGTTTCAGTCAATA[C/G]CCAATTTACACATAAACCTATTTCAATATCTGCTGTTGTCTC AACAGAAGAGCACTAAGCACCCTCAATATTTAGAACAAACATATATACATTATATAATTACA TTTATCATCCCAAAGTACCACAACCAATAGACCCTATTGAAAGTTGAAACTGTTTACCTT AGCAGCAAAGTGAAATTAGTTTCAGCAAAAGAGTT (SEQ ID NO: 283)
common_1524	ACACTAAGGGACACCAAAATAATAAATTTCCCTTCATTATTTTCAGATACAATGGGGATT TAGTCCCCCCTCCACACACATATACCCAATCACTTGAGGGCATGCAGTGCCCGTCAA TGATAATTGTTTCAGAAGACTAATGGAAATAAAGTCCATTCTTACCCACATCTGTATAGGT TGATTAAAAATGAATTT[G/A]GCAAGGTACTCCATTTCTGGATCTGCTCGCATCCATAACAA GGGCGAGTCCATACTGTGAAAGAAAAGTTTGACACACGCATGCAAGACACAAAGAAAAT AATAATTAGATCCTCCTATCCAAATAAAGTTTAAACAGAGCTTAAAGAAATGAACCTACCTGG AGCGCATATCTAGAGGAAGGTACCATCTCCATTGTCA (SEQ ID NO: 284)
GBScompat_ common_504	AAATCAAACAGATAAACCTACAAGCATACTAGTCAAACCTTTTCAGAAACAACCAACCAC CATTTTTGCTTTATAAATATACAATGACCCCCAATGCACGAAAAACCACTCGGTTTAGGA GGAAAGAAATATCAACAAATAGTCAAAATAGATAAAGATTAGAACGATATTATCAGAAAGGTA AGGAAACAGAGCTGTGAA[T/C]GCACCGCACCTTATTCTAAAAATCGTGAATTGCTGCTT CTATGGATTCTCAATGGTGGCATTCAATAATCCGAGTGATCTTGATCTTTCTGCTGCCCA CTGCGTCTCCAAAAGCCCTCCTCCACCAAAGTTCGAACCTCTTTGTATGTTGCGGCAAT CTTCAGCAGAGTTCAAGAATCGGTTAAACCAAGCAAGCAA (SEQ ID NO: 285)
SNP at position 99811627 on NC_044370.1	ATGAATCATGAGAGAATAGTTCTGATGATTTTGAAGATGTTTCATCCAGAC[A/G]TGCCATAA GATGATCAGGACATGCCCGCGATGGTCTGCTAGAAAGAAA (SEQ ID NO: 286)

TABLE 17

Targeted sequencing primers (5' to 3') for the SNPs identified in Table 15, as described in this Example 6.				
SNP	Forward Primer 1	Reverse Primer 1	Forward Primer 2	Reverse Primer 2
common_517	GGATTGCTTTCACG ACAGCC (SEQ ID NO: 95)	CCCAAAACCACTTT CGCCAG (SEQ ID NO: 96)	GGATTGCTTTCACG ACAGCC (SEQ ID NO: 95)	TCCCAAAACCACTT TCGCCA (SEQ ID NO: 287)
common_507	TCAGGAGCAGATTC TCAGGT (SEQ ID NO: 288)	CGAGGAAGTGTGT CTGGTCC (SEQ ID NO: 289)	TCAGGAGCAGATTC TCAGGT (SEQ ID NO: 288)	ACGACGAGGAAGT GTGTCTG (SEQ ID NO: 290)
common_294	GTACTGGAAGGCA ACCGGAG (SEQ ID NO: 291)	TGTCAGATAGCACA CGTGGG (SEQ ID NO: 292)	GTACTGGAAGGCA ACCGGAG (SEQ ID NO: 291)	TATAAGCCGCCAC GTGTCAG (SEQ ID NO: 293)
GBScompat_ common_56	CGTAAGTGTCGG TTCCCAT (SEQ ID NO: 294)	GCAGCTGGTTTGT GAGTGG (SEQ ID NO: 270)	TTCGTAAGTGTCG GTTCCC (SEQ ID NO: 269)	GCAGCTGGTTTGT GAGTGG (SEQ ID NO: 270)
GBScompat_ common_36	GCCTTCGCCTTTTT CGTCAG (SEQ ID NO: 295)	CACGTTAGTGCCAC ATCAAAA (SEQ ID NO: 296)	TTTCGATTTCATGGC CTTCGC (SEQ ID NO: 297)	CACGTTAGTGCCAC ATCAAAA (SEQ ID NO: 296)

TABLE 17-continued

Targeted sequencing primers (5' to 3') for the SNPs identified in Table 15, as described in this Example 6.				
SNP	Forward Primer 1	Reverse Primer 1	Forward Primer 2	Reverse Primer 2
common_480	TGCCACTTGATCTG CAGAACT (SEQ ID NO: 298)	AAACCACTCGTGC CATCGAT (SEQ ID NO: 299)	TGCCACTTGATCTG CAGAACT (SEQ ID NO: 298)	AACCACTCGTGCC ATCGATT (SEQ ID NO: 300)
common_561	TTTGGGTCGGGTG AGTCTTG (SEQ ID NO: 301)	CGCCACAAGACG CCATTAC (SEQ ID NO: 302)	GTTTGGGTCGGGT GAGTCTT (SEQ ID NO: 303)	TCGCCACAAGAC GCCATTA (SEQ ID NO: 304)
common_685	ACCTCAGGGCTGA AGACTGA (SEQ ID NO: 305)	TCCAGGTCTGCACA CTTACC (SEQ ID NO: 306)	ACCTCAGGGCTGA AGACTGA (SEQ ID NO: 305)	CCAGGTCTGCACA CTTACCT (SEQ ID NO: 307)
common_1025	TTCCGAGTGGATG CATGTGC (SEQ ID NO: 308)	AGCAACACCCACA GATACGT (SEQ ID NO: 309)	CGAGTGGATGCAT GTGCTTT (SEQ ID NO: 310)	AGCAACACCCACA GATACGT (SEQ ID NO: 309)
common_2949	AGCTGAGGCAAGA GCTTTCT (SEQ ID NO: 311)	ATGGCAATGTGGG CTCTGAC (SEQ ID NO: 312)	GCAACAACAAGCT GAGGCAA (SEQ ID NO: 313)	GGCTCTGACCCAAT TTGCTC (SEQ ID NO: 314)
rare_7	GCGGGGATCTCTG ATCACTG (SEQ ID NO: 315)	TGGGCCTAGTACAT CTCCGA (SEQ ID NO: 316)	CGGGGATCTCTGA TCACTGC (SEQ ID NO: 317)	TGGGCCTAGTACAT CTCCGA (SEQ ID NO: 316)
common_119	GCTAAACCAACGAA CGAGCG (SEQ ID NO: 318)	TGCTTCACTCTGGAG CATGGT (SEQ ID NO: 319)	GCTAAACCAACGAA CGAGCG (SEQ ID NO: 318)	GAGCATGGTGGAG CAAGACT (SEQ ID NO: 320)
common_528	TTACTGGTGAAGC CCATCG (SEQ ID NO: 321)	GGTGAGGCTGAAC AAGAACC (SEQ ID NO: 322)	TGCAAGCCCATCG AAGTGAA (SEQ ID NO: 321)	GGTGAGGCTGAAC AAGAACC (SEQ ID NO: 322)
GBScompat_ common_139	TGCAATTGTTAGCC AGCTGC (SEQ ID NO: 322)	CACTGGGTTCATTA GGAGGACA (SEQ ID NO: 323)	TGCAATTGTTAGCC AGCTGC (SEQ ID NO: 322)	ACTGGGTTCATTAG GAGGACAC SEQ ID NO: 324)
GBScompat_ rare_186	GTTTGATCGAGCCC TTTCGC (SEQ ID NO: 325)	TAAACGAAGGTGA GGCAGCT (SEQ ID NO: 326)	TTTGATCGAGCCCT TTCGCA (SEQ ID NO: 327)	TAAACGAAGGTGA GGCAGCT (SEQ ID NO: 326)
common_489	TGCCTTGGAGGAT CATTGGT (SEQ ID NO: 328)	ATCCACATTGCAAA CAGGGC (SEQ ID NO: 329)	CCTTGGAGGATCAT TGGTGCT (SEQ ID NO: 209)	CCACATTGCAAA GGGCAG (SEQ ID NO: 210)
GBScompat_ common_883	ATGAGGTAAACCCC AGCACT (SEQ ID NO: 330)	GGGTGCTTAGTGC TCTTCTGT (SEQ ID NO: 331)	ATGAGGTAAACCCC AGCACT (SEQ ID NO: 330)	TGAGGGTGCTTAGT GCTCTTC (SEQ ID NO: 332)
common_1524	CCAATCACTTGAGG GCATGC (SEQ ID NO: 333)	GCGCTCCAGGTAG GTTTATT (SEQ ID NO: 334)	CCCTCCACACACA TATACCC (SEQ ID NO: 335)	GCGCTCCAGGTAG GTTTATT (SEQ ID NO: 334)
GBScompat_ common_504	AATGACCCCCAATG CACGAA (SEQ ID NO: 336)	CTCTGCTGAAGATT GCCCCA (SEQ ID NO: 337)	ATGACCCCCAATGC ACGAAA (SEQ ID NO: 338)	CTCTGCTGAAGATT GCCCCA (SEQ ID NO: 337)
SNP at position 99811627 on NC_044370.1	CCTATGGTCAAGA CGATGAAT (SEQ ID NO: 339)	ATTGGTCTGACATT AGCCATT (SEQ ID NO: 340)	GAAGACGATGAATC ATGAGAG (SEQ ID NO: 341)	TTTATTTATCGCGG GTGAAGAT (SEQ ID NO: 342)

SEQ ID NO: 343: Nucleic acid sequence from position 99811627 to 99817925 on chromosome NC_044370.1 (LOC115696400) showing the polymorphism as position 99811627 in square brackets (position 1 of SEQ ID NO: 343).

[A/G]TGCCCTAAAGATGATCAGGACATGCCCGCCGATGGTGCTGCTAGAAAGAAAGATAATCTTGATGAG

TCAGATCCTGTGGTCACAAATGGCCAAAGAAATTACAAATAGCTAAAAATAAAGGTGTCCCTACCATTTTG

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SEQUENCE LISTING

Sequence total quantity: 343

SEQ ID NO: 1 moltype = DNA length = 403

FEATURE Location/Qualifiers

source 1..403

mol_type = other DNA

organism = Cannabis sativa

variation 202

allele = n = t/c

SEQUENCE: 1

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ctccatgttt cataaatgaa ataaaccaga tagtaaatat atgcatacgt aaagagataa 180
ccacctgatt agactgaata gngtgtccag tcacgtctgt gtaaacagta ggtaccacct 240
aaaacataaa cagcaaccct tgatctgaga gctaatagat ttcccttata tcgtctagta 300
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aatactgata catccacta ggtgtggcct gcgaccattg cac 403

SEQ ID NO: 2 moltype = DNA length = 403

FEATURE Location/Qualifiers

source 1..403

mol_type = other DNA

organism = Cannabis sp.

variation 202

allele = n = t/a

SEQUENCE: 2

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ccgctcatga tcatatggtc gcttctaatac gaagtattt tcacttctgc acttcatacc 360
gatattcttg atggtattgc ctgcgtatta cttcttgaaa tga 403

SEQ ID NO: 3 moltype = DNA length = 403

FEATURE Location/Qualifiers

source 1..403

mol_type = other DNA

organism = Cannabis sativa

variation 202

allele = n = t/c

SEQUENCE: 3

tgtgtgtag atcacaaaga ccaaccgtac gagctgttac agatttgcat tgaagatcaa 60
tgccctatct acaaatgata tagcattggc caaaaaatca acccaaaagt gctgaaggat 120
ttgcttcacg acagcccgct gacgggtggtc ggagtgggta tagagaatgt ggcgaggcgg 180
ttggaggagg atcacggttg gntcatgcct aaagtgggtg aactgcgaga ttggcgggcc 240
caggccacag cccaaaccaa ggcccaaaag gcagatattc agggtagtac tagcggatcat 300
gataatggaa agaagatcat gacgaagaag gtgatgatga tgaaaagaaa cttagccga 360
tttggtatag cgaagctggc gaaagtgggt ttgggaaaag aga 403

SEQ ID NO: 4 moltype = DNA length = 403

FEATURE Location/Qualifiers

-continued

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source          1..403
                mol_type = other DNA
                organism = Cannabis sp.
variation       202
                allele = n = c/g
SEQUENCE: 4
tctgccgtga tatttttgaa atttttaggt ttgtttccgt aaatttgatc gttgtacata 60
at ttgttagc ttaaagttta tgattttttg ctgaacaagc ttgggtgata cagctgctgc 120
agcgaatag gatgagtttg gtgtattgac ggagagggtt ggattgaagc ctcaaggtaa 180
atcggctcca atggcggcct cnaagagatc tgtcaatagt aacgatggcc agaattggaa 240
ttcgggggtt gattcgggtg taaaccccaa cttttcatct ttttcagacg atcggaacgg 300
tttttatcaa tccggcaaag ataaggtagc tcagaacagc ggaagcttga atgattatga 360
tgatatattt ggtggaccta gtaagcaatc gggtagtcac ggt 403

SEQ ID NO: 5    moltype = DNA length = 403
FEATURE        Location/Qualifiers
source          1..403
                mol_type = other DNA
                organism = Cannabis sp.
variation       202
                allele = n = a/g
SEQUENCE: 5
atttgttact tcttcgtgtc ttgatgccaa gatagcgaaa aagtacatag atcattcggt 60
cattttgaga ttgctagatt tgcttgattc cgaggatcct agagaaagag actgcttaaa 120
gactattcct cataggacct atgggaaatt tatggttcat aggccattta taaggaaagtc 180
tataaacaat gtattttacc gntttgtgtt tgagactgag aaacataatg ggattgctga 240
ggtattggag atctatggga gtatcattag tggatttgca ttacctttaa aggaggagca 300
taagatatcc ttgcggaggg ttttgattcc tctgcatagg ccaaagtctt tgggggttta 360
cttcaacaa cttcctact gtgttacaca atttatagag aag 403

SEQ ID NO: 6    moltype = DNA length = 403
FEATURE        Location/Qualifiers
source          1..403
                mol_type = other DNA
                organism = Cannabis sp.
variation       202
                allele = n = c/t
SEQUENCE: 6
cccatatgat acatatgaat gtaaaacact taccgctctc tcaggaaagg aacttgagaa 60
tccgaatatt ttaccagaaa atacatttga agtttttccg ttttgagtgc ttgaaccaag 120
ttgcactctc tttggagctt tattcttata acttactcga cccttacttt ttttattgac 180
agcagtaacg ttaactctgct gngatgattt tgcgtcagtt gccataatgt tgacatcagt 240
cgtgttgatt tctgggtttt ttcccttgcg tccgcccagt ggtaattggc ttccaattcc 300
acctcttatg ctaccagata aaggattaga tggcatgac ggatgaacag cagagactct 360
agcttgattt agactggtca atcctgtaat tgctccttca aga 403

SEQ ID NO: 7    moltype = DNA length = 403
FEATURE        Location/Qualifiers
source          1..403
                mol_type = other DNA
                organism = Cannabis sp.
variation       202
                allele = n = c/t
SEQUENCE: 7
tcgcgattgt tattcccacc tcgtctttta cttegaatct cagtgtctcg tagaagttta 60
ttagagctgc cttcgccgcc tacacattag tgtctcgtaa atatcaagtg atgatatgtt 120
ttattggctt aagcaacaca tatgttttaa tcaattactc acagcatata agctcattct 180
aggcagaggt aaccagttct cnactgaggc attaactatg actctaccat tgctttggta 240
taagtaagga agagccacaa atgttggata aacatttccc caaaagttaa tgcctgaaa 300
acccaaaaag ataaattaca aaaattaaga gtgttcacgg attggtttag tctagatttt 360
taatttttcc aaaataaacc actattacgg ttttttaaaa att 403

SEQ ID NO: 8    moltype = DNA length = 403
FEATURE        Location/Qualifiers
source          1..403
                mol_type = other DNA
                organism = Cannabis sp.
variation       202
                allele = n = c/a
SEQUENCE: 8
tttccaagtg tatacgaaac ttgcactct tatcaatgtt gtggtccatt gttatctaaa 60
cgtttatcat gttgtgcttt tgcagtacta ttctttggat attacatatt ttaagtcttc 120
tcttgatagc catctactgg atctgctatg gaacaagtac tgggtgaata cactttcttc 180
ttcaccctta ctgggtaatg gngattatgt tgcaggacaa atttcagact taggtacctc 240
aattgttatg ataccagatt ttttttttaa atgctcgtct agctttctgt aatgtttatc 300
atggtataaa aatcattaac tgtgtgtgac tttttatcat ttatgcagct gaaaagtgtg 360

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agcaagcaga gaaccaattg tcacattccc gctttgggcc att 403

SEQ ID NO: 9 moltype = DNA length = 403
 FEATURE Location/Qualifiers
 source 1..403
 mol_type = other DNA
 organism = Cannabis sp.
 variation 202
 allele = n = a/g

SEQUENCE: 9
 ttgacttcat aacttattag taactgatgt tattttctca cttccaacat cttgtggttg 60
 cttattatga tgggtggtgt tctttctgtc ctaacaggag agcaagattc tgaagttttt 120
 aggctttatg tggaatcctt tgcctgggtt catggaagct gctgctataa tggetattgc 180
 cttggcaaat ggtgggtggaa gncctcggga ttggcaggac tttgttggtta ttattgtcct 240
 gctgatcatc aactcaacta tcagttttat tgaagaaaaa aatgctggta atgcagcagc 300
 tgctctcatg gctgggtcttg ctccaaagac taaggtaaaa caacagcctc accatcacca 360
 tcagacttgt ttggttttta gcttttatgg tatttttttt taa 403

SEQ ID NO: 10 moltype = DNA length = 403
 FEATURE Location/Qualifiers
 source 1..403
 mol_type = other DNA
 organism = Cannabis sp.
 variation 202
 allele = n = a/g

SEQUENCE: 10
 ctactctctc ttctccattt ccactctccag ctccagttct agttccatct ctttctagat 60
 actcaactgt atacatatat catcacaaat ttatatctta aatcaaaacc atgaaacttt 120
 aatggctatg attcagcttc agagggtgtt gtgttcggcc acctcaacca caaccacaac 180
 cacagtttct gcttctttct cnaacccttc tacttcatca gctctgcaac tccgcccttt 240
 catggctaag gttcttaaac tccatttcat tatcttcttc ctgctttctt aaatttatta 300
 tgggtttcat tattatatta aacttcttct cttcaatact tagctataac aatttttttt 360
 ttcggctgct gtcgtaacac tctctaatg aggaacttta cat 403

SEQ ID NO: 11 moltype = DNA length = 403
 FEATURE Location/Qualifiers
 source 1..403
 mol_type = other DNA
 organism = Cannabis sp.
 variation 202
 allele = n = a/g

SEQUENCE: 11
 tacaattaca tttacaaatc ccataaaggc ctaaaacagg ccatgatgaa acgacgtcgt 60
 cgtcgttgat gggtaaaacg cttctcgaag cttgtgttac ttttgatat caaagctatg 120
 aatctccatg agccgttttt gcttgagag aaagcggcgt tggaagtagc cttggaaagt 180
 ttggctcgtc gaggtacgga gnaagaagcc gttagcccaa gcgaaatgga tcgatgtcac 240
 aaagaagcat atggggtcca tcacgtccca actcagctcc caaaagggtta gtctcatggc 300
 cgcaagtgtt tgggccagta cgaagcccag cccaatataa agctcccac ggaccaattc 360
 tcttgcttcc tggtcgatca tcgatttttg tttctccatt tcc 403

SEQ ID NO: 12 moltype = DNA length = 403
 FEATURE Location/Qualifiers
 source 1..403
 mol_type = other DNA
 organism = Cannabis sp.
 variation 202
 allele = n = a/g

SEQUENCE: 12
 tcttcaaaaca gtttctgtcc ataaatgaaa aaatgtttga aaagagaaaa actatgttta 60
 attttgatc tttacaaaaa caaatgagta aatagatacc tgaacaagtc tctctttttt 120
 aaaattgctg gccaaagtga ttctgctaaa gcttgtgaaa agacaagcaa ttcaaacac 180
 ttgttatcct cgtggactgg anctcccat tctttatcgt ggaataaagt ataaattggg 240
 tctgtgattc attacaaaaa gaaaaggaaa aaaaagatcg atcgtgtaag cattataaat 300
 tataattgaa ccaccccgac aaggtcaaac gtgccatttt tggacaaaac taagacatga 360
 gccacgaata tctcattaca attatctatt caacaagtta cat 403

SEQ ID NO: 13 moltype = DNA length = 403
 FEATURE Location/Qualifiers
 source 1..403
 mol_type = other DNA
 organism = Cannabis sp.
 variation 202
 allele = n = c/a

SEQUENCE: 13
 atccaattag taatccgtgc ccttggtaaa gagaagtttc agtaacgcca aagatcctca 60
 aagtatatac atcaggtaca gagaggaaac aaagagttgt cattgtcac tagaaccagc 120

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cacatacga acaacgactg tgaatcatc aagctttccc ccatagtagc ggaaccagc 180
atcttgagca gctgtggaaa anggtgtctg cctgtccttg tcctgtgctc gttgacgcgc 240
caatgctgct attttttgag ctgttgcttg aggccttaag ccagctcgca tggcatgcac 300
tactactgcc gttatctcat tgtgtacaa gttgtcaaa agtccatcag tcccagcaat 360
gatgacatcc ccgggagcta caggactgt aaaaacctga atc 403

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SEQ ID NO: 14      moltype = DNA length = 403
FEATURE           Location/Qualifiers
source            1..403
                  mol_type = other DNA
                  organism = Cannabis sp.
variation         202
                  allele = n = a/g

```

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SEQUENCE: 14
catccacttc gaaccgtact ttgccacaca gtaagcgctc gtacaccctt taccgtcgat 60
ggtcttgacc ggaacaagat tctttgccc aatgataccg agttccactg tgccaacagg 120
aggcttcag agttgcctag ccgtggggcg gtagtcgctg ctcatgtgtg caacctcctc 180
catcacgtga tatccgcat cnaaacagag ccttaggtga acacgtccgt tataagttct 240
tcttttctca tctgaatat ttccaaggct gagccaccgt gacgtactt ttctgtcctc 300
cactctcttc tctacaacag taagagggaat tattgccgtt cccaatgtag ttgtccctt 360
tgatgtcgt atctcaagag tgattacaat gtgttcagag aaa 403

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SEQ ID NO: 15      moltype = DNA length = 403
FEATURE           Location/Qualifiers
source            1..403
                  mol_type = other DNA
                  organism = Cannabis sp.
variation         202
                  allele = n = g/c

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SEQUENCE: 15
aaatggaggg ctcttttagc aggggctata gctggacctt ccatgcttct tactggagtt 60
aatacgcagc acactagttt ggctatttac atacttatgc gagctgcggt tttagcgctg 120
cgttgtggga ttaaaagcaa acgggtttgg agatattgta agccactcac atgggctcat 180
ggcgacattt tcttatgtg tntctcttct tcccaaatc tgaactata tcttttcaaa 240
ttggattgtt atgttttctt acatctaaact ttgttttgtt tttatttctc tcttagtttc 300
gttttgattt ctttctaatt ggggattcgc aaaatgtatt tggtttatgg gcattgtatt 360
gaattttcga gtttggttg ttcccctgtc tttatggcg tg 403

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SEQ ID NO: 16      moltype = DNA length = 403
FEATURE           Location/Qualifiers
source            1..403
                  mol_type = other DNA
                  organism = Cannabis sp.
variation         202
                  allele = n = t/c

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SEQUENCE: 16
acattgtaaa tccaaaaatt caattgcaac cctgtttaag caagtaaat tcaactgatt 60
taatcaagat ttcttgagct ataaacgcag ttcccttatt caaaaagtca aatttcctct 120
taacatttta atttgatata aatccacagt ttctgaattg ggtccggaag atacaagcaa 180
gaacatagta gagatcatat tncaatcaag ctggtctcaa aaggaagccc caatatgcaa 240
aatcgaccga atactgaaag ttcaacaata ccagaaaaac atatccaagt tcgaggagta 300
tcgagaagcc attaaagcca aggccaccaa gctcccaaag aagcaccctc gatgcatcgc 360
cgacgggaac gagctcctcc gcttccattg cggcaccttc atg 403

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SEQ ID NO: 17      moltype = DNA length = 403
FEATURE           Location/Qualifiers
source            1..403
                  mol_type = other DNA
                  organism = Cannabis sp.
variation         202
                  allele = n = c/t

```

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SEQUENCE: 17
tatatataca tatatataat catttctatc tttctctatc atcatcgta tcagtttctc 60
tgtctgcgga aaaatccggt tgttttttct gtaaattgat tagatgggaa aggacgacta 120
caacaattac gatgagagag acagagggct ctctctcat cttggtagtg catactattc 180
ccagcagcac cactctggat cntaccctcc gcccccgcgc tctgcatatc caccgcgcgc 240
tcatcatcat caagcctacc ctctccacc cggtaccaca tctgctggat accctcctcc 300
gtgcctcac gccgcatacc caccgcccta tactcctccc ggtggatacc ctcttcctgc 360
ttatcctcat cctcctctc ctgctccata tccatattct gca 403

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SEQ ID NO: 18      moltype = DNA length = 403
FEATURE           Location/Qualifiers
source            1..403
                  mol_type = other DNA
                  organism = Cannabis sp.
variation         202

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                                allele = n = g/a
SEQUENCE: 18
tttcaccatt cctgtctctt tgctcaattt ccaactacaac catcctccct ccattccgat 60
cataataaca cccacatcct cattcgtcgt aaagcagcct catccaacgg tcgacatttc 120
acagctcgct cttctagccc ggactcaatc cctccttctc cttctcctcc tccttcttcc 180
tcattacttc aacagggtcat tntcagctca gcggtgacca tcgcattcgc cgttgccaac 240
cgcgttcttt ataagctagc tctcgttccc atgaaagaat atcctttctt tttagctcag 300
ttggccactt ttgggtgagt aagtagtggt cattaattta taaagaaaat aaaaaataaa 360
aatttcggac tcaatgatga tgattaataa ataaaatctc cct 403

SEQ ID NO: 19      moltype = DNA length = 403
FEATURE           Location/Qualifiers
source            1..403
                  mol_type = other DNA
                  organism = Cannabis sp.
variation         202
                  allele = n = t/c
SEQUENCE: 19
aatacgtcaa cttgcataac acgataatac aaattgccac ctttatgaga ggtgatccac 60
aggcagcacg tttctatgtc ctttaagggc tacaaaatta tcattttgat tcaatagtgt 120
tttcaaagtt gaaagacatt gatattgttt acccccccatt gggtttgtgc agttgttgga 180
atttgacgag tcaaaaagcct cncaagaaga tgaagatgac aaagttagca gcaagaagga 240
ggaatagttt gagaagctca taacctaaaa cttccctttt tagatgaata cttttttttt 300
tttattatca actttcttct ttttcttatt tccaaggtcc caaaacggcc ttagagaacg 360
cattttggct aattcttatg ctgctccaca ccccatgtaa cag 403

SEQ ID NO: 20      moltype = DNA length = 403
FEATURE           Location/Qualifiers
source            1..403
                  mol_type = other DNA
                  organism = Cannabis sp.
variation         202
                  allele = n = t/c
SEQUENCE: 20
ctgtctagac ctgacatcac cgttgctgtc aactgtctca gccagtttat ggccaatcca 60
tagacaactc atctccaagc agttcatcat ctactacact accttaaaga aaatcctggt 120
caagggtctt tctattcacc taactcagcc ttacactac gcggtttttc tgactttgat 180
tgggcatctt gccctgttac tngatgttca acaacagggg attgtatctt cctcggagac 240
agtctcatct cttggagagc aaagaagcag tcaataatat ccaaaagctc tgcagaagtc 300
gagtatagag ccttggatgc aacacaaggt aaaatgacat ggctgcaata tcttctccat 360
gacttcagag taccacagcc acatcttgcc tttatttact gtg 403

SEQ ID NO: 21      moltype = DNA length = 403
FEATURE           Location/Qualifiers
source            1..403
                  mol_type = other DNA
                  organism = Cannabis sp.
variation         202
                  allele = n = a/g
SEQUENCE: 21
gaagacagat tttcaatttg caagctcaga gggcggttag tgggtggatta aggagatgtc 60
aggaggtgga tgtagcatag tatggttcag gagagatcta agggtagaag ataatccagc 120
tttagcagct ggtgttagag caggtgctgt gattgctgtc tttgtctggg cacccgagga 180
agaaggtcat tactatcctg gnagggatc aagatggtgg cttaagcaaa gtttggtcca 240
tcttgatttc tctttcgcaa gttttggcac ttctctcatc accaaaagat ctgctgatag 300
tgtttcttct cttcttgagg ttgtcaagtc cactgggtgcc aaacaaatac tcttcaacca 360
cttatatggg cagtattttc tttcgttata tatttttcac agc 403

SEQ ID NO: 22      moltype = DNA length = 403
FEATURE           Location/Qualifiers
source            1..403
                  mol_type = other DNA
                  organism = Cannabis sp.
variation         202
                  allele = n = c/g
SEQUENCE: 22
aacaaaaaaa gttatctcta gtcaacataa gaaagagcaa cgacaacagc tcttatctta 60
gcataaaaaa aacataagcc actaccctac ttccctgggc gactgtcctc agactcggcc 120
atgccggcag cgaccttgcc tatatctttg gcagtggcat caggcttttt ggtagtgtat 180
agtgtgaaat atccaatggg ancgatgatg agtaagccac caatggccat gatggctggg 240
ctgtagggaa gcttctctct ctgatggagg acacccccag gattgtggcc tggtaggctc 300
tgagaccctt ctactccagc tgcttttatt tctctttgac tcattttgtg ggtgtctgcc 360
attttctctc actcttcggt cttttccatt ttccctttct aga 403

SEQ ID NO: 23      moltype = DNA length = 403
FEATURE           Location/Qualifiers

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source          1..403
                mol_type = other DNA
                organism = Cannabis sp.
variation       202
                allele = n = g/a

SEQUENCE: 23
gaaatacaag ccaaacccca taaaatattc tcaatcaaat aagtagtttc atcttcattg 60
ggaggcagca gcagcagcaa cagcagaatc gttgggtttg ggcaaagggg tattaacttg 120
gattctacca tcaatacagg aagagcactt ggtttcagcc catgtctcca agtcaaaatt 180
tccatagccc atgattctct tncctgagca caaccgaccc accattactg ccattgctcc 240
taagatgaga atagcaacta agacaccaac aacaggacca acagaagatg aacctgggct 300
cactctctgc tgetgcgggc acactactac tacactcagt ggtggagcct gtgatgacat 360
ttcgttaaaa aaagagatta gttaacacaa gctgaccact aca 403

SEQ ID NO: 24      moltype = DNA length = 403
FEATURE           Location/Qualifiers
source            1..403
                mol_type = other DNA
                organism = Cannabis sp.
variation         202
                allele = n = c/g

SEQUENCE: 24
agaaacctta atcttcaatt taagaccatt gttcttatta ttctgttctt tcgtcagccc 60
atccccctcc tgaacaccga atctccgaat accctccacg tacttcatgt acatgtcgtt 120
gtacggaagc ttcattgccc cgccaccccc accgtcgtca cgggggttcg gaggggtcgg 180
cgaaacacca aaacccattc cnacctgacc aacgctggcg aataagccac cggggtgata 240
ataccctctc tgggaatttc attgagaaga ccctaaaccg ctgacggagt aaaacccatc 300
tttcgtctca tcaaatagtt gaactcctct cttaccattt aatctcaaaa aataaaaagac 360
acaaaccttc ggacacgctc ttttttattt gctttttctt ctt 403

SEQ ID NO: 25      moltype = DNA length = 403
FEATURE           Location/Qualifiers
source            1..403
                mol_type = other DNA
                organism = Cannabis sp.
variation         202
                allele = n = a/t

SEQUENCE: 25
tgtccaatca atctcttccc aactctctct ggttccatta caaagaggcg aaaccgtcgt 60
cgttttatgc tccccaggcg tggaaactcg ggaaatcata ttcggttgcc aaagagctgg 120
ccttttgacc gtaccatttt ctccaccaga cccaacttca gacaactgcc accatcttat 180
aagagtctta tcccaaacca ancccaaaag cgcaatagct caccaaaact acataaccaa 240
agttcaccaa ttcattgacct ctctctctc ggccacttca aaacttgcaa gcttgttaaa 300
aaaaactaaag tggatctcaa ctcaacatgt caagcagtag tccaaaacga catctttgaa 360
gttattcgat tctgggttgt cgtataaagg gtgtaatgca gaa 403

SEQ ID NO: 26      moltype = DNA length = 403
FEATURE           Location/Qualifiers
source            1..403
                mol_type = other DNA
                organism = Cannabis sp.
variation         202
                allele = n = a/g

SEQUENCE: 26
atatcgtgct taatatataa gaacaataaa taacatgtta ttatagctaa ttaatatagc 60
acttaataaa ttcttatact ataatggtac gtacataatg tgggaagcta gcaattaagt 120
gatgatggat gagttattaa ttacctgtgt tgatcattat tatgatcatt aggaaatcct 180
ctaagggtgt tgcaggcaga gncagagcct ctgtaattgg gtgtagggaa ttggtcagat 240
tcaattttga ggaggctctg cacaaggatt tcgtagtgtc tcttgacttc gtctggagtc 300
ttcccaccca ctgctttggc cacattttgc catcgggtcag ggggtgtcctt gtcatacaca 360
gctaaagcct tttcaaattg cttgttttgc ttagacgtcc aag 403

SEQ ID NO: 27      moltype = DNA length = 403
FEATURE           Location/Qualifiers
source            1..403
                mol_type = other DNA
                organism = Cannabis sp.
variation         202
                allele = n = c/t

SEQUENCE: 27
gggtgaagata catgatcttt ctggctcagc tgttgccgca gcgttcatga caactccgtt 60
tgtgccttcc acaggtttgtg attgggttgc caagtccaac ccaggagctt ggttgattgt 120
tcggcctgat gcttgaggcg cagagagctg gcagccatgg ggtaaacctg aagcttggcg 180
tgaacgtggt atgagggact cngtttctgt tagatttcgt ctcatgtctg aaggctcatc 240
ggaggcaggg gagcttctca tctctgagat ttatatcaat gccgaaaggg gaggggagtt 300
tttcatagac acggatagcg aggcgcttgc aacggcagca actccaatcc caagcccgca 360

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aagcagcgga gactttgcag gactcagccc agttgttgga ggc 403

SEQ ID NO: 28 moltype = DNA length = 403
 FEATURE Location/Qualifiers
 source 1..403
 mol_type = other DNA
 organism = Cannabis sp.
 variation 202
 allele = n = a/g

SEQUENCE: 28
 agccccaaca ccaccccac caccaacacc actgcctcca cctatgctca gaccaagaag 60
 atcacgcgtc atgggcggaac tcgcgaacgt caacgatggg cccatgacta gatctgtcaa 120
 gccagaattt ccaccaggct gaagcccaag cccaagccca gctgccaaag tcgacgaacc 180
 actttccggc ttgatttgag tncgtttcca ctgcgtgcca agactgtctt taacggaggc 240
 tgaagatggt gatgttgtca agccaaaact gcacaacagc gaggaatttg aagccgaagc 300
 acccatttgg gctgcttttt gaagcaatgc cgttgccagc attgtggag gctgcagtgg 360
 tgatgggctg aagttgcggt gctcgccctg gtcttgtgtt tga 403

SEQ ID NO: 29 moltype = DNA length = 403
 FEATURE Location/Qualifiers
 source 1..403
 mol_type = other DNA
 organism = Cannabis sp.
 variation 202
 allele = n = a/t

SEQUENCE: 29
 caagaacttt tgcagcgatc gattccgaag gctcttcgtc aaccttgaaa tcgaaaactt 60
 ccatagctag acgaggagag gtgtttgttc cgcactccag aaatgaactg gctggagtca 120
 aaagtcaact gggaatcgtc aactagtgtt catatggatc ggggtgagga gtactttaga 180
 gagaagaact cgggagtgtg cngcgggaag aggaatttgg tgggtgagaa tcgtgcccta 240
 atttgtactg tactaatcca gttatagaag agatcagctg actttacttg tacttgtaact 300
 ttgagaagta aagtgatgat atgaaacag agcaaatatg atcattattt ttttgtgtgt 360
 gtatgattag cgtgtttggt aaatcagatt ttcgtgtttg cat 403

SEQ ID NO: 30 moltype = DNA length = 403
 FEATURE Location/Qualifiers
 source 1..403
 mol_type = other DNA
 organism = Cannabis sp.
 variation 202
 allele = n = a/t

SEQUENCE: 30
 agaatttttag agacgggtgga gtttggttgt gtgaaaatgc gccaggcttg cttagcaagc 60
 aacgcctggt tgaagtgaac aagagatcgg aacccaagcc cgccttctct tttagacctt 120
 caaagttttt tccaactttg ccaatgaatc ttggggcaat tgtgtgcctt gaatccccc 180
 cagaaattag ccataatgga cncagagag tggcaagtgg ccacagggag acgaaagcag 240
 gccatagtat aggtgggaat tgccctgaatg acagacttga gaaggatttc cttacctccc 300
 ttagagaaga ctttattttt ccaattatga aggtggggccc aaactttgtc atgaagatag 360
 tgaaaaaat tatttttttc gttcggccaa tatattgagg gag 403

SEQ ID NO: 31 moltype = DNA length = 403
 FEATURE Location/Qualifiers
 source 1..403
 mol_type = other DNA
 organism = Cannabis sp.
 variation 202
 allele = n = g/a

SEQUENCE: 31
 caacccaaag aaagaaaaaa aaaaaatcta acaacgcatt atatgagttg tattgctaaa 60
 ataactccca atcccaaatc caaaaccatc aatagaagaa gaaggagaag gagatgatga 120
 taatgaagat gaagatccaa ttctcaaaac cctaagcaaa tccaaacaca tactcctcac 180
 tctgtgaccc tcacagtacc cncctgtcat caccatcaga atcttcgcaa ttccattgct 240
 gtaccaccac gcctccttcg tctctctctc cccaataaag cagcacacgc tccaaaccac 300
 cgccaccgcg tcctctgtcg cgcacctga caccctcatc attctctcca cgattgccac 360
 cgcacacctt cagtctactc cgattgtgta tcttcctcc gct 403

SEQ ID NO: 32 moltype = DNA length = 403
 FEATURE Location/Qualifiers
 source 1..403
 mol_type = other DNA
 organism = Cannabis sp.
 variation 202
 allele = n = a/c

SEQUENCE: 32
 tgaaacttca aagaatgcat cttggtatcc ttgaaatgat ccggaaaaga atttttttga 60
 aggagccaaa aagtactaat catttcattg gttaccctgt aattggttga gaaccatttc 120

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agactcatta tctttgtcat cattgtcatc ggtattgaaa ctaccttctt caaatgctac 180
ttcattgtca ttgtttagcgt cnagtgataa gccggcaatc gatgcatacc actcacotcc 240
taaatagcct gaatccctca tcaatctaac caaaacttct ttgctttcga gatcattttc 300
ttgtcaccat agatcgagaa aagcagatac gtagacaggt ttgggectcc agttgttcga 360
accatcggtt gcaaaaagctt ttttccagta agtcaaaagcc tcg 403

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SEQ ID NO: 33      moltype = DNA length = 403
FEATURE           Location/Qualifiers
source            1..403
                  mol_type = other DNA
                  organism = Cannabis sp.
variation         202
                  allele = n = a/g

```

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SEQUENCE: 33
ggatagaaaag accacgttat catgagggcc acaacttagt agtgtaaaaa attcagggaa 60
aaagtttaatt gttctttaga aaaagcaata atgggagcct aaattttgca aattagaatt 120
caatttagca tttttctcac catgagaaga ttccacgtcc aaaatcagat caagtgcata 180
gtcataatat gggacttgac tntctaaaac acaaagggtt aaatcatctt gaatgtattc 240
atcatcgact tcgcaaaaga actcatttcc tcgcaaaattg caaaaccatg aaatccaaga 300
agtgctcatct ccatcagagc cacttacatc ggattcttca ctatccgtct cagattcctc 360
tgcaaacacac gacacaacac gtcaataaagc aagcttagag aat 403

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SEQ ID NO: 34      moltype = DNA length = 403
FEATURE           Location/Qualifiers
source            1..403
                  mol_type = other DNA
                  organism = Cannabis sp.
variation         202
                  allele = n = a/g

```

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SEQUENCE: 34
tttgggacag ataaaaactat ttgggtacca atcatgtcaa gtccaaaaga acttggttct 60
actagggacc cgtcaccagg ctttgtccag atgtgccttg aaagtatcaa tgtcacaaac 120
tggtggacca gaaaatgtta atatgaaaat caatgacgtc aaggagaaat tacagaaagc 180
cattcctgac tcagttaaagg ancttgaatg ggagaaaggg gctgatatac tgggtcgaca 240
attgctttct cttggacaaa aggcattcaa atgggttact gttgttttga ttgctgtgag 300
ctttttatca gatgttatat tcactatctc taggaaccag gaattagtca tgccatttgg 360
tctcttggtt ggttgtttat tggtgattt ctttaaagaa aca 403

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SEQ ID NO: 35      moltype = DNA length = 403
FEATURE           Location/Qualifiers
source            1..403
                  mol_type = other DNA
                  organism = Cannabis sp.
variation         202
                  allele = n = c/t

```

```

SEQUENCE: 35
ttgtccgcct acaaaacaat tatgcaatat cagaacctta tcaccacta caacaccatg 60
ttctattcaa atcaacatat cttagataaa caaaagctgt agacatggac tgacctgaga 120
ccagtatgag atccaaatgc cgcggatctc atgggtgaca cgtgggtcag ataacgcac 180
aatccatcgt ttgatgaatc gntcttgctt gaaaaaatca aggtgtgaaa gccaaagtcaa 240
gattagtgc tgcaataaga ttgagaaatt ttgattttac ttgcagaaa atagaggagg 300
ataagtacct gtctggtgta aaggatcggt atctctctcc aggttgcttg aagttgttct 360
ccttctcgat aatgcactac actcaccaaa taaatacaca cat 403

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SEQ ID NO: 36      moltype = DNA length = 403
FEATURE           Location/Qualifiers
source            1..403
                  mol_type = other DNA
                  organism = Cannabis sp.
variation         202
                  allele = n = c/a

```

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SEQUENCE: 36
aagcaattac agaagcagca gaagggtggtg caaatgcagg tggatatgat gccccatgat 60
cgggtttctt gttcgggtgc tcaggtaggt ttggttgctt ttgtaaaatc cgtttcaa 120
taacagaaac agcattatct tctggaaatg tctgatttct tcgtttacct agttccaaat 180
caaggcttat tccagcatcc gnttcgagtt catttaaacc attgccctcc aaggactgct 240
tcgcgggagg agggctacaa gggcattcgg aggtgagtat cttagtcca aaaaacctga 300
tactattact gtccatcact gttcctcctg ttatacatga actctccgga ccataagata 360
tgaaccttct gttatcattt tgcaagtcct ttaatgccac ctt 403

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SEQ ID NO: 37      moltype = DNA length = 403
FEATURE           Location/Qualifiers
source            1..403
                  mol_type = other DNA
                  organism = Cannabis sp.
variation         202

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                                allele = n = g/c
SEQUENCE: 37
gagcttcttg ttagattcat caatcgataa attcaacaaa gctgttactg catgctcttg 60
aatctttgaa tctggatagg ataggagctg tactaatggt gggattcctc catgtggggc 120
aatcaaaaatt ctgttctcag ggttctcttt ggagagcaac cgaaccttct taaccgcctt 180
tcgctgcgtt tctaaatggc tngaagatag tcttcaaca agggacaaga tctcctcttt 240
gagatcagct gaggagcttt catggctctc tggattatcc tttaggggaa gtttaaaatt 300
gttcttctca ctccactgca agattatggt tttaagagca taatttgacg caagtgtcag 360
atgaggaaga gtttgctggt ttttgggaca agtctctgtg tta 403

SEQ ID NO: 38      moltype = DNA length = 403
FEATURE           Location/Qualifiers
source            1..403
                  mol_type = other DNA
                  organism = Cannabis sp.
variation         202
                  allele = n = g/a
SEQUENCE: 38
aggaagacaa aagatccaaa taaagaaatt ggaaaacaag agtaacaaac atgtcacttt 60
ctctaaacgg cgctctggtc tcttcaagaa agctggcgag ctccagccttc ttgtggggtc 120
caagggtggc gtcacgtctt tctctccac tggcaaacctc ttctgcttcg gtcacccatc 180
tgtggacgac gtcgttctgt cntatctcga ttataataac caaacagtag tagtattacc 240
tggtagtaat gataacatta ctgctactac tactactagt actagtactt ttgtaaaagc 300
tatgactgag ttggaagagg ccgagaaaaa gaagaagcaa ttggtgagac aagcgagatt 360
ggcctcgatt ttagataata atagttgcgc gtggtgggag gaa 403

SEQ ID NO: 39      moltype = DNA length = 403
FEATURE           Location/Qualifiers
source            1..403
                  mol_type = other DNA
                  organism = Cannabis sp.
variation         202
                  allele = n = g/c
SEQUENCE: 39
atccccagca gaacaccatc aaagccgtca atatctcatc acatcagatc catcagttct 60
ccatgcagat ctcatccggt gatttcccag ctccagggacg gcgttgctga gcttcaactg 120
tgggtaaccc aaccggaatg tcggacctcc gtttggctga ccaacggggt gacccgactg 180
agagacctcc acgactgtct cnaagatatt cttcagcttc cacagaccca agaactcttc 240
cgccgattac ccagctccgt cgtagagaat cttctagaag atttctcccg cttcgttagac 300
gtgtacggta ttttccagac ctcgattctg gtaatcaagg aagaacaatc agcggcgcaa 360
gtgggtttac gaaagagaga cgaagttaag gtttctcttt acg 403

SEQ ID NO: 40      moltype = DNA length = 403
FEATURE           Location/Qualifiers
source            1..403
                  mol_type = other DNA
                  organism = Cannabis sp.
variation         202
                  allele = n = c/a
SEQUENCE: 40
tcagtaacag tctaagtctc tccattatta tgggctgcgc caactcgaag ctcaacgac 60
ttccggcggg ggcttctgtc cgggaccgat gtaaatacct agacgaagct ctccgccata 120
gccaagactt cgctgaagct cagcccgctt acctcgactc cctcaaggca ttgggtcccg 180
cttttgatcg tttcttcagt anaaaagtac cagtcaacaa tgaaacttta aaaaagtctt 240
cttctttatc tccggcgggc gtggcttctt cttcccccgt gcataaatca aactcagcag 300
actcgccact tcgattccct tccgattctt ctgaggacga ggatagtgat aatcataata 360
aactcgggca taattgcctc aacgaagagg aatcccaagg ctt 403

SEQ ID NO: 41      moltype = DNA length = 403
FEATURE           Location/Qualifiers
source            1..403
                  mol_type = other DNA
                  organism = Cannabis sp.
variation         202
                  allele = n = c/a
SEQUENCE: 41
tcatgacctg aacttgctt tttttaattt aaaagtcgtt ttggaagaag ttaggatttg 60
tatattaatc tttttcggct tcctatgttt ggaccaaacc atctgctgca gggttgtgga 120
gagtggtggg gaagatgtga atgaagtagt tgaaggagat agagtgatcc caacatttct 180
ttcagattgt ggggagtggt tngactgcaa atcaagaaga agtaacctct gttegaattt 240
ccccctcaag gtctcacctt ggatgcccag atatgagtc accagattca cagacctcaa 300
aggagacctt ttgtaccatt ttctgtttgt ttcaagtttt agtgagtata ctgtggttga 360
tatctccaat cttacaaaaa ttagctctga aatccctcca aac 403

SEQ ID NO: 42      moltype = DNA length = 403
FEATURE           Location/Qualifiers

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source          1..403
                mol_type = other DNA
                organism = Cannabis sp.
variation       202
                allele = n = t/c

SEQUENCE: 42
cttctctaata  atagtatagc  tatgcttatt  ggataatcct  aaagctccat  ttttaattatg  60
ctatagctaa   gtctactttt  gccacactgc  atatttattg  ccagaagtgg  ctagtgtcca  120
ctcccttcca   gcagggcacc  aagagccctc  cccaatcttc  atacaaacat  tctcaccaat  180
tactgcagaa   tagaggttag  gntgagcttc  caggattcta  atggatgac  ggctgtgaat  240
gtcctgacgc   tttcgaattt  caatctgtgg  agtagaaata  aaacataaag  tttcaccaaa  300
gcatgaataa   tttacttggt  acactatcaa  aacaaaaact  agtgtcctac  cagcttcaca  360
atttgatcat   gaatagagct  tccccaatcg  taaaaatggt  cat          403

SEQ ID NO: 43   moltype = DNA  length = 403
FEATURE        Location/Qualifiers
source         1..403
                mol_type = other DNA
                organism = Cannabis sp.
variation      202
                allele = n = g/t

SEQUENCE: 43
gatattctact  tgcagatgta  cgctaaagat  ttgtttcaga  aagactttct  ttagaggaag  60
aagaaaatga  agatgaaaga  gacaagagtg  agtgattatt  aagagaacca  aactcaaaac  120
tcaagagatt  acaaaaaaga  aaagaaaagt  tgctagtgtg  acacgccaaa  gtaaaaaaag  180
agtgtagaga  tagagtgtgt  tnggggtgat  aattaattac  tatgaagacg  acttctgatt  240
atattatcagt  atatagattt  aaaagttttg  gcagaaaaag  gtaattatgt  attgaccgtt  300
acatccacgc  cgtccaatcc  ctatttcttt  tgttctcagc  tttggtcaga  ctaaaattta  360
ctccctctct  ttaattactg  tcttatcttt  ggaaactagcc  cat          403

SEQ ID NO: 44   moltype = DNA  length = 403
FEATURE        Location/Qualifiers
source         1..403
                mol_type = other DNA
                organism = Cannabis sp.
variation      202
                allele = n = g/c

SEQUENCE: 44
ggcagacaaac  ataacccttc  tatttcattt  tcctttcttc  ttgtttttgg  tctctctctc  60
tcttccacat  acaatccaag  ttctgtgaaa  gtacacaaat  aagaaaataa  caaagattta  120
taataacgta  ctatagaggt  ttaatatata  gttataatct  gatgatgatg  cggaaaagtga  180
tgggttcato  ttcttctctc  tnttgttttt  attatcgtea  cataataata  acaacaatag  240
tattgttatt  ttttattaat  ccatcgtggg  gtttgataaa  gttaccgcca  aatcagacag  300
ttccggcagt  gatcgtgttc  ggagactcga  tcatggacac  agggaaacaac  aacgctctta  360
agactctggt  caagtgtgat  ttcgctcctt  atggtgaaaa  ttt          403

SEQ ID NO: 45   moltype = DNA  length = 403
FEATURE        Location/Qualifiers
source         1..403
                mol_type = other DNA
                organism = Cannabis sp.
variation      202
                allele = n = t/c

SEQUENCE: 45
tttgatatatt  taggtacgga  tcgggggaag  agaggaagtt  gttggagatg  gcgccgacta  60
cgattcggaa  agcgattggg  gcgggtgaag  accagactag  tattggaata  gctaagggtg  120
ctagcaacat  ggcgccggag  cttgaggttg  cgattgtcaa  ggcgactagt  cacgacgatg  180
atccggctag  cgagaagtac  anaagagaga  tcctgaatct  cacatcttac  tctcgagggt  240
acgttcacgc  ctgtgttacg  gcgatttcga  agcgtttggg  gaagacgcgc  gactggattg  300
tggcactcaa  ggctttgatg  ctgatccatc  gtttgcttaa  tgaaggggat  cctttgtttc  360
aggaagagat  cttgttcgct  accagaagag  ggaccaggct  tct          403

SEQ ID NO: 46   moltype = DNA  length = 403
FEATURE        Location/Qualifiers
source         1..403
                mol_type = other DNA
                organism = Cannabis sp.
variation      202
                allele = n = c/g

SEQUENCE: 46
atctcaaatg  aattggattg  acaatgatgg  gttatttctc  ttgtccaaca  agtctcaatt  60
cgctttcggg  ttcaccacca  ctaaccaaga  tgtcacaaaa  tttctgctag  taatcgtcca  120
catgggaagc  caacgagtta  ttggacagc  gaatatagac  accccagttg  ccaattccga  180
taagtttggt  ttcgatgaaa  anggtagagt  tttcttgcaa  aaaggagcaa  gtgtggtttg  240
gtccattgat  actggtgcca  aaggagcttc  tgctatggag  ttgatggatt  caggtaattt  300
ggttctgttt  ggagaggacg  agaatagtaa  aatatgggag  agttttgacc  atccaaccga  360

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taccctttta tggggacagg aatttgttga agggatgaga ctt 403

SEQ ID NO: 47 moltype = DNA length = 403
FEATURE Location/Qualifiers
source 1..403
 mol_type = other DNA
 organism = Cannabis sp.
variation 202
 allele = n = t/c

SEQUENCE: 47
agtggatcat ctactgaaca gaaggttctg ttatccatag taaaatttgt tatgagagca 60
ctgaattttt cttcgtttct ttcatTTTgt tgatgtggcg gtgacaaaca tttatttaat 120
gggtactctg tcgtatctga tgctgttgac atttttttta tgctttcatc agctatctgg 180
aaagtctgag gtagctatgc cngaggacct tacgtgagta gtgatttgtc ctttccaatt 240
gttcttatct tgagcttatt gatattgata atttgttctt tttctcttta ctataatata 300
atctttttct tcaaatcttt ttatagttag tggcttagct atataatctt tctttattcc 360
ccttcctcat tatggtttgg tggcaaatTT gacagaatcg att 403

SEQ ID NO: 48 moltype = DNA length = 403
FEATURE Location/Qualifiers
source 1..403
 mol_type = other DNA
 organism = Cannabis sp.
variation 202
 allele = n = c/t

SEQUENCE: 48
aattaaaaat ggcattcttc gcctcagaaa ctctcacttt catcttcttc atctctctac 60
tccatttggg ttcacggggg aggattcttt ccgacgagtc tgaccaaacc caacagcctc 120
ttccctttca ataccataat ggccctcttc tgtttggaaa aatctccatt aacttaatat 180
ggtaacggaaa cttcaaacca anccaacgag ccattgtctc cgacttcatt acctccctca 240
cttcatcttc taaaaccaac acagaccaac catcagtcaa cacgtggtgg aagaccatcg 300
aaagctacca ataccaccac aagctgagca actccgtctc gttaggatcc cagttcatcg 360
acgagaacta ctccctgggg aaatcgttga ccagtcaaca aat 403

SEQ ID NO: 49 moltype = DNA length = 403
FEATURE Location/Qualifiers
source 1..403
 mol_type = other DNA
 organism = Cannabis sp.
variation 202
 allele = n = t/c

SEQUENCE: 49
gagtatTTtc aagtttaaga tctctatgac atatttgctg ccagaagtca tttattaaat 60
aattagacaa tgacaaagtt gataatcaat gcagttcttt agagatatgt atgtatatct 120
atcacaaaag gatagttcaa acaaaattct taccattgaa tgacagtaac tgactcctga 180
tattagtTgt tgaaagaaaa ancttgctg attcaataga acaaaaagat gcttaagtgc 240
ttacattcat taagatttca tttggtgtaa aaacataagt ttagttagaa atttctctca 300
ttacctcatc ttactaaat ctaccagcag tgcataattct ttcaaagagc tctccaccac 360
cagcatactc catcacaatg gccagatgag ttggagttaa aag 403

SEQ ID NO: 50 moltype = DNA length = 403
FEATURE Location/Qualifiers
source 1..403
 mol_type = other DNA
 organism = Cannabis sp.
variation 202
 allele = n = g/c

SEQUENCE: 50
cttttttatt atctgagatt tagtaagtgc ccatatgttt gtagatcgtg ggatttagtc 60
gattagggta ctagtTTTga ggtttttgtt ttatcaatta gttgcttcaa gaaactcact 120
ttcttttttc gtttttatgc tagtagaaat ggacttgaac taggatctcc aagcaggcgt 180
tgtaggaatc actgagagtg tncatgtcgc tcttgttgag gaagctccaa cggaaggtag 240
ccaggctatg gttctcgttg taatgcctca gcaaggggca actctagaaa ttgtcgtaga 300
ggatgaacct actacagagg aaggaccctc caaaaaggac aatggaaaaa agaggggccaa 360
gacaccccca gccgcacagg cttacaatga ctccatttgg gag 403

SEQ ID NO: 51 moltype = DNA length = 403
FEATURE Location/Qualifiers
source 1..403
 mol_type = other DNA
 organism = Cannabis sp.
variation 202
 allele = n = c/g

SEQUENCE: 51
gtgggtttac atacccagc acgcatctga ttgagccac cattgggtatg aactagaagg 60
taacctcgag acccggcagg ggctgatata acatcaaaga taattaatac aattcacata 120

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tctgaataga aaaataatac cattatgttc cttacgtgta taattaagac ttgggtttgc 180
acatggcacg aaatctcgat tngatggagg ttcccatagc tttccattt ccaaaattcc 240
actcgcctta tccaactgtc acgtctaaag gagtgaaaat gggaaaaaaa tgcattgattt 300
gttcgcatta ttacaaaaga agggtagtta taaatacga tacccttctc ttctttttaa 360
tcaaaatttt cccttcttat tatgggaagtc tctcctacta aag 403

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SEQ ID NO: 52      moltype = DNA length = 403
FEATURE           Location/Qualifiers
source            1..403
                  mol_type = other DNA
                  organism = Cannabis sp.
variation         2
                  allele = n = c/g

```

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SEQUENCE: 52
agctcacaaac aaagtgaaga atgcggggac caccttggat tccacgtcga caatattcga 60
caatgcctac  tacaactac  ttctgcaagg caagagtatt ttctcttcag accaatctct 120
actcaccacc  ccaccaacta  aggccttagt ctccaaattc gctgcttcta aacaagactt 180
cgataaagcc  ttcgttgagt  cnatgatcaa gatgagtgc  atccatggta gtggccaaga 240
aattaggctc  aactgcaagg  tcgtttaatta attaattaat tagttttttt tctttcaata 300
tattattata  aatttctcca tcaatcgatc gtttaataat tgaagaagag gagtgataag 360
atgaacaaaa  ccaattaaga  tgcattgtaat gattagtaag ttt 403

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SEQ ID NO: 53      moltype = DNA length = 403
FEATURE           Location/Qualifiers
source            1..403
                  mol_type = other DNA
                  organism = Cannabis sp.
variation         202
                  allele = n = t/c

```

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SEQUENCE: 53
gaaccttgtc  tgaaaaatag  ttcacatgta attagagata ggaaaaacaaa gttccagcag 60
tgaataaaga  ggaaaattac  aaactcaaat tgatctcttg aataccatac cttctgacca 120
ccaacaggct  tcacacaatg  cacaccgaag aatttcgtca caagggaatt ttcataccga 180
cagacatggt  gatgataact  anaaagcatc cgtaaatagca cctgaaataa tattacaatt 240
aagttatttt  acctgacctc  tcaaaacccg aaattatgct actcaaatgg cttttgagac 300
agcataagct  ccataataaa  gcaaaagtc  aagaagtagg caacaaacct tgacttctga 360
tttcttcacc  gtctttatca  taaacctatc atcttgtgtc aag 403

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SEQ ID NO: 54      moltype = DNA length = 403
FEATURE           Location/Qualifiers
source            1..403
                  mol_type = other DNA
                  organism = Cannabis sp.
variation         202
                  allele = n = t/g

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SEQUENCE: 54
ctataaatat  actttcccaa  aagggaactat tcgctttaat ctaattgtaa tagtacaaat 60
atgtttgttt  cacttttttt  ctttattgta gaaaaaagta ccccttttat ccgttgaagc 120
attgcaccaa  ttgcataatc  ttatcttcgt cctagccatt gtccatgtca cttctgtgtg 180
tctcactggt  gtgtttggag  gngtaaaggt aagtatatct tccaaattca tatgcttctc 240
tctcttttat  actctactac  ttgttttttg ttcgtttatg agaaatattt ctacacaact 300
agatacgtca  atggaaaagt  tgggaggatt ctattgcaaa tgagagttat gacactgaac 360
aaggtaaagta  ttctcagtac  cacctttgac aatattatag aaa 403

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SEQ ID NO: 55      moltype = DNA length = 403
FEATURE           Location/Qualifiers
source            1..403
                  mol_type = other DNA
                  organism = Cannabis sp.
variation         202
                  allele = n = c/a

```

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SEQUENCE: 55
aatgtcaatc  tctatcgga  cgccgcggtt tgatatctta gccattgctg acacaggcag 60
cgatctgacg  tggactcagt  gcagcccttg caaaaaatgt tacaagcaag tggtctctct 120
cttcaaaccc  aactcttcaa  aaacatacac agatgctacc tgtgattcct ctgtttgtaa 180
gtccgccacc  ggagctaaaa  cntcttgtct ctccctcgac gattcatgcc aatactccgt 240
atcttacggc  gaccaatctt  tctccaacgg taacattgct actgacgttc tcaccctctc 300
ttccacctct  ggaagaccgc  tcaccttccc caatttcac  atcggttgca gccacataag 360
tgatggtata  tatataatat  taattatggc gtgataacct tat 403

```

```

SEQ ID NO: 56      moltype = DNA length = 403
FEATURE           Location/Qualifiers
source            1..403
                  mol_type = other DNA
                  organism = Cannabis sp.
variation         202

```

-continued

allele = n = g/t

SEQUENCE: 56

cccttggttg	aaccgctgag	caagctctac	agttgcactt	ttgtcaccat	cagtaggcca	60
gccaatctca	ccgatgatta	tggacaagtt	tccaaatcca	ttttctgca	gagccatac	120
cagtggtgca	tggtttgcat	ccaagacatt	ctggtagatt	tttccattgt	cgtttatggc	180
agcgggaatag	ccatcaaaga	angcgaatc	aacagggaaa	ttgggatcgt	tgtggaggct	240
gataaaagga	tagatgttta	cagtaaagga	acctccattg	tcgcttaaga	acttaacgat	300
ggccaccatg	agatotttta	tgtctgttct	gaagtcaccg	tcggaagggt	tctcactcga	360
gctgccatat	acatcagcgt	ttaaagggat	agtgacttta	act		403

SEQ ID NO: 57 moltype = DNA length = 403

FEATURE Location/Qualifiers

source 1..403

 mol_type = other DNA

 organism = Cannabis sp.

variation 202

 allele = n = t/c

SEQUENCE: 57

agtaatttta	aaagtctctg	ttgttttcag	gaagaaaatt	ggtgtaataa	aagtaggcaa	60
tgagagttac	agaagagaat	cagagtcaga	tgagttaaat	acaaaaaaag	caaatctttc	120
agcttcaagg	aaggaaaagg	ttcagctacc	aataatccca	aattatgagg	gtaagaaatt	180
ccctattggt	gaatttttaa	gncaaccctg	tgcaattgaa	gccctcctca	ataccaatgc	240
cttaaaaagt	ttccaatgtc	ttctccttaa	cacttacagg	taggctacct	ttcaacttca	300
attcaaccct	ttttctttgt	aaaatctaac	ttagtatatc	aaagtagttt	aagctagaat	360
ataatagaac	tattttcacc	attattagaa	gaagttggga	ttg		403

SEQ ID NO: 58 moltype = DNA length = 403

FEATURE Location/Qualifiers

source 1..403

 mol_type = other DNA

 organism = Cannabis sp.

variation 202

 allele = n = t/c

SEQUENCE: 58

gtttgaatat	aaaagctagc	agcaccttca	atgattatta	tgttctaacc	agaaagaata	60
gctacaaact	ctgaactttt	caagtacaat	gtatgtgtat	gtttgactaa	tcaatcttac	120
caaaatattt	tgtggcaatg	tggaacgaat	tcctagtgtg	ctgctctttc	cccaagtgtc	180
acggttgagt	tttcctttca	tnaatgattc	aggagtgtca	tttctaaaaa	cctcctgagc	240
attggcgtaa	gcagttgcgg	tatgtcact	tactgttttc	atcaatccaa	tcgcacata	300
gctcctctct	acagggctca	cgtaataccc	catatcctgt	agaagaagtt	tccacagaat	360
tagcgaaaaa	attatttccc	atcaaagcca	gaaagttag	gtg		403

SEQ ID NO: 59 moltype = DNA length = 403

FEATURE Location/Qualifiers

source 1..403

 mol_type = other DNA

 organism = Cannabis sp.

variation 202

 allele = n = g/a

SEQUENCE: 59

taagctttct	tacaatgtgc	tgtccaatga	aagccgcaac	agtagccaca	gcaacaaagt	60
aagcagctgc	aaataaaatca	catcacgttt	tagtaaaagta	ccatattcgg	ttctgctgta	120
aagggtgttca	acagaacaaa	tgaaaattag	ataatgatca	aagaagagta	tatgtaaagg	180
acactaacca	tagggaactg	gnaaacgttt	tagaaggtaa	tattctacaa	cagacataga	240
agaggagaac	gtcattgcaa	atgtggctgt	tgcactcgag	acctacacag	aagtaaccaa	300
ctaaattata	actgattcaa	ttgcataatt	atcaataaat	taactcttta	aggttgaatg	360
ctattttgct	tttactatgc	tagggaaaaa	ctcaatcccc	tta		403

SEQ ID NO: 60 moltype = DNA length = 403

FEATURE Location/Qualifiers

source 1..403

 mol_type = other DNA

 organism = Cannabis sp.

variation 202

 allele = n = a/c

SEQUENCE: 60

attcaagaac	ttgggatgaa	gatattattc	gagacctttt	tgaggataga	gatcagaccc	60
tcattttttc	tattcagcta	agtgaaaatg	ccttgaggga	tcattggtgc	tagaagtttg	120
agaacgatgg	aggttacatg	gtcaacaatg	cgtacagaca	tcttcagggtg	ctcaagggag	180
cttgccctgc	tgcaaacccg	anaaatcttt	ggcatactct	ctggagctct	aaagtgcccc	240
caaaggctcg	caattttttg	tgaggggcag	cgctcagggtg	tttgccact	tgtgtacaat	300
tgcaagaagag	acatgtacca	gtgagtataa	tctgcccgtg	ttgcaatgtg	gatgatgaga	360
ctattttcca	tgctcttggt	gattgtcctg	tcgcgagatc	gtg		403

SEQ ID NO: 61 moltype = DNA length = 403

FEATURE Location/Qualifiers

-continued

```

source          1..403
                 mol_type = other DNA
                 organism = Cannabis sp.
variation       202
                 allele = n = g/a
SEQUENCE: 61
ccgcacaaaaa gaaaattata ctcataaagt tgatgaaaaa gaaagggtcc aaagcagtat 60
aattcatgtt aattaagttg tgatattatt gttttacatt ctcacccaca cagcgaaagt 120
gtactgataa gcagaacttg ttatgctcca ctctgactcc agggatggat tttgtttctt 180
ccacaaaaat agtgtacacc tntatgcatg cacatcttta taattattta tgtctatata 240
tatataggag agaaaattaa attataaact aattaattaa ttataattac ctcatcaatc 300
attggtgaaa attctgccgc aggttgatac ataactgcat tcttaatctg agtatcaatt 360
aaaaatataa aataaacatt aaaattttta tgtaaggtc aac 403

SEQ ID NO: 62      moltype = DNA length = 403
FEATURE           Location/Qualifiers
source            1..403
                 mol_type = other DNA
                 organism = Cannabis sp.
variation         202
                 allele = n = c/t
SEQUENCE: 62
accaacaaag agtgccctcag catcggggct aaaggatatt ccagcaatct cgccaaatat 60
gtctatctct tgccctttag agtaccctga ctgtgtatcg atgatgtgaa caaagtctgc 120
aggttcagcc atggccagaa accggccatc gttagtgaac ctaacagctc ttattgcccc 180
cattcttccc ttcaggacgg cnaaggactc tgacaggttc cttatgtccc acaatctgca 240
ggtggtgtct tggttcccgg tagccaaaat acgtccatct ggatgccaaag cagaggcaaa 300
agagtagtcc aagtgccctt tgaggctccc agtgatctga gaggaaggaa attagaatca 360
ggaacagaga ggaatggaga ccaaacaaaa tcaaggataa agt 403

SEQ ID NO: 63      moltype = DNA length = 403
FEATURE           Location/Qualifiers
source            1..403
                 mol_type = other DNA
                 organism = Cannabis sp.
variation         202
                 allele = n = c/t
SEQUENCE: 63
gccaaaaaact cgcaacaaag tcaagtttgt gtactcggat gacattaaca cgaagaaaat 60
aatggaggat cactttgata tggatcaact ggagtctgca tttggtggaa atgatactac 120
aggtttcgat attaataaat atgcagagag aatgaaagag gatgacaaaa agatgcctgc 180
tttctggact aaaggaaatc cncacgctcc agcaacctca gaacctgtcc cgaacaatga 240
cactccatct tcggactcta ccatcaagtt agaagatgat gcagctgatt ccatcgagaa 300
gagaaatgga tcggaagggg tgctcccagc ccccaatcac actatgctca ctgttgacag 360
tagcagaaat cctactaaag aggttcagta attaaggatc ctt 403

SEQ ID NO: 64      moltype = DNA length = 403
FEATURE           Location/Qualifiers
source            1..403
                 mol_type = other DNA
                 organism = Cannabis sp.
variation         202
                 allele = n = t/c
SEQUENCE: 64
agtttccgcc tccagggtag agtaggcagt ggtgaattct gatctcatag cccatctacc 60
cgtatcacca ttaattacca ccgcagcaag accagctgct ccttccctcc aagaggcatc 120
agtcatcaag acaatcgagg cgttgctgac tctccgacaa ggattaaagg cagtcgaacc 180
aggaggatct gcagcagagg cngaaataca gaagtccttc cattgagcag caacctgttg 240
gaggagcaaa gggaggaata ctgaacttcc tttgtataac aacttggtcc tttcagccca 300
aactgcacta aaaagatagc ccatgtagt gatgacagag cttcggtcag gagtgggaaa 360
gatggatact aaattctcca cactactctg cattgtctca cca 403

SEQ ID NO: 65      moltype = DNA length = 403
FEATURE           Location/Qualifiers
source            1..403
                 mol_type = other DNA
                 organism = Cannabis sp.
variation         202
                 allele = n = a/g
SEQUENCE: 65
tccttgccct gtgggtaatg cctcaatcag atcaatctcc catattataa atgggatatg 60
ggaggtcacg atttcaact cggttggttg tatgcaaggt atatgtgaaa acttttggca 120
tatcgcatth ctcaacaaat tcgtgtgtgt ctttgttgat tattggctag taatatcctt 180
aacggataat cttcttccat anactctgcc aacctacatg atccccacag aagtcttcat 240
gaatttcctt gattatttca ttgcttcta ctggtaacac acatcgtaag agaggcatgg 300
aataacgtct tctgtataac ttcccctcga ccattatata gcgaggtatt gagtataaga 360

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gtttgtgtgc ctcggttttg tctgtaggaa gttctccact aac 403

SEQ ID NO: 66 moltype = DNA length = 403
FEATURE Location/Qualifiers
source 1..403
 mol_type = other DNA
 organism = Cannabis sp.
variation 202
 allele = n = c/g

SEQUENCE: 66
attaggaagt ctcceaagct gcagaacagc atgttagtga aattgggaaa ttctctctcc 60
aatacttcggt tactacccat cgtcgtctga gttctcagaa cattcttcca tcgcttcagc 120
agtcataaga gcagccacag ccttgtgaat tgetacatct gctctgtaaa ccattctctg 180
agctctttct ctcttaagct tngcaatgtt aatagcatgt tcagcagcgc aagaagcatc 240
acgcaacctg aactcatcaa gatctggacc atcaaatttc tcaatgatat gcctctgaga 300
tctctctagg tggaaatgct gtggactctg ccactcagaa gaaccaatat tccactgatg 360
atgctcatgt cttttttcta acattctacg atcatataaa gcc 403

SEQ ID NO: 67 moltype = DNA length = 403
FEATURE Location/Qualifiers
source 1..403
 mol_type = other DNA
 organism = Cannabis sp.
variation 202
 allele = n = c/t

SEQUENCE: 67
tttcccgttt cttaataact atgaatctgc ctaaaatttg tctttgattg ctcaccgaga 60
atcatcacaa acccgacaag agaagtatat ttgtgtaaca tgtgaaaagt agaaatagaa 120
ttgtttgttt agtatgtaga atcgcaaaaag gatctctgag agcttaccta tcattcacat 180
ggactcttct cgtgtcaaga anttctagta aaggaagcaa tggcatctag attttccttc 240
atgaaccatt catgaggata gacagagtta gctgattcaa gatgaagatg ccaaaccatg 300
attctagagc aatcttgctc cagtccttcc atttggtctc catcttccac aatttgttca 360
ataaatgtgc ttacctgaaa atgcaaatca actagattag tcc 403

SEQ ID NO: 68 moltype = DNA length = 403
FEATURE Location/Qualifiers
source 1..403
 mol_type = other DNA
 organism = Cannabis sp.
variation 202
 allele = n = a/t

SEQUENCE: 68
tgacacaacc tgcattgtga ttgatatatt acctcaagag aagccacctg ctccgctgcc 60
cccacctaag aagcaggga aaggagtgtt taagtccatg ttctgtaaaa agtcaactga 120
atcaacttct tatgttgaca aagagtacat agagccagac gtggtggagg aattatttga 180
agagggtctc gctatgcttt cngaaaaggt ttgtgcttgaa aaagtcaact tcaacttaac 240
atgattgcca tgagaaaagt ctctctctct ctctctctct ctctctctct ctctctctct 300
ctctctctct tttcttttc tttttagtgt cacttttctt tctttattct tgttctgttt 360
tcgttcttcc tgtttaactt ttggggggtt gacctatgaa cag 403

SEQ ID NO: 69 moltype = DNA length = 403
FEATURE Location/Qualifiers
source 1..403
 mol_type = other DNA
 organism = Cannabis sp.
variation 202
 allele = n = g/a

SEQUENCE: 69
gtcgcgtcta agcaaaaact acccacctga tcttttcttt catgggtttc ttcacgaaat 60
caacaaaaac ccattctctcc aatcttccat tttattcata gcatgcgcgc aacgagccca 120
cgccgccact tgcttgcccc ttcaacacaa actcgtaagt aaagagttgg cctggcggtta 180
gctgtctcca tggctattgt cnactcggag cgggtgtttc ttcttctctc cgtggttgtt 240
tcggatatgt tgggtgctgag tatggggggt tacgtgtccc cgctgagctc agagaaatct 300
tacgtgtcag ccgttgagga ccaggaatg aagagcccaa atgtcagagt tggtttagaa 360
gcttggaact tctgcaatga agttggagct gaagctcctg gta 403

SEQ ID NO: 70 moltype = DNA length = 403
FEATURE Location/Qualifiers
source 1..403
 mol_type = other DNA
 organism = Cannabis sp.
variation 202
 allele = n = c/a

SEQUENCE: 70
caataatttt gacacaaaac acattttctt tttgttgttg ttttttaggc ggagaatctt 60
atgcttgaga ggggagaaaa caaagaggta aaatccaagt agtcgaaatg tttgttgact 120

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tgaattctat cccctcttgat gatagttact aaatgtgatt tgtttacgta tatagtacct 180
tccaattgaa gggtttggcgg cnttcaataa ggcaacggca gaattattgt ttggagcaga 240
taaccaggta atcaaagaac aaagagtaag cttgaatttc gaatttgagt caagaaaaat 300
taattctttt catggtagct tgattatttt tgtcatattt tctctgcttt atatcagggt 360
gcaaccgttc aaggactttc aggaacaggt tctctccgtc tcg 403

```

```

SEQ ID NO: 71      moltype = DNA length = 403
FEATURE           Location/Qualifiers
source            1..403
                  mol_type = other DNA
                  organism = Cannabis sp.
variation         202
                  allele = n = t/c

```

```

SEQUENCE: 71
atgagataat taagcaatga gcatgcatgt taaatcaatt ttattaatta tacatactga 60
atttgttggt atcatcgtaa tataatagat attaattaat ttaaaaaata aattataata 120
ataataataa tatttacagg gatatacact ttacttggtg caagcccatc gaagtgaact 180
aaggcccca aaatgcgtag anaatattaa tcaccaatgc gtagaagcaa ccacaggaga 240
ggcctcaatc ctatacgtgg ggctctacct tgttgctctg gggactggag gaatcaaagc 300
agcactccca tcattgggtg ctgaccagtt cgatgagagg gatccaaagg agtcgagtc 360
attgtcgagt ttcttcaact ggttcttggt cagcctcacc att 403

```

```

SEQ ID NO: 72      moltype = DNA length = 403
FEATURE           Location/Qualifiers
source            1..403
                  mol_type = other DNA
                  organism = Cannabis sp.
variation         202
                  allele = n = a/g

```

```

SEQUENCE: 72
aattacccta aaaactatat gtttgagaaa gaaaaaataa aacaataagg tgtgtgagtg 60
tgacaagact tattcaaatg tgcaggctct tattctaggg ggactgata caacaacagt 120
aacaatgaca tgggcattgg ctctacttgt caacaaccaa gacacgttga gaaaagcaca 180
agaagaatta gaccaagtgg tngggagaga aagacaagta aagcaatcgg acataaacia 240
gctggtttat ctccaagtgg ttatcaaaga aacaatgcgc ttatacccag ccgcaccact 300
cgactcccc caccaatcgg ttgaagactg tactgtgagt gggtaccacg ttccagctgg 360
cacacgcctc ctctcaacc ttccaagct acaacgagat cca 403

```

```

SEQ ID NO: 73      moltype = DNA length = 403
FEATURE           Location/Qualifiers
source            1..403
                  mol_type = other DNA
                  organism = Cannabis sp.
variation         202
                  allele = n = c/t

```

```

SEQUENCE: 73
tttattatat gtactactgt aatgccatga gtttaatatg taaatttttc caggcattat 60
acatgcttaa gaaatttaaa caaggagtcc ccttatttta tcttttttca gtaggatttg 120
tgctggttgc caaagtgaga ttgggttttg acggatctcg aactgtttta atgcattttg 180
gcatccagaa tgcttccgtt gncgtgcttg ccatctaccc atttctgatt atgaggtatt 240
tttaaacctg tataatatga gctgtctaag caatgggtgca ttggtataag acttctagtc 300
atttaactgg tggttatttt cctttttgca gttttctaca tctgggaact atccttacca 360
taaaacctgt tataaagaga actatcatcc gaaatgtgat gtc 403

```

```

SEQ ID NO: 74      moltype = DNA length = 403
FEATURE           Location/Qualifiers
source            1..403
                  mol_type = other DNA
                  organism = Cannabis sp.
variation         202
                  allele = n = t/a

```

```

SEQUENCE: 74
atatatatat agataagaag acagctaatt gttttttttt cttcttcttc ttcttcttc 60
ttcaccgccg taagtgtgtt cgattgggtt aaaagataaa ggcaaaagta gtgataaaag 120
tttgatgata tgggtaagaa gaagaagcag aatcagaaaa caaaggaaat tttagtagcc 180
atagctgaag cagcttcacg tntaaaagga gaacctcatc atcatcatca tcatcatcat 240
catcagtcga cgccgagaaa gagaggtagg cctcggaaaa ttagcattga aaaaagtcta 300
cagaaaaaag aagaagaaaa agaaatagta gttgggtgat atgatggat cattattatt 360
agccagagtc aatccaagaa agctaaaatc gatgaacttg aag 403

```

```

SEQ ID NO: 75      moltype = DNA length = 403
FEATURE           Location/Qualifiers
source            1..403
                  mol_type = other DNA
                  organism = Cannabis sp.
variation         202

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                allele = n = a/c
SEQUENCE: 75
tcctgaattg ttgtccactg aaaccgtccg gcaactgcat gctaccatcg aaaaagaatg 60
ggatgccctt cgaagggtcag catgtcaaac ggctgctgga cgagcactgt ggaacatgt 120
cactcatgac cctttggcag ccttacttgc tggagagact tacctaagaa gccttcatga 180
gaagataaag aaagactgtg cnaacaatgc aagtgaatc tctggagtta ttcttgacgt 240
tagaactctc tggttcgatt caaaacttga agcagccctt cattccttac atggcacaga 300
aacacaagtt gttctccttg gtgcaggtaa atgataaaca ccaaatttta catatatata 360
cagatagcat aattgtactt ttacatatct taattagtag tac 403

SEQ ID NO: 76      moltype = DNA length = 403
FEATURE           Location/Qualifiers
source            1..403
                  mol_type = other DNA
                  organism = Cannabis sp.
variation         202
                  allele = n = t/g
SEQUENCE: 76
atatcataat gcaacattac taattatttt gatgactaat ttatagatac aagtaactga 60
aattgattta ttgtattctc agaattctag ctgtgtacca gtaaaactata tggaaaatca 120
accgatataa aatgaaaata tgagaggggt tctgattgat tggctttagg aggtacttaa 180
attgctgctt tctgttgggt antgtactca atgaattcct cttaattgga attctctgac 240
catttagtgc aaattacagg ttcaactaca gtttgaatta atggatgaga cgttgtatct 300
catggtcaac ttgattgata gattcttagc tgttcaatct gtgccaagga ggaacttca 360
gttggttggg gttgtgccc tacttctagc ctgcaaatac gaa 403

SEQ ID NO: 77      moltype = DNA length = 403
FEATURE           Location/Qualifiers
source            1..403
                  mol_type = other DNA
                  organism = Cannabis sp.
variation         202
                  allele = n = a/t
SEQUENCE: 77
tagacgaaag gaaagtagta ggtacaaatt tatagaacaa aaaaaaagaa cataatgatg 60
cgaatgtatg gttataatgt ttccaataaa ctagaaaata caatactcac ttgataagcc 120
agtgaactc ttctccttgc tcagaacaaac cataatctga agtccacgag tggcctgaat 180
tttttttttt tttggacaaa tnattatagc aaacaatcag ctgcagtttg aacttatgca 240
cacatttaat tagttgtgtt aaaaggtgta tatagactaa cctatggtaa acttgtggaa 300
tcgcagcatg tccattacac caacatgagc tagtgccacg ccaaaaagat caggtctctg 360
taacaaaatg aaaaaatttg ttttgaatc aaccacgcat aag 403

SEQ ID NO: 78      moltype = DNA length = 403
FEATURE           Location/Qualifiers
source            1..403
                  mol_type = other DNA
                  organism = Cannabis sp.
variation         202
                  allele = n = c/t
SEQUENCE: 78
caatattggg tgtgatctat ctttgcgatt gggctctatg agtggtgaga acaagcaacc 60
ccaggctatt gcagacgtta atccccgaga gggaggcgaag ttttgtgacc aattaccatt 120
attggataaa gggtgctcca tgtttctctc tagaggcaat gcttgccagg cattaagttc 180
atacaaaagt gtagaagcag cngtgcgaaa acgaaagcca tcttttgata atggaatcga 240
cgatcagcag ttttgaattt tgaagccaag tcctcctttt ctccacttaa atggaaaaat 300
gagaaatgct ggatcataag ttctgatttt ttcttacttg gttaactttg tggtgcaatt 360
ttgtgtactt cccggatgtg ttatgcaccc aagagctacc aga 403

SEQ ID NO: 79      moltype = DNA length = 403
FEATURE           Location/Qualifiers
source            1..403
                  mol_type = other DNA
                  organism = Cannabis sp.
variation         202
                  allele = n = a/g
SEQUENCE: 79
gtacataaaa gacagaaaca cctttttttt ttttgaaaaa tacacaagta aaaaaaatt 60
gtagaaaaaa aaaatccata aaaagaaaaa aaatatatat attataaga aaaaatacaa 120
ttttaagtag tgtaaaagat atttataagt aacctggggt ttgaaattat gcaatcgcaa 180
ttaaatagag tatggcagca gngcagattg attgactcag gtatgtcatc ttctaaaaat 240
tcaataacta ctgatttctt ttaaatgaaa ccacacaatc caactcaaat aaacaaataa 300
aaaaatacat ataatgtaat ggtcaactac tctacagct atctagaaaa tgatacatca 360
aacatggctg catctccaga tcaagagtgg tcatcctccc agc 403

SEQ ID NO: 80      moltype = DNA length = 403
FEATURE           Location/Qualifiers

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source          1..403
                mol_type = other DNA
                organism = Cannabis sp.
variation       202
                allele = n = c/a
SEQUENCE: 80
cgaatcctct tcttctctcaa cgtctcagag cttcacacaa tggcgctttcc cactacccaaa 60
ctgtccaggtt cgtaatatca atatacgatc cgaatccgat ttaacacccg aggaaagtga 120
tcagactggtt caccattacg gcccgccacc tatttctccc acgaacctcc aagaagtgtt 180
ccacgcagct  gagctccaac tnagcgtcgg gtcggattcg gatcagttat ctgccctaca 240
acttctagag agatctctgg ttcccaaccc gccgaccgat cctgaatgcc cgccggagtt 300
gatgcgcggt gtagtgggga gtttaatttc tcaggttggg gcgaaacctg cgtcgaagat 360
tttgctagct ctctgcttgg cggagcgaaa ccgagagatt gcc 403

SEQ ID NO: 81      moltype = DNA length = 403
FEATURE           Location/Qualifiers
source            1..403
                mol_type = other DNA
                organism = Cannabis sp.
variation         202
                allele = n = g/a
SEQUENCE: 81
tgatcttcag aagcaggttg ccgagatggc tggaaatgag aaacagattc ttagtattga 60
agaagaaata agaagattac agaatttggg ttggtgatgt ctacaagatc ctggactaga 120
agatcaaatt tcttctggca gtaacattga gtgcttagaa atgttgctga ggaagcttct 180
agaaaattat gcaaaaatttt cnaccataag acctgtactt ggtgggtggaa ttgatgagct 240
gcagactgac gtgatgactg tagaggcagc taagaactta agcaaaaccc atgctgggga 300
gtccgatgaa gttatcatga agaaagagct tgaggaagct ttgcacgagt tgattcttgt 360
gaaggaggag agaggtgtat ttgttgagaa gcaacaatct ttg 403

SEQ ID NO: 82      moltype = DNA length = 403
FEATURE           Location/Qualifiers
source            1..403
                mol_type = other DNA
                organism = Cannabis sp.
variation         202
                allele = n = g/a
SEQUENCE: 82
tgtgcaccaa accaaggaaa tggagttgaa gaaagacaag tttgtcaggt gaactccttt 60
acttcattat ctttctttta caatcatgca tgcttaactg gggtttttca ttttctccaa 120
ttgtagattt taccatgaag aaaacaataa agatagtaac ttagagaagc cgttaccagt 180
gttcaaahta gcagcagcag ancccttggg tgaggcagaa ggaggtagaa gtatgtagag 240
tagaaacaga atttatattc caaaaatttg aagggtttaa gttttccag aaaatgagaa 300
tccatggagg aagaagattc ttgaccctgg aagtgatatt ttcttgcaat ggaacagagt 360
ttttctcttc ttttgcttgg tagcactttt tgttgatcca ctg 403

SEQ ID NO: 83      moltype = DNA length = 403
FEATURE           Location/Qualifiers
source            1..403
                mol_type = other DNA
                organism = Cannabis sp.
variation         202
                allele = n = a/t
SEQUENCE: 83
aaagctcaag aatttccaat cctggcgctg aaactctggt atcctcccta caaacgggtt 60
actctcaacg ttttaactca caagcttagt caacctcaca atggagttgg gcaatctccc 120
atcaaaacta tttctgtcca agtaaaactt cttcaagtaa accattcctt caaaactatc 180
gtccggtatg tcccctgaaa antcattgta cgacaagtac aaggctctca accctcgaag 240
cactttgaat tcaggatttc cccccccaaa gtggttgctt atcacgctga ggctgcgcag 300
agtgttggtg agcttgagca atgtgtagac atcgatgggt ccacctagtc ccatgttttc 360
aagcttaaat ccgtaaaggc ttccattcac gcatgtcaac cct 403

SEQ ID NO: 84      moltype = DNA length = 403
FEATURE           Location/Qualifiers
source            1..403
                mol_type = other DNA
                organism = Cannabis sp.
variation         202
                allele = n = a/t
SEQUENCE: 84
atcaagtaac tgaagttggg gaatagacca gagtgaacca ggaagagcac catcgagttt 60
attgtcactc agaaccacaa cccgcagctg agaaaggctt gaaaacaagc cctttggcaa 120
tggaacctct agaccattag cactaatatc aataaccttc aagctcctca aaccctgtaa 180
ttcttttaga aatgaccacg tnaaaaaatt cttgctacaa tttagctcaa caagtcgaga 240
aagtttatcc aactgaacag gtatagaagc ggtaagacta ttatcagaag tacttaaaaa 300
ctcaagcttc gagaggttac ctaaacctga tggaaatcgc ccagaaaaga aatttgagga 360

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gaggtcaaga gtagtaagat ttcgaagtga ggtaaaactcg aac 403

SEQ ID NO: 85 moltype = DNA length = 403
FEATURE Location/Qualifiers
source 1..403
 mol_type = other DNA
 organism = Cannabis sp.
variation 202
 allele = n = g/a

SEQUENCE: 85
tgaaagataa aactagcatt aaaaaataa aaattaaaaa tattcttatg atatcgaact 60
gcaggcagaa tgtctgttgg tgaagatcga tatccattgc cgtattcaag gggatgttgt 120
tcttgagtgt atccacttag acgaagatct tgtacgtgag gagatgatgt ttagagttat 180
gttccataca gcattttgtgc gntcaaatat tttgttgcta aatcgtgatg aaattgatgt 240
tttgtgggat gccaaagacc agtttcccag ggactttaag gcagaggtag gtattactat 300
tcttctgtgg tttctttttg tgaagctgtt ggagttgaat catactcata tctgtatcag 360
gtactttttt tggatgctga tgcggtgtgc cctgatatca cca 403

SEQ ID NO: 86 moltype = DNA length = 403
FEATURE Location/Qualifiers
source 1..403
 mol_type = other DNA
 organism = Cannabis sp.
variation 202
 allele = n = t/c

SEQUENCE: 86
ggtacgtgca ataaaaatatt taaacttatt ttcagcacga taaattatatt aaaattttat 60
aaaatttaac aataaaattca aacaaaatga taaataacag tgtataatat tttgatacat 120
aatttttagaa cacactttat ttgaaagctg aaaattaatt attaatcagt acttaagata 180
gttagagaa caatccaaca tnttggaatg ggcctcattt gcagattttg atgctgcttc 240
atcatccaag ttgtacctca ctgtccaacc atgttcaacc tttgggataa tctcaacatg 300
gctctcaacc tataccattg ccaaccattc attaatattt gcatacaatt aattaatata 360
attaaaactt attcatttat acaattcata tcaattatta tca 403

SEQ ID NO: 87 moltype = DNA length = 403
FEATURE Location/Qualifiers
source 1..403
 mol_type = other DNA
 organism = Cannabis sp.
variation 202
 allele = n = g/t

SEQUENCE: 87
aaaatcatct tctgcttctt cttcagctgt tagaagaccc aaattatcga tctctgatca 60
acaatctaaa attaacgggtg gtgatcgcac tgttaaatcc tctttggact cgtatgaccgt 120
cgatgggtac cttcgcactg aagcctcagc tatcactgct gagtctactc ttttagatgc 180
acaaattact ctgacgcacc cnaectccgac aacctccatt tctgctgctg cgggtggcagt 240
ggctcctgga gatttgaatt ccggcagcat tggctcatct agtgctccta agactgttga 300
tgaagtgtgg cgggagatcg tgtctggaga taggaaggag tgtaagagg aggagcagga 360
tatggtgatg acgcttgaag actttttact tgccaaaact ggg 403

SEQ ID NO: 88 moltype = DNA length = 403
FEATURE Location/Qualifiers
source 1..403
 mol_type = other DNA
 organism = Cannabis sp.
variation 202
 allele = n = t/g

SEQUENCE: 88
gagcacacta ttgttatggt actattttat gacattttta atgaatgtga taagtattat 60
tcaaatcaat tgccatgaat agcttctgaa tctccatgcc gatgattttc atcgagggtta 120
ggattatttg tctgtatttg tggcggggtt gcaggtagtt gatccatctg agatgtttct 180
acggaagcct gctgttggtt tncctgagag cgtcgttgtc tttccattt tgaatctcg 240
acaagtatag aatcccaca catagtaacc ccgaatccgg taaatgtagc taacagaaca 300
gatagaactc cttgcatatg aagctgtaga ttacatgata gcaacatttt aaacagtcta 360
tactgaagat ctttaattgaa aatagttttc ttttgaataac caa 403

SEQ ID NO: 89 moltype = DNA length = 403
FEATURE Location/Qualifiers
source 1..403
 mol_type = other DNA
 organism = Cannabis sp.
variation 202
 allele = n = t/g

SEQUENCE: 89
caaagtctag ctgtaagata gaactacttc tgaattttga ttcaacttct gtagttctaa 60
ctatatcttc taacactggc actgaacgtt cagagtggta tagcaaacag agctgtggga 120

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gagaccgta tccacaatta acttttacaa tataatgaca gtagattcct ctgtattctc 180
tttggggtag tctagattta tntgaagacc atattttcct tgtttctgac cttattataa 240
atagaaatga acattaccag cagcagaagt cattgtgttt acatattcac tctccagcaa 300
gaattgaaga aagataatcg gtacaacatc tatttgtgtt tgtatcaggt tttacataa 360
actaatattg aactttgaaa atctactgac aggtcgagaa cta 403

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SEQ ID NO: 90      moltype = DNA length = 403
FEATURE           Location/Qualifiers
source            1..403
                  mol_type = other DNA
                  organism = Cannabis sp.
variation         202
                  allele = n = a/g

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SEQUENCE: 90
gttgtatcga aacctattcg cccagctttg gcaaattcca agggaaactt ttgtttgtag 60
caaccatttc tgtagagaa agcagcttgg caattacact cattcaaaca gctttgttta 120
caagatcctt tgtcaaccag tgagaggaat tcgtaagtgt ccggttccca tctcaaaccg 180
tctaattcag ctatagaaac cnccttgatc cagccttcta tgggtgaatt tctgttgacg 240
cctaaagtct tctgggtttt atcaatgtaa tctaaaccag gaagacaaag acaaacaggt 300
tcttgattta taagctgaca atatgaattg atgccacata atccaattgg agcacaaga 360
tggttagttg aagaccactc acaaaaccag ctgctgttct gaa 403

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SEQ ID NO: 91      moltype = DNA length = 22
FEATURE           Location/Qualifiers
source            1..22
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 91
tggctaaagt catgctccat gt 22

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SEQ ID NO: 92      moltype = DNA length = 23
FEATURE           Location/Qualifiers
source            1..23
                  mol_type = other DNA
                  organism = synthetic construct

```

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SEQUENCE: 92
cacacctagt gggatgtatc agt 23

```

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SEQ ID NO: 93      moltype = DNA length = 20
FEATURE           Location/Qualifiers
source            1..20
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 93
ctgtcgaggc ccatctcaaa 20

```

```

SEQ ID NO: 94      moltype = DNA length = 21
FEATURE           Location/Qualifiers
source            1..21
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 94
agcgaggcaa taccatcaag a 21

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```

SEQ ID NO: 95      moltype = DNA length = 20
FEATURE           Location/Qualifiers
source            1..20
                  mol_type = other DNA
                  organism = synthetic construct

```

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SEQUENCE: 95
ggatttgctt cagcacagcc 20

```

```

SEQ ID NO: 96      moltype = DNA length = 20
FEATURE           Location/Qualifiers
source            1..20
                  mol_type = other DNA
                  organism = synthetic construct

```

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SEQUENCE: 96
cccaaaacca ctttcgccag 20

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```

SEQ ID NO: 97      moltype = DNA length = 20
FEATURE           Location/Qualifiers
source            1..20
                  mol_type = other DNA
                  organism = synthetic construct

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```

SEQUENCE: 97

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agcttgggtg atacagctgc	20
SEQ ID NO: 98	moltype = DNA length = 21
FEATURE	Location/Qualifiers
source	1..21
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 98	
acccgattgc ttactaggtc c	21
SEQ ID NO: 99	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 99	
tgcttgattc cgaggatcct	20
SEQ ID NO: 100	moltype = DNA length = 21
FEATURE	Location/Qualifiers
source	1..21
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 100	
gactttggcc tatgcagagg a	21
SEQ ID NO: 101	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 101	
acacttaccg gctcttcagg	20
SEQ ID NO: 102	moltype = DNA length = 21
FEATURE	Location/Qualifiers
source	1..21
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 102	
agaggtggaa ttggaatgcc a	21
SEQ ID NO: 103	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 103	
tattagagct gccttcgccg	20
SEQ ID NO: 104	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 104	
aaccaatccg tgaacactct	20
SEQ ID NO: 105	moltype = DNA length = 22
FEATURE	Location/Qualifiers
source	1..22
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 105	
agtcctctct tgatagccat ct	22
SEQ ID NO: 106	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 106	
ccaaagcggg aatgtgacaa	20
SEQ ID NO: 107	moltype = DNA length = 22
FEATURE	Location/Qualifiers

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source	1..22 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 107		
tggtgtggtt tctttctgtc ct		22
SEQ ID NO: 108	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 108		
tgatggtgat ggtgaggctg		20
SEQ ID NO: 109	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 109		
agcttcagag gggtttgtgt		20
SEQ ID NO: 110	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 110		
aggagtgtta cgacagcagc		20
SEQ ID NO: 111	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 111		
aggccatgat gaaacgacgt		20
SEQ ID NO: 112	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 112		
tattgggctg ggcttcgtac		20
SEQ ID NO: 113	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 113		
ttgctggcca agtgagttct		20
SEQ ID NO: 114	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 114		
ccttgtcggg gtggttcaat		20
SEQ ID NO: 115	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 115		
tccgtgccct tggtaaagag		20
SEQ ID NO: 116	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 116		

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acggcagtag tagtgcattgc	20
SEQ ID NO: 117	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 117	
tcgaaccgta ctttgccaca	20
SEQ ID NO: 118	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 118	
gaaaagtagc gtcacgggtg	20
SEQ ID NO: 119	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 119	
ggttttagcg tcgcgttg	20
SEQ ID NO: 120	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 120	
acacgcccac aaagacagg	20
SEQ ID NO: 121	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 121	
tgggtccgga agatacaagc	20
SEQ ID NO: 122	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 122	
atgaaggtgg cgcaatggaa	20
SEQ ID NO: 123	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 123	
tcagtttctc tgtctgccga	20
SEQ ID NO: 124	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 124	
ggagtatagg gcggtgggta	20
SEQ ID NO: 125	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 125	
tcctccctcc attccgatca	20
SEQ ID NO: 126	moltype = DNA length = 20
FEATURE	Location/Qualifiers

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source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 126		
actcaccctaa aagtggccaa		20
SEQ ID NO: 127	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 127		
ccacaggcag cacgtttcta		20
SEQ ID NO: 128	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 128		
ctaaggccgt ttggggacct		20
SEQ ID NO: 129	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 129		
gacctgacat caccgttgct		20
SEQ ID NO: 130	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 130		
actcgacttc tgcagagctt		20
SEQ ID NO: 131	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 131		
aggagatgtc aggagggtga		20
SEQ ID NO: 132	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 132		
ttgtttggca ccagtggact		20
SEQ ID NO: 133	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 133		
taccctactt ccttgggcga		20
SEQ ID NO: 134	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 134		
aggaaaatgg cagacaccca		20
SEQ ID NO: 135	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 135		

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tcttcattgg gaggcagcag	20
SEQ ID NO: 136	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 136	
aggctccacc actgagtgtgta	20
SEQ ID NO: 137	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 137	
tctccgaata ccctccacgt	20
SEQ ID NO: 138	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 138	
gcgtgtccga aagtttgtgt	20
SEQ ID NO: 139	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 139	
tcttcccaac tcctcctgggt	20
SEQ ID NO: 140	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 140	
tgcaagtttt gaagtggcgcg	20
SEQ ID NO: 141	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 141	
tctaaggtgg ttgcaggcag	20
SEQ ID NO: 142	moltype = DNA length = 21
FEATURE	Location/Qualifiers
source	1..21
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 142	
tggacgtcta agcaaaacaa g	21
SEQ ID NO: 143	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 143	
caagtccaac ccaggagctt	20
SEQ ID NO: 144	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 144	
ggctgagtc tgcaaagtct	20
SEQ ID NO: 145	moltype = DNA length = 20
FEATURE	Location/Qualifiers

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source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 145		
caccactgcc tccacctatg		20
SEQ ID NO: 146	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 146		
caaatgggtg cttcggttc		20
SEQ ID NO: 147	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 147		
ttttgcagcg atcgattccg		20
SEQ ID NO: 148	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 148		
ggcacgattc tacaccacca		20
SEQ ID NO: 149	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 149		
acgcctggtt gaagtgaaca		20
SEQ ID NO: 150	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 150		
acaaagtgtg ggcccacctt		20
SEQ ID NO: 151	moltype = DNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 151		
cccaatccca aatccaaaac ca		22
SEQ ID NO: 152	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 152		
ggtggcaatc gtggagagaa		20
SEQ ID NO: 153	moltype = DNA length = 24	
FEATURE	Location/Qualifiers	
source	1..24 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 153		
tggttacct gtaattggtt gaga		24
SEQ ID NO: 154	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 154		

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cttttgcagc cgatgggttcg	20
SEQ ID NO: 155	moltype = DNA length = 22
FEATURE	Location/Qualifiers
source	1..22
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 155	
tgagggccac aacttagtag tg	22
SEQ ID NO: 156	moltype = DNA length = 21
FEATURE	Location/Qualifiers
source	1..21
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 156	
tggctctgat ggagatgaca c	21
SEQ ID NO: 157	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 157	
caggctttgt ccagatgtgc	20
SEQ ID NO: 158	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 158	
acaaccagcc aagagaccaa	20
SEQ ID NO: 159	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 159	
tggactgacc tgagaccagt	20
SEQ ID NO: 160	moltype = DNA length = 21
FEATURE	Location/Qualifiers
source	1..21
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 160	
gcaacctgga gagagatacc g	21
SEQ ID NO: 161	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 161	
atccgggttc ttgttcggct	20
SEQ ID NO: 162	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 162	
atggtccgga gagttcatgt	20
SEQ ID NO: 163	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 163	
ggattcctcc attgtgggca	20
SEQ ID NO: 164	moltype = DNA length = 20
FEATURE	Location/Qualifiers

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source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 164		
acttgtccca aaaccaggca		20
SEQ ID NO: 165	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 165		
gcgctctggt ctcttcaaga		20
SEQ ID NO: 166	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 166		
aggccaatct cgcttgcttc		20
SEQ ID NO: 167	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 167		
gcagaacacc atcaaagccg		20
SEQ ID NO: 168	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 168		
cgtctacgaa gcggaggaaa		20
SEQ ID NO: 169	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 169		
ctctccgcca tagccaagac		20
SEQ ID NO: 170	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 170		
gccttgggat tcctcttcgt		20
SEQ ID NO: 171	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 171		
ctgcagggtt gtggagagtg		20
SEQ ID NO: 172	moltype = DNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 172		
gtttggaggg atttcagagc t		21
SEQ ID NO: 173	moltype = DNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 173		

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gttcactttt gccacactgc a	21
SEQ ID NO: 174	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 174	
tgtgaagctg gtaggacact	20
SEQ ID NO: 175	moltype = DNA length = 24
FEATURE	Location/Qualifiers
source	1..24
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 175	
gcagatgtac gctaaagatt tggt	24
SEQ ID NO: 176	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 176	
gcgtgggatg taacgggtcaa	20
SEQ ID NO: 177	moltype = DNA length = 22
FEATURE	Location/Qualifiers
source	1..22
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 177	
tggtctctct ctctcttcca ca	22
SEQ ID NO: 178	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 178	
tgatcgagtc tccgaacacg	20
SEQ ID NO: 179	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 179	
cggggaagga gaggaagttg	20
SEQ ID NO: 180	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 180	
gcatcaaagc cttgagtgcc	20
SEQ ID NO: 181	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 181	
cggtttcacc accactaacc	20
SEQ ID NO: 182	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 182	
gggtatcggt tggatggtca	20
SEQ ID NO: 183	moltype = DNA length = 20
FEATURE	Location/Qualifiers

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source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 183		
tgttgatgtg gcggtgacaa		20
SEQ ID NO: 184	moltype = DNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 184		
tgccacaaaa ccataatagg ga		22
SEQ ID NO: 185	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 185		
tccgacgagt ctgaccaaac		20
SEQ ID NO: 186	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 186		
caacgatttc cccagggagt		20
SEQ ID NO: 187	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
source	1..25 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 187		
tgacaaagtt gataatcaat gcagt		25
SEQ ID NO: 188	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 188		
tatgctgggtg gtggagagct		20
SEQ ID NO: 189	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 189		
ctaggatctc caagcaggcg		20
SEQ ID NO: 190	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 190		
gtcattgtaa gcctgtgcgg		20
SEQ ID NO: 191	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 191		
tttacatacc ccagcacgca		20
SEQ ID NO: 192	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 192		

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cgtgacagtt ggatgaagcg	20
SEQ ID NO: 193	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 193	
accaccttgg attccacgtc	20
SEQ ID NO: 194	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 194	
cgaccttgca gttgagccta	20
SEQ ID NO: 195	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 195	
cttctgacca ccaacaggct	20
SEQ ID NO: 196	moltype = DNA length = 22
FEATURE	Location/Qualifiers
source	1..22
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 196	
cagaagtcaa ggtttgttgc ct	22
SEQ ID NO: 197	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 197	
cogttgaagc attgcaccat	20
SEQ ID NO: 198	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 198	
tcccaacggt tccattgacg	20
SEQ ID NO: 199	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 199	
atctgacgtg gactcagtg	20
SEQ ID NO: 200	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 200	
cactattgtg gctgcaaccg	20
SEQ ID NO: 201	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 201	
ccagccaatc tcaccgatga	20
SEQ ID NO: 202	moltype = DNA length = 20
FEATURE	Location/Qualifiers

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source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 202		
gtgagaaacc ttccgacggt		20
SEQ ID NO: 203	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 203		
agttctgggt tgtttcagga aga		23
SEQ ID NO: 204	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 204		
tgaggagggc ttcaattgca		20
SEQ ID NO: 205	moltype = DNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 205		
agctagcagc accttcaatg a		21
SEQ ID NO: 206	moltype = DNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 206		
aggagctatg atgcgattgg a		21
SEQ ID NO: 207	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 207		
gtccaatgaa agccgcaaca		20
SEQ ID NO: 208	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 208		
taggtctcga gtgcaacagc		20
SEQ ID NO: 209	moltype = DNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 209		
ccttgaggga tcattggtgc t		21
SEQ ID NO: 210	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 210		
ccacattgca aacagggcag		20
SEQ ID NO: 211	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 211		

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aagaaagggt ccaaagcagt	20
SEQ ID NO: 212	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 212	
atgtatcaac ctgcggcaga	20
SEQ ID NO: 213	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 213	
ccaacaaaga gtgcctcaga	20
SEQ ID NO: 214	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 214	
tctgcttggc atccagatgg	20
SEQ ID NO: 215	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 215	
tggagtctgc atttgggtgga	20
SEQ ID NO: 216	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 216	
gcatagtgtg attgggggct	20
SEQ ID NO: 217	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 217	
cctccagggt agagtaggca	20
SEQ ID NO: 218	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 218	
tgcagtttgg gctgaaagga	20
SEQ ID NO: 219	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 219	
aactcggttg gtggtatgca	20
SEQ ID NO: 220	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 220	
accgaggcac acaaactctt	20
SEQ ID NO: 221	moltype = DNA length = 20
FEATURE	Location/Qualifiers

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source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 221		
cgg tactacc catcg tctgc		20
SEQ ID NO: 222	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 222		
tggcagagtc cacagcattt		20
SEQ ID NO: 223	moltype = DNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 223		
tgctcaccga gaatcatcac a		21
SEQ ID NO: 224	moltype = DNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 224		
agagccaaat ggaaggactg g		21
SEQ ID NO: 225	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 225		
atagagccag acgtggtgga		20
SEQ ID NO: 226	moltype = DNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 226		
ctgttcata g tcaaccccc a		21
SEQ ID NO: 227	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 227		
attcatagca tgcgcgcaac		20
SEQ ID NO: 228	moltype = DNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 228		
tcagctccaa cttcattgca g		21
SEQ ID NO: 229	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 229		
aggtttggcc gccttcaata		20
SEQ ID NO: 230	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 230		

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cgagacggag agaacctgtt	20
SEQ ID NO: 231	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 231	
tgcaagccca tcgaagtga	20
SEQ ID NO: 232	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 232	
ggtgaggctg aacaagaacc	20
SEQ ID NO: 233	moltype = DNA length = 21
FEATURE	Location/Qualifiers
source	1..21
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 233	
ggctcttatt ctagggggca c	21
SEQ ID NO: 234	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 234	
gaaagggttgaggaggcg	20
SEQ ID NO: 235	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 235	
gtgctggttg ccaaagtga	20
SEQ ID NO: 236	moltype = DNA length = 25
FEATURE	Location/Qualifiers
source	1..25
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 236	
acatcacatt tcggatgata gttct	25
SEQ ID NO: 237	moltype = DNA length = 22
FEATURE	Location/Qualifiers
source	1..22
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 237	
tcttcttctt tcttcacccc ga	22
SEQ ID NO: 238	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 238	
gcctacctct ctttctcggc	20
SEQ ID NO: 239	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 239	
ggcaactgca tgctaccatc	20
SEQ ID NO: 240	moltype = DNA length = 20
FEATURE	Location/Qualifiers

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source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 240		
acctgcacca aggagaacaa		20
SEQ ID NO: 241	moltype = DNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 241		
aggggttctg attgattggc t		21
SEQ ID NO: 242	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 242		
gcaggctaga agtagggcag		20
SEQ ID NO: 243	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 243		
cacgcgtggc ctgaattttt		20
SEQ ID NO: 244	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 244		
tttgctgtg cactagctca		20
SEQ ID NO: 245	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 245		
gggaggcaag tttgtgacc		20
SEQ ID NO: 246	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 246		
tgcataaacac atccgggagt		20
SEQ ID NO: 247	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 247		
atggcagcag agcagattga		20
SEQ ID NO: 248	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 248		
gctgggagga tgaccactct		20
SEQ ID NO: 249	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 249		

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agagcttcac acaatggcgt	20
SEQ ID NO: 250	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 250	
taaactcccc actacaccgc	20
SEQ ID NO: 251	moltype = DNA length = 21
FEATURE	Location/Qualifiers
source	1..21
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 251	
ccgagatggc tggaaatgag a	21
SEQ ID NO: 252	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 252	
aacttcacgc gactccccag	20
SEQ ID NO: 253	moltype = DNA length = 21
FEATURE	Location/Qualifiers
source	1..21
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 253	
tcatgcatgc ttaactgggg t	21
SEQ ID NO: 254	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 254	
actctgttcc attgcaggaa	20
SEQ ID NO: 255	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 255	
atcctggcgt ggaaactctg	20
SEQ ID NO: 256	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 256	
ccaagctccc aaccactctg	20
SEQ ID NO: 257	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 257	
tcactcagaa ccacaaccgc	20
SEQ ID NO: 258	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 258	
ctgggtcgat tccatcaggt	20
SEQ ID NO: 259	moltype = DNA length = 20
FEATURE	Location/Qualifiers

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source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 259		
tgccgtattc aaggggatgt		20
SEQ ID NO: 260	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 260		
atcagggaca acggcatcag		20
SEQ ID NO: 261	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 261		
tggaatgggc ctcatttgca		20
SEQ ID NO: 262	moltype = DNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 262		
aggttgagag ccatgttgag a		21
SEQ ID NO: 263	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 263		
cgttggtgat cgcactgtta		20
SEQ ID NO: 264	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 264		
gcgtcatcac catatcctgc		20
SEQ ID NO: 265	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 265		
gcttctgaat ctccatgccg		20
SEQ ID NO: 266	moltype = DNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 266		
acagcttcat atgcaaggag t		21
SEQ ID NO: 267	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 267		
gcaaacagag ctgtgggaga		20
SEQ ID NO: 268	moltype = DNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 268		

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agttctcgac ctgtcagtag a 21

SEQ ID NO: 269      moltype = DNA length = 20
FEATURE            Location/Qualifiers
source              1..20
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 269
ttcgtaagtg tccggttccc 20

SEQ ID NO: 270      moltype = DNA length = 20
FEATURE            Location/Qualifiers
source              1..20
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 270
gcagctgggt tgttgagtg 20

SEQ ID NO: 271      moltype = DNA length = 403
FEATURE            Location/Qualifiers
source              1..403
                    mol_type = other DNA
                    organism = Cannabis sp.
variation           202
                    allele = n = a/t

SEQUENCE: 271
ttaatctgaa cacactcatt tcttcaggag cagattctca ggtaaactt 60
ccttggtcaa tttcaacatc ttaactagca tagttaacat aatatacatg catctcttaa 120
ttatcttgag gatgtgacat attctagtag tccattttct cagattatcg catggagctc 180
tgataaattgg gaaagacaaa anagcacgtt ctgacagact ccagctggaa ggacaggaat 240
gtcagacaca caagtgcagt ttcacagga ccagacacac ttcctcgtcg tacatgagac 300
tcagctatct atatacgaag cctcgaaact cgaacgcata ttgcaggtag aacaatcaaa 360
cattgcttca aaaacagtca tgaatctaaa aatttattac aat 403

SEQ ID NO: 272      moltype = DNA length = 403
FEATURE            Location/Qualifiers
source              1..403
                    mol_type = other DNA
                    organism = Cannabis sp.
variation           202
                    allele = n = a/t

SEQUENCE: 272
ggctaacttac ggtgagaaaag agtgggtattt cttttcgccg cgtgatcgga aatacccgaa 60
cgggtcgaga ccgaatcggg cagctggaac cgggtactgg aaggcaaccg gagcggataa 120
accgatcggg aatccgaaac cgggtgggat taagaaggct ctggtttttt acgccggaaa 180
agctccgaaa ggagagaaaa cnaattggat tatgcatgag taccggcttg ctgacgtgga 240
cagatcagct cgtaaaaaaa acagcttaag ggtacatttt ctaattaaac tccatgatat 300
tagcaggggt agtttcgtca ttccacagca gtccaagct gtggtggtag taaatgggtc 360
ccacgtgtgc tatctgacac gtggcggctt atagcgctag att 403

SEQ ID NO: 273      moltype = DNA length = 403
FEATURE            Location/Qualifiers
source              1..403
                    mol_type = other DNA
                    organism = Cannabis sp.
variation           202
                    allele = n = t/c

SEQUENCE: 273
ggtagagtta agataattaa tttagttata cagaactgca taccgacgta ttttatagtg 60
tatacccttc ctcgtacgct ttagaagttc tcctcctctg caaacaaagt tcaatgctgt 120
gaaagctttc gattcatggc cttcgccctt ttcgtcagcc aatgcaaatg tagtacttcc 180
acagcccctt tgaagccttg tncctagaat agcagctcct cttaaggctg tagtatacac 240
aaagaaaact ctcctcataa attttattca aagaagctat tgtttaatta gttttttaca 300
tgctaatacta ttgttatgag taaaaataaa taggattgtc ctctttaaac tatgactatt 360
gtagtaataa atattttgat gtggcactaa cgtgacgtta atc 403

SEQ ID NO: 274      moltype = DNA length = 403
FEATURE            Location/Qualifiers
source              1..403
                    mol_type = other DNA
                    organism = Cannabis sp.
variation           202
                    allele = n = c/g

SEQUENCE: 274
caaggagtta tctcactaga gctacatcac taacaaaatt gatttcccac tctcataaat 60
aaaactaaaa gactgtaacc aacacatatt attgtactct tgtttagttt tgggtgagaat 120

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aaagagatga gaatgaacta cacccttatt gccacttgat ctgcagaact tttcttcaca 180
acacctctag tgcataactc anttaacaac tccaaagcct cagttgcaaa tccttcataa 240
gctagacctt caatgagaat ggtatatgtc acttcagtggt gtttacatcc tttggatacc 300
atatgatcca gaaaatcgat ggcacgactg gtttgcttag tcttacaag tcccaacatg 360
atagagtga aggtgattgc attagcctt attccaagtc ctt 403

```

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SEQ ID NO: 275      moltype = DNA length = 403
FEATURE            Location/Qualifiers
source             1..403
                   mol_type = other DNA
                   organism = Cannabis sp.
variation          202
                   allele = n = a/g

```

```

SEQUENCE: 275
atgggttcgta cattgttgct atatttattc tctttttaac ttaaattctt tcttgtgaaa 60
actttccgat tccggtttgg gtcgggtgag tcttgtcttg tgacatgttt gagagttagt 120
atgtgtgttt gtgcgtccga ttacggttgt ttatgatcta atggcgtcta gtagtagaac 180
tatggaagat ttgggacatc antgggctga tctacaagtg gaagatgaga atgaagtggg 240
gcttttattt gatcaggagg aggtgttggt tgatgagttt gatgggaggt ggtgtttagt 300
cgggaaatta ctatctgata ggccggtcga ttttgactct ctaaggaatg taatggcgct 360
tttgtggcga ccggggaaag ggatgtacgt caaagagttg gag 403

```

```

SEQ ID NO: 276      moltype = DNA length = 403
FEATURE            Location/Qualifiers
source             1..403
                   mol_type = other DNA
                   organism = Cannabis sp.
variation          202
                   allele = n = g/a

```

```

SEQUENCE: 276
tacaccagaa gcaaacttcc caacaagatc atcttcagtc atatcaagaa cctcagggct 60
gaagactgag ccattgtcat aaacagattg gacaacaaga ccatatgaaa atggcctgat 120
gccaagcttg gcaagaaggg cagccctcaga ggaaccaacc ttgtcaccct tcttgatgag 180
ctcaactggg gtgataatth cnacagtacc cttgttaatc ttagtggga tattaagcac 240
ctgttgagcc caacgggtgtt agtcataaat tctaaggaaa tacaatgaat gaaaaaatat 300
agggttagta agtgtgcaga cctggaagaa agaagtctga gatggatcga gaccagtgtt 360
gccaggagga acaaccacgt caatcggggc aatcaaacc aca 403

```

```

SEQ ID NO: 277      moltype = DNA length = 403
FEATURE            Location/Qualifiers
source             1..403
                   mol_type = other DNA
                   organism = Cannabis sp.
variation          202
                   allele = n = t/a

```

```

SEQUENCE: 277
ttagtcttcc gagtggatgc atgtgctttt acctatagct atggaaagag gagtttgcat 60
aatcacaaac atgggtgcaa gtaagattac taaataaaat tttatgcaat caacaaactt 120
ttttttttct taaaagaaat agagtttaga taaatgttct tattattgtt tctttttctg 180
ggcagtggtt ccatgtgggt cncaggagaa agttgtagaa atagccagca gcctgggact 240
gaacgtatct gtgggtgttg cttatgaggt tttgtcagc aaaacaggta ttttagtca 300
cttttaggag gatgaattgg agatgatacg tttgagattt tataaacact aaatgcttca 360
caacattcaa ttaaatcaag gattgtttta tgatttaatc atg 403

```

```

SEQ ID NO: 278      moltype = DNA length = 403
FEATURE            Location/Qualifiers
source             1..403
                   mol_type = other DNA
                   organism = Cannabis sp.
variation          202
                   allele = n = t/c

```

```

SEQUENCE: 278
gaactcttct tcaaatcatc atattgcttc ttctatacta ctagtactac tacttgact 60
tgtaatgagt tttgggcaac aacaagctga ggcaagagct ttctttgtgt ttggagattc 120
tcttgggac aatggcaata acaattattt agccaccaca gcaagagctg actctcctcc 180
ttatgggatt gactacccta cncatagacc cactggccgt ttctctaagt gcctcaacat 240
gcctgacctt atcagtaaat ctctcattaa ttttctcttt aattatctaa gtatacatat 300
atatgtatgt atctctgatt tgtgtatata tctattaact tgaataaata ggtgagcaaa 360
ttgggtcaga gcccacattg ccatacttga gtccggagct cac 403

```

```

SEQ ID NO: 279      moltype = DNA length = 403
FEATURE            Location/Qualifiers
source             1..403
                   mol_type = other DNA
                   organism = Cannabis sp.
variation          202

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allele = n = g/t

SEQUENCE: 279

tagtcaactt	ggatggccaa	gagtggtgat	ttgtatggag	ttagaagatt	atgcattaag	60
tgtattgaac	ttaaatagct	gactggtttt	caatctatat	aatcaacata	gaactctcac	120
tatcatgcgg	ggatctctga	tcactgctta	tgattttatac	tctcagatgt	tttaatccat	180
tattgactta	aatgggtttt	tngatatagg	tattcttgaa	gggcccatcc	ttgtatgcct	240
ttaggggect	ggctggccgg	ttgctccta	ttggagtcca	tatagcaatg	ctactaataa	300
tggcaggtgg	aactotaagt	gcagctggga	gcttcaaagg	aacagtcaca	gtgccgcagg	360
ggctgaattt	tgtggtcgga	gatgtactag	gcccaactgg	ggt		403

SEQ ID NO: 280 moltype = DNA length = 403

FEATURE Location/Qualifiers

source 1..403

 mol_type = other DNA

 organism = Cannabis sp.

variation 202

 allele = n = t/c

SEQUENCE: 280

aaaaagcaca	ggaagagatt	ttgagatcca	tgtttgagacc	aaaggaaagg	agtactctct	60
ctctctctct	ctgaactgtc	tgtttgagct	ttggtgagct	aaaccaacga	acgagcgagc	120
ttcaagctca	acaggccttc	tttgagtatg	tatgtaatgt	acccgtcttc	accttcttct	180
catgaaacgc	cttatgaaat	gncatccaac	tctactacca	ctgtcattac	ccataggata	240
tcaacctctt	tcccgctctc	tcacttcttc	aacctccaca	tatcaacgta	aatggaaacc	300
aaagaccact	ctcatcatta	ccaacccaac	tttattactc	atggagctct	gtccaccat	360
gtccagatg	aagcaaattc	aagctcaa	gactctcact	ggt		403

SEQ ID NO: 281 moltype = DNA length = 403

FEATURE Location/Qualifiers

source 1..403

 mol_type = other DNA

 organism = Cannabis sp.

variation 202

 allele = n = a/t

SEQUENCE: 281

tcatctgttt	gtgacattat	tctttgctta	tttttgctta	agaaaacttc	gttcgattta	60
ttaaattata	tgaagttc	gtcctttatg	tggaaatgat	attttccttt	cctgcagaca	120
aatttgcaat	tgtagccag	ctgctacttg	cggaacagtc	aggcatactc	tgcgtaacct	180
attttgaaag	gtacatatgt	cntaaaatat	atatcttttag	cttactatgg	ttacggttta	240
ctatttttct	attttgcttg	agtttgatta	tttttttctc	ttacactatt	atctgaagga	300
acacaattgg	ctcaatcccg	ctacttgctt	gcaatgtctt	gctttcgaat	ggatcttctt	360
aatgaagctg	aaacagcatt	gtgtctctct	aatgaaccca	gtg		403

SEQ ID NO: 282 moltype = DNA length = 403

FEATURE Location/Qualifiers

source 1..403

 mol_type = other DNA

 organism = Cannabis sp.

variation 202

 allele = n = g/a

SEQUENCE: 282

tcgatctgct	accctagtat	gatgagtgta	taaaagtaca	gtatagagat	tttcatacgt	60
ataaataata	tcgtggtctg	acagtcacata	atatattttt	ttatatatat	acaaatacaa	120
ttgagagtac	ataatattta	acgtatgtcc	aagttagttt	gatcgagccc	tttcgcagct	180
tcgaagcgcg	tgacgcgtgt	tntgtcaagt	cttgctagcc	gacaaaggtc	tgctttgtca	240
gtttatgagg	actttacttg	gcagttgcca	ccttcactca	cggactcaat	aaatcttact	300
ttgcacgcgc	ttcaacgtgg	tcggtgaaat	cttttcttat	tttgccaagt	ggacttagtc	360
aaagctgcct	caccttcggt	tatttatatta	gagaaattta	cat		403

SEQ ID NO: 283 moltype = DNA length = 403

FEATURE Location/Qualifiers

source 1..403

 mol_type = other DNA

 organism = Cannabis sp.

variation 202

 allele = n = c/g

SEQUENCE: 283

aaatgaggtt	aaccccagca	ctaaacagtc	taacttaaaa	ataagaacta	cctatcacca	60
tccctacaa	acttacaa	tttaattatg	cactctagta	taaaacgaaa	aaggcaatca	120
ttgtcataga	caaaacacaa	ggcatttaaa	tttaataaag	tagcaactgc	agcaccagga	180
caatacaagt	ttcagtcagt	anccaattta	cacataaacc	tatttcaata	tctgctgttg	240
tctcaacaga	agagcactaa	gcacctcaa	tatttagaac	aaacatatat	acattatata	300
attacattta	tcatocacaa	agaccacaa	ccaaatagac	cctattgaaa	gttgaaactg	360
tttaccttag	cagcaaagtg	aaaattagtt	cagcaaaaga	ggt		403

SEQ ID NO: 284 moltype = DNA length = 403

FEATURE Location/Qualifiers

-continued

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source          1..403
                mol_type = other DNA
                organism = Cannabis sp.
variation       202
                allele = n = g/a

SEQUENCE: 284
acactaaggga acacaaaaa taataaatt ccccttcatt atttccagat acaatgggga 60
ttagtcccc cctccccc cacaataacc caatcacttg agggcatgca gtgcccgtc 120
aatgataatt gttcagaaga ctaatggaat aataactgcc attcttacc acatctgtat 180
aggttgatta aatgaattt cngcaaggta ctccatttct ggatctgctc gcattccataa 240
caagggcgag tccatactgt gaaaagaaaa gtttgacaca cgcattgcaaa gacacaagaa 300
aataataatt agatcctcct atccaaataa actttaacag agcttaaaga aatgaaccta 360
cctggagcgc atatctagag gaagagtacc atctccattg tca 403

SEQ ID NO: 285      moltype = DNA length = 403
FEATURE            Location/Qualifiers
source             1..403
                  mol_type = other DNA
                  organism = Cannabis sp.
variation          202
                  allele = n = t/c

SEQUENCE: 285
aaatcaaaca gataaaccta caagcatact agtcaaactc ctttcagaaa caaccaacca 60
ccatttttgc tttataaata tacaatgacc cccaatgcac gaaaaaccac ctcggtttag 120
gaggaaagaa tatcaacaaa tagtcaaata gataaagatt agaacgatat tcatcagaag 180
gtaagggaac agagctgtga angcacgcga ccttattcta aaaatcgtgg aattgctgct 240
tctatggatt ctcaatgggtg gcattcataa tccgcagtga tcttgatctt tcctgctgcc 300
cactgcgtct ccaaaaagcc ctctccacc aaacttcgaa ccctctttgt atgttcgggc 360
aatcttcagc agagttcaag aatcggttaa accaagcaag caa 403

SEQ ID NO: 286      moltype = DNA length = 101
FEATURE            Location/Qualifiers
source             1..101
                  mol_type = other DNA
                  organism = Cannabis sp.
variation          51
                  allele = n = a/g

SEQUENCE: 286
atgaatcatg agagaatagt tctgatgatt ttgaagatgt tcatccagac ntgcctaaag 60
atgatcagga catgcccgcc gatggtgctg ctagaagaa a 101

SEQ ID NO: 287      moltype = DNA length = 20
FEATURE            Location/Qualifiers
source             1..20
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 287
tccaaaaacc actttcgcca 20

SEQ ID NO: 288      moltype = DNA length = 20
FEATURE            Location/Qualifiers
source             1..20
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 288
tcaggagcag attctcaggt 20

SEQ ID NO: 289      moltype = DNA length = 20
FEATURE            Location/Qualifiers
source             1..20
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 289
cgaggaagtg tgtctggtcc 20

SEQ ID NO: 290      moltype = DNA length = 20
FEATURE            Location/Qualifiers
source             1..20
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 290
acgacgagga agtgtgtctg 20

SEQ ID NO: 291      moltype = DNA length = 20
FEATURE            Location/Qualifiers
source             1..20

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SEQUENCE: 291	mol_type = other DNA organism = synthetic construct	
gtactggaag gcaaccggag		20
SEQ ID NO: 292	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 292		
tgtcagatag cacacgtggg		20
SEQ ID NO: 293	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 293		
tataagccgc cacgtgtcag		20
SEQ ID NO: 294	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 294		
cgtaagtgtc cggttcccat		20
SEQ ID NO: 295	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 295		
gccttcgcct ttttcgtcag		20
SEQ ID NO: 296	moltype = DNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 296		
cacgttagtg ccacatcaaa a		21
SEQ ID NO: 297	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 297		
tttcgattca tggccttcgc		20
SEQ ID NO: 298	moltype = DNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 298		
tgccacttga tctgcagaac t		21
SEQ ID NO: 299	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 299		
aaaccagtcg tgccatcgat		20
SEQ ID NO: 300	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 300		
aaccagtcgt gccatcgatt		20

-continued

SEQ ID NO: 301	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 301		
tttgggtcgg gtgagtcttg		20
SEQ ID NO: 302	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 302		
cgccacaaag acgccattac		20
SEQ ID NO: 303	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 303		
gtttgggtcg ggtgagtctt		20
SEQ ID NO: 304	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 304		
tcgccacaaa gacgccatta		20
SEQ ID NO: 305	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 305		
acctcagggc tgaagactga		20
SEQ ID NO: 306	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 306		
tccaggtctg cacacttacc		20
SEQ ID NO: 307	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 307		
ccaggtctgc acacttacct		20
SEQ ID NO: 308	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 308		
ttccgagtgg atgcatgtgc		20
SEQ ID NO: 309	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 309		
agcaacaccc acagatacgt		20
SEQ ID NO: 310	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	

-continued

SEQUENCE: 310	mol_type = other DNA organism = synthetic construct	
cgagtggatg catgtgcttt		20
SEQ ID NO: 311	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 311		
agctgaggca agagctttct		20
SEQ ID NO: 312	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 312		
atggcaatgt gggctctgac		20
SEQ ID NO: 313	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 313		
gcaacaacaa gctgaggcaa		20
SEQ ID NO: 314	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 314		
ggctctgacc caatttgctc		20
SEQ ID NO: 315	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 315		
gcggggatct ctgatcactg		20
SEQ ID NO: 316	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 316		
tgggcctagt acatctccga		20
SEQ ID NO: 317	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 317		
cggggatctc tgatcactgc		20
SEQ ID NO: 318	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 318		
gctaaaccaa cgaacgagcg		20
SEQ ID NO: 319	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 319		
tgcttcactc ggagcatggt		20

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SEQ ID NO: 320	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 320		
gagcatggtg gagcaagact		20
SEQ ID NO: 321	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 321		
ttactggtgc aagcccatcg		20
SEQ ID NO: 322	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 322		
tgcaattgtt agccagctgc		20
SEQ ID NO: 323	moltype = DNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 323		
cactgggttc attaggagga ca		22
SEQ ID NO: 324	moltype = DNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 324		
actgggttca ttaggaggac ac		22
SEQ ID NO: 325	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 325		
gtttgatcga gccctttcgc		20
SEQ ID NO: 326	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 326		
taaacgaagg tgaggcagct		20
SEQ ID NO: 327	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 327		
tttgatcgag ccccttcgca		20
SEQ ID NO: 328	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 328		
tgccctggag gatcattggt		20
SEQ ID NO: 329	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	

-continued

	mol_type = other DNA organism = synthetic construct	
SEQUENCE: 329 atccacattg caaacagggc		20
SEQ ID NO: 330 FEATURE source	moltype = DNA length = 20 Location/Qualifiers 1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 330 atgagggttaa ccccgact		20
SEQ ID NO: 331 FEATURE source	moltype = DNA length = 21 Location/Qualifiers 1..21 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 331 gggtgcttag tgctctctg t		21
SEQ ID NO: 332 FEATURE source	moltype = DNA length = 21 Location/Qualifiers 1..21 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 332 tgagggtgct tagtgctctt c		21
SEQ ID NO: 333 FEATURE source	moltype = DNA length = 20 Location/Qualifiers 1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 333 ccaatcactt gagggcatgc		20
SEQ ID NO: 334 FEATURE source	moltype = DNA length = 20 Location/Qualifiers 1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 334 gcgctccagg taggttcatt		20
SEQ ID NO: 335 FEATURE source	moltype = DNA length = 21 Location/Qualifiers 1..21 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 335 ccctcccaca cacatatacc c		21
SEQ ID NO: 336 FEATURE source	moltype = DNA length = 20 Location/Qualifiers 1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 336 aatgaccccc aatgcacgaa		20
SEQ ID NO: 337 FEATURE source	moltype = DNA length = 20 Location/Qualifiers 1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 337 ctctgctgaa gattgccga		20
SEQ ID NO: 338 FEATURE source	moltype = DNA length = 20 Location/Qualifiers 1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 338 atgaccccca atgcacgaaa		20

SEQ ID NO: 339	moltype = DNA length = 22		
FEATURE	Location/Qualifiers		
source	1..22		
	mol_type = other DNA		
	organism = synthetic construct		
SEQUENCE: 339			
cctatggctcg aagacgatga at			22
SEQ ID NO: 340	moltype = DNA length = 22		
FEATURE	Location/Qualifiers		
source	1..22		
	mol_type = other DNA		
	organism = synthetic construct		
SEQUENCE: 340			
atttggctcag cattagccat tt			22
SEQ ID NO: 341	moltype = DNA length = 21		
FEATURE	Location/Qualifiers		
source	1..21		
	mol_type = other DNA		
	organism = synthetic construct		
SEQUENCE: 341			
gaagacgatg aatcatgaga g			21
SEQ ID NO: 342	moltype = DNA length = 22		
FEATURE	Location/Qualifiers		
source	1..22		
	mol_type = other DNA		
	organism = synthetic construct		
SEQUENCE: 342			
tttattttatc gcgggtgaa g at			22
SEQ ID NO: 343	moltype = DNA length = 6294		
FEATURE	Location/Qualifiers		
source	1..6294		
	mol_type = other DNA		
	organism = Cannabis sp.		
variation	1		
	allele = n = a/g		
SEQUENCE: 343			
ntgcctaaag atgatcagga	catgcccgcg	gatgggtgctg	ctagaaagaa agataatcct 60
gatgagtcag atcctgtggt	cacaatggcc	aagaaattac	aaatagctaa aaataaagg 120
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1. A method for characterizing a *Cannabis* spp. plant with respect to a hermaphroditism trait, the method comprising the steps of:

- (i) genotyping at least one plant with respect to a hermaphroditism QTL by detecting (a) one or more polymorphisms associated with the hermaphroditism trait as defined in any one of Tables 1, 3 to 11 and 15; and/or (b) a polymorphism causal for the hermaphroditism trait which is an A/G SNP at position 99811627 on chromosome NC_044370.1 with reference to the cs10 reference genome; and
- (ii) characterizing the plant as having a hermaphroditism presence QTL or a hermaphroditism absence QTL based on the genotype at the polymorphism.

2. The method of claim 1, wherein the polymorphism is selected from the group consisting of “common_491”, “common_512”, “common_517”, “common_518”, “common_525”, “rare_57”, “rare_50”, “common_534”, “common_511”, “GBScompat_common_56”, “common_54”, “common_294”, as defined in any one of Tables 3 to 11 and 15, the A/G SNP at position 99811627 on chromosome NC 044370.1 with reference to the cs10 reference genome, and combinations thereof.

3. (canceled)

4. The method of claim 1, wherein the genotyping is performed by PCR-based detection using molecular markers, sequencing of PCR products containing the one or more polymorphisms, targeted resequencing, whole genome sequencing, or restriction-based methods, for detecting the one or more polymorphisms.

5. The method of claim 4, wherein the molecular markers are for detecting polymorphisms at regular intervals within the hermaphroditism QTL such that recombination can be excluded; or wherein the molecular markers are for detecting polymorphisms at regular intervals within the hermaphroditism QTL such that recombination can be quantified to estimate linkage disequilibrium between a particular polymorphism and a hermaphroditic phenotype; or wherein the molecular markers are designed based on a context sequence for the polymorphism in Tables 1, 3 to 11 or 15 or are selected from the primer pairs as defined in Table 12 or 17.

6. (canceled)

7. (canceled)

8. The method of claim 1, wherein the hermaphroditism QTL is a quantitative trait locus having a sequence that corresponds to nucleotides 28544332 to 35677966 of NC_044370.1 with reference to the CS10 genome and is defined by one or more polymorphisms associated with hermaphroditism as defined in any one of Tables 1, 3 to 11 and 15, or a genetic marker linked to the QTL.

9. A method of producing a *Cannabis* spp. plant having a hermaphroditism trait of interest, the method comprising the steps of:

- (i) providing a donor parent plant having in its genome a hermaphroditism QTL characterized by (a) one or more polymorphisms associated with the hermaphroditism trait of interest as defined in any one of Tables 1, 3 to 11 and 15; and/or (b) a polymorphism causal for the hermaphroditism trait of interest which is an A/G SNP at position 99811627 on chromosome NC 044370.1 with reference to the cs10 reference genome;

- (ii) crossing the donor parent plant having the hermaphroditism QTL with at least one recipient parent plant to obtain a progeny population of cannabis plants;

- (iii) screening the progeny population of cannabis plants for the presence of the hermaphroditism QTL; and

- (iv) selecting one or more progeny plants having the hermaphroditism QTL, wherein the mature plant displays the hermaphroditism trait of interest.

10. The method of claim 9, further comprising:

- (v) crossing the one or more progeny plants with the donor recipient plant; or

- (vi) selfing the one or more progeny plants.

11. The method of claim 9, wherein the screening comprises genotyping at least one plant from the progeny population with respect to the hermaphroditism QTL by detecting one or more polymorphisms associated with the hermaphroditism trait of interest as defined in any one of Tables 1, 3 to 11 and 15 and/or the polymorphism causal for the hermaphroditism trait of interest, and optionally wherein the method further comprises a step of genotyping the donor parent plant with respect to the hermaphroditism QTL by detecting one or more polymorphisms associated with the hermaphroditism trait of interest as defined in any one of Tables 1, 3 to 11 and 15 and/or the polymorphism causal for the hermaphroditism trait of interest, prior to step (i).

12. (canceled)

13. The method of claim 11, wherein the genotyping is performed by PCR-based detection using molecular markers, sequencing of PCR products containing the one or more polymorphisms, targeted resequencing, whole genome sequencing, or restriction-based methods, for detecting the one or more polymorphisms.

14. The method of claim 13, wherein the molecular markers are for detecting polymorphisms at regular intervals within the hermaphroditism QTL such that recombination can be excluded or such that recombination can be quantified to estimate linkage disequilibrium between a particular polymorphism and the hermaphroditism trait of interest, optionally wherein the molecular markers are designed based on a context sequence for the polymorphism in Tables 1, 3 to 11 or 15 or are selected from the primer pairs as defined in Table 12 or 17.

15. (canceled)

16. The method of claim 9, wherein the polymorphism is selected from the group consisting of “common_491”, “common_512”, “common_517”, “common_518”, “common_525”, “rare_57”, “rare_50”, “common_534”, “common_511”, “GBScompat_common_56”, “common_54”, “common_294”, as defined in any one of Tables 3 to 11 and 15, an A/G SNP at position 99811627 on chromosome NC 044370.1 with reference to the cs10 reference genome, and combinations thereof.

17. (canceled)

18. The method of claim 9, wherein the hermaphroditism QTL is a hermaphroditism absence QTL and the hermaphroditism trait of interest is a hermaphroditism absence trait.

19. The method of claim 9, wherein the hermaphroditism QTL is a quantitative trait locus having a sequence that corresponds to nucleotides 28544332 to 35677966 of NC 044370.1 with reference to the CS10 genome and is defined by one or more polymorphisms associated with hermaphroditism as defined in any one of Tables 1, 3 to 11 and 15, or a genetic marker linked to the QTL.

20. A method of producing a *Cannabis* spp. plant comprising a hermaphroditism trait of interest, the method comprising introducing into a *Cannabis* spp. plant a hermaphroditism QTL:

- (a) characterized by one or more polymorphisms associated with the hermaphroditism trait of interest as defined in any one of Tables 1, 3 to 11 and 15, wherein said hermaphroditism QTL is associated with the hermaphroditism trait of interest in the plant; and/or
- (b) comprising a polymorphism causal for the hermaphroditism trait of interest which is an A/G SNP at position 99811627 on chromosome NC_044370.1 with reference to the cs10 reference genome.

21. The method of claim **20**, wherein introducing the hermaphroditism QTL comprises crossing a donor parent plant having the hermaphroditism QTL with a recipient parent plant.

22. The method of claim **20**, wherein introducing the hermaphroditism QTL comprises genetically modifying the *Cannabis* spp. plant.

23. The method of claim **22**, wherein genetically modifying the *Cannabis* spp. plant is by targeted mutagenesis of guanine to adenine (G>A) at position 99811627 on chromosome NC_044370.1 with reference to the cs10 reference genome.

24. The method of claim **20**, wherein the hermaphroditism QTL is a quantitative trait locus having a sequence that corresponds to nucleotides 28544332 to 35677966 of NC

044370.1 with reference to the CS10 genome and is defined by one or more polymorphisms associated with hermaphroditism as defined in any one of Tables 1, 3 to 11 and 15, or a genetic marker linked to the QTL, optionally wherein the hermaphroditism QTL is a hermaphroditism absence QTL, and the hermaphroditism trait of interest is a hermaphroditism absence trait.

25. (canceled)

26. (canceled)

27. (canceled)

28. (canceled)

29. A *Cannabis* spp. plant comprising a hermaphroditism QTL:

- (a) characterized by one or more polymorphisms associated with the hermaphroditism trait of interest as defined in any one of Tables 1, 3 to 11 and 15, wherein said hermaphroditism QTL is associated with the hermaphroditism trait of interest in the plant; and/or
- (b) comprising a polymorphism causal for the hermaphroditism trait of interest which is an A/G SNP at position 99811627 on chromosome NC_044370.1 with reference to the cs10 reference genome, optionally wherein the plant comprises a hermaphroditism absence QTL and displays a hermaphroditism absence trait.

30.-37. (canceled)

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